FALCON AUTOTECH

FALCON's Response to Flipkart for Supply of 160K + 20K Loop CBS system (Kalash)

November 19, 2023

www.falconautotech.com

Kind Attention –

M/s Flipkart

Offer Ref: F22-403 Option 2; Date: 19-11-2023

Subject – Techno -Commercial Offer of Loop Cross Belt Sorter 160K + 20K PPH

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have done an in-depth data analysis and evaluated various solution options best suited for your requirements, along with the information collected during the meetings and discussions with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Intra-logistics Automation Technologies and references.

To conclude, I would like to add my personal commitment on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

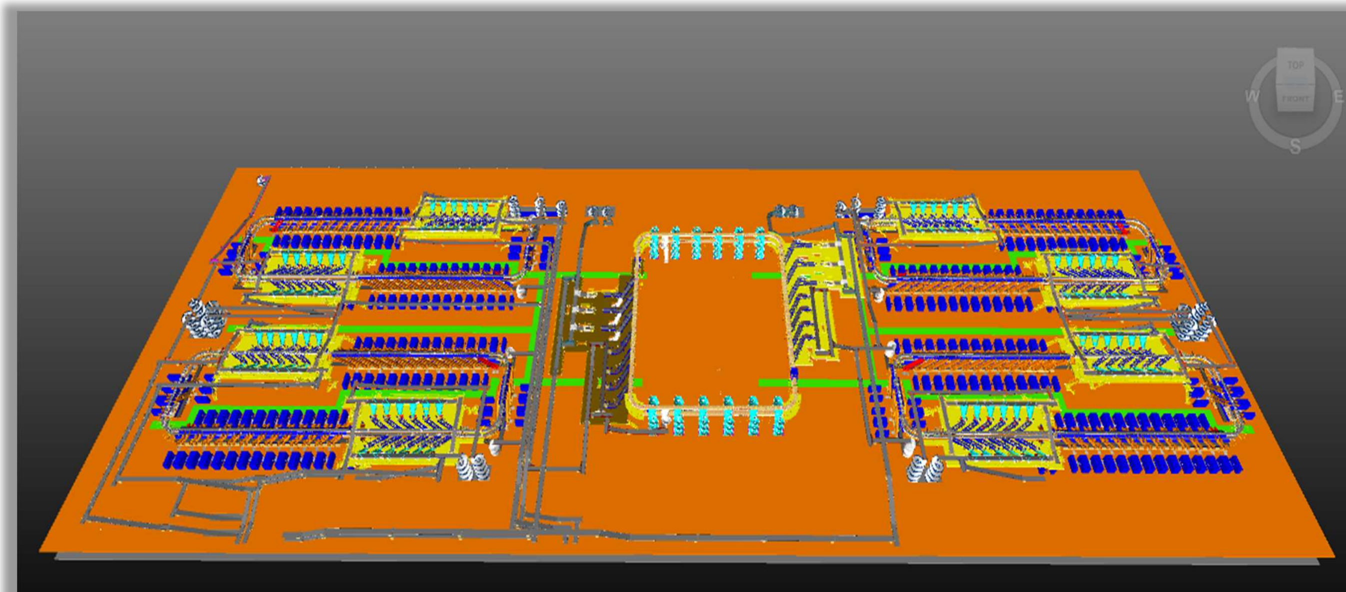
Best Regards,

Sandeep Bansal

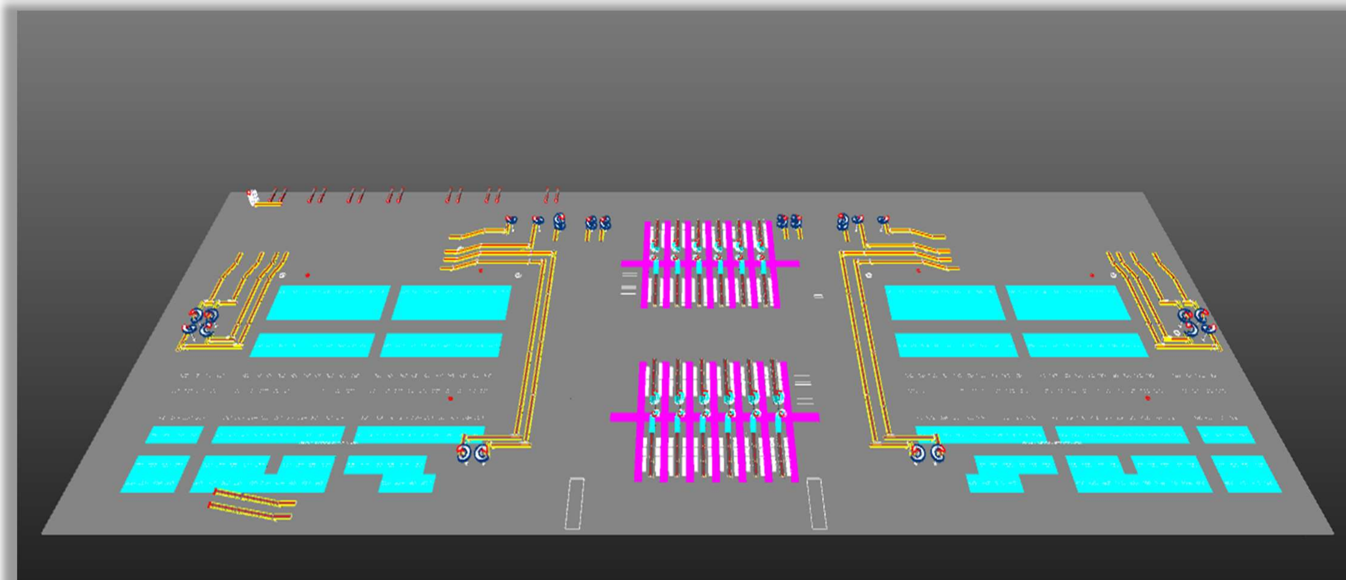
Chief Business Officer

Response to RFP for Loop Cross Belt Sorter 160K + 20K PPH Throughput

Proposal Reference – F22-403 – Option 2



G+1 Level



G Level

Contents

1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Shipments Per Hour
3	ICR	Intelligent Character Recognition
4	MEZZ	Mezzanine
5	FWD	Friction Wheel Drive
6	ECDS	Empty Carrier Detection System
7	AC	Alternating Current
8	DC	Direct Current
9	PLC	Programmable Logic Controller
10	IT	Information Technology
11	BOQ	Bill Of Quantity
12	I/O	Input/ Output
13	PDP	Power Distribution Panel
14	PC	Personal Computer
15	UPS	Uninterrupted Power Supply
16	CBS	Cross Belt Sorter
17	MDR	Motor Driven Roller
18	DAP	Design Approval Phase

2. Executive Summary

Falcon Autotech is pleased to confirm its great interest in responding to this Flipkart RFQ of Loop CBS for Kalash North. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening, understanding your needs, and ensuring this project's success.

Following the same objective for the Kalash CBS system, we are happy to offer a compliant solution meeting all technical and operational requirements at competitive price, delivering key results.

Our solution is based on the following key characteristics:

Shipment Sorter 160K PPH:-

- *Four double deck loops of shipment sorter, employing our cross-belt technology, have been designed to handle a throughput of approximately 20K packages per hour on each loop, making the total capacity of the system 160K sorts per hour.*

- *Each loop is equipped with its own infeed conveyor system, comprising of:*

    - *Infeed Conveyor Line – Conveyor based infeed lines are provided to transport the FC volume to the VDS loop.*
    - *Powered Spiral Line – Spiral conveyors followed by powered conveyors are present to transport the Marketplace load into the VDS loop.*

  Furthermore, to cater the priority line/load an additional Spiral conveyor is incorporated in the system.

- *For each loop there are two induction zones featuring six induct lines and provision for irregular shipment takeout in each zone.*
- *Every double deck system has 60 secondary chutes (L-Type), 60 Direct bagging chute (L-Type) and four rejection chutes (L-Type).*
- *Bagging conveyors are planned below the loop to transport the bags to the Bag & Semi-Large Sorter.*
- *Provision of irregular shipments take out & segregation via PTL setup is provided near each induction zone..*

Bag & Semi-Large Sorter 20K PPH:-

- *Two loop of shipment sorter(Double Deck), employing our cross-belt technology, have been designed to handle a throughput of approx. 10K packages per hour on each loop, making the total system capacity of 20K PPH*
- *Each Loop is equipped with its own infeed conveyor system, comprising of:*
    - *Bag Infeed Conveyor – Conveyor based infeed line followed by spiral conveyor is provided to transport the Cross dock volume to the dump VDS loop & separate Spiral conveyors are present to transport bags from shipments sorters for further sortation in Bag sorter.*

- *Semi-Large Infeed conveyor – Two Spiral conveyors are present to transport the Cross dock load into SWEDI based VDS loop & one conveyor based infeed line is present to transport Semi large shipments from Fulfilment center.*

- *Within each loop one induction zone featuring seven induct lines. Where three automatic inducts are dedicated for Semi-large shipments & four semi-auto inducts are dedicated for Bags.*
- *In loop, there are 24 Spiral chutes, and two rejection spiral chutes.*
- *The system layout has been meticulously planned to facilitate smooth operational flow, ensuring efficient and safe man movement*
- *Furthermore, additional conveyor automation is implemented, ensuring refeeding of bags automatically when collected in the rejection chute.*
- *Spiral chutes are present to transport bags & SL Shipments to Ground (G) level, which are further connected to small sections of conveyors delivering the bags into the Trolley.*

This solution has been crafted especially keeping Flipkart's technical and operational requirements, as listed in the RFP document, making it a tailor-made system delivering a faster TAT and an efficient material flow.

*The Falcon Autotech Cross Belt Sorter, proudly **MADE IN INDIA**, incorporates in-house designed and developed system components and associated peripherals.*

1. Falcon's reliable Shipment sortation systems

Falcon Autotech has over 50+ Cross belt sorter loops installed globally which are used by most innovative brands such as Flipkart, Amazon, Delhivery, Asendia, Fastway and many more. The main and critical components of the Falcon Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from industry leading suppliers in the world, such as SEW, Siemens, SICK, Vahle, Forbo etc. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. Commitment to quality systems

Demonstrating Falcon's clear commitment to the Flipkart satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

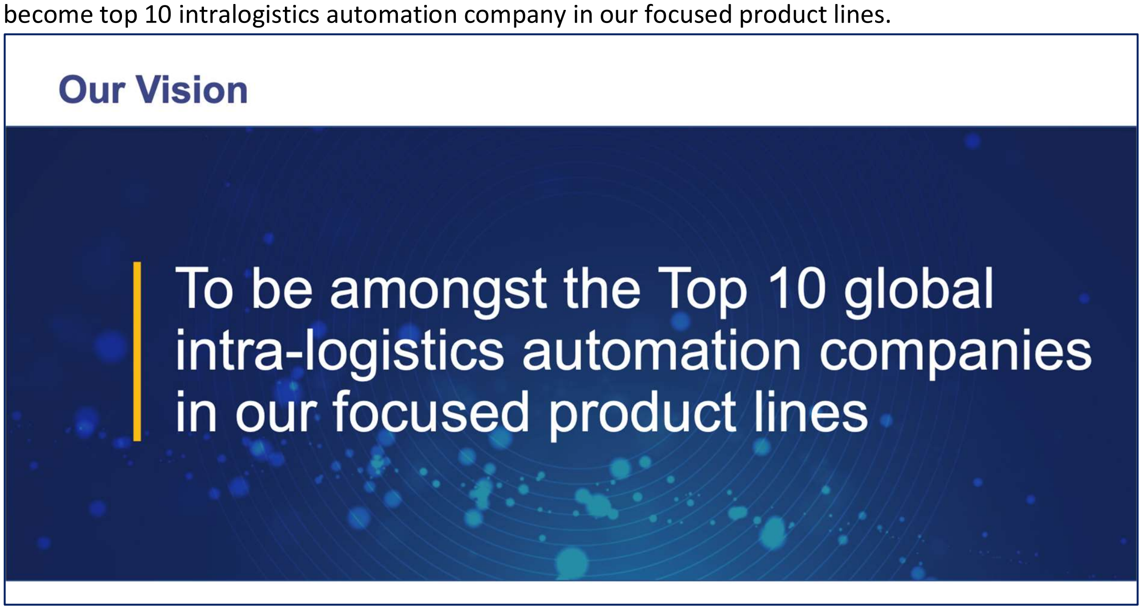
3. A strong and already successful partnership between Flipkart & Falcon

Flipkart has been Falcon's customer since long and using multiple Falcon Systems and now a new shipment sortation system of 20,000 PPH is being installed at the Ulluberia location of the Flipkart network. For each of these projects, several advanced tools to improve the service's quality and efficiency are provided: Service Desk, Hotline support, remote access, regular governance meetings and monthly reports.

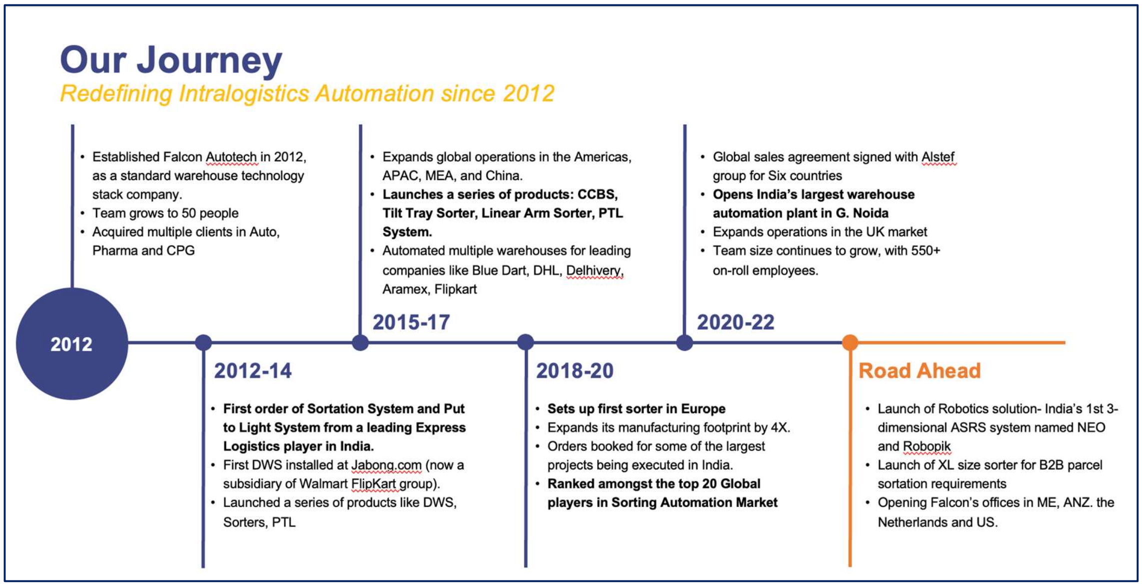
3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

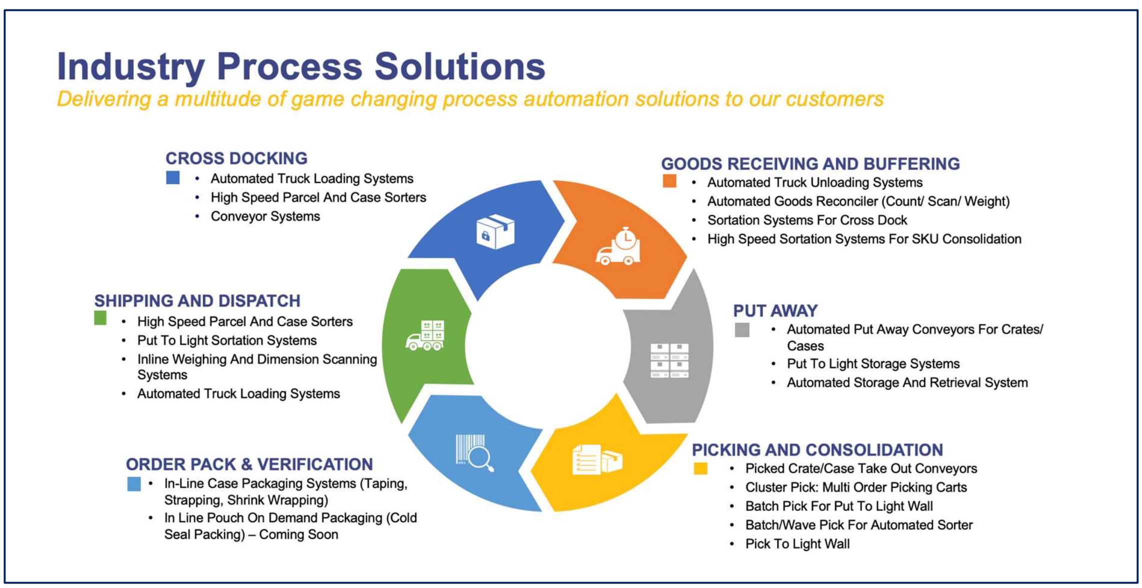
Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.



The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chai

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

Solution Specifications –

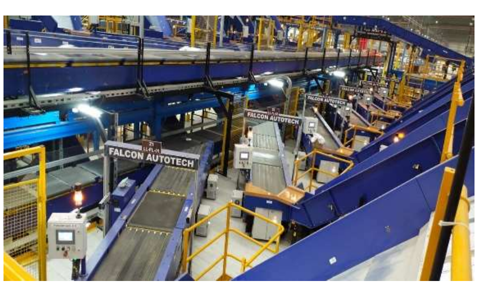
- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- 2 Sets of Double Decker CBS Sorters
- Mezzanine Structures (one large mezz on to install a double decker Sorter).
- Automated Singulators
- Fully Automatic Inductions
- Semi-Automatic Inductions
- Telescopic belt conveyors
- PVC Belt Conveyors
- Modular Belt Conveyors
- Spiral Chutes with Braking rollers
- 5-Sided Scanning Tunnels
- High speed weighing conveyors
- Direct Bagging Chutes
- Put to Light Chutes
- Volume Distribution systems
- High Availability Server Systems
- WCS

Site Pictures -



5.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

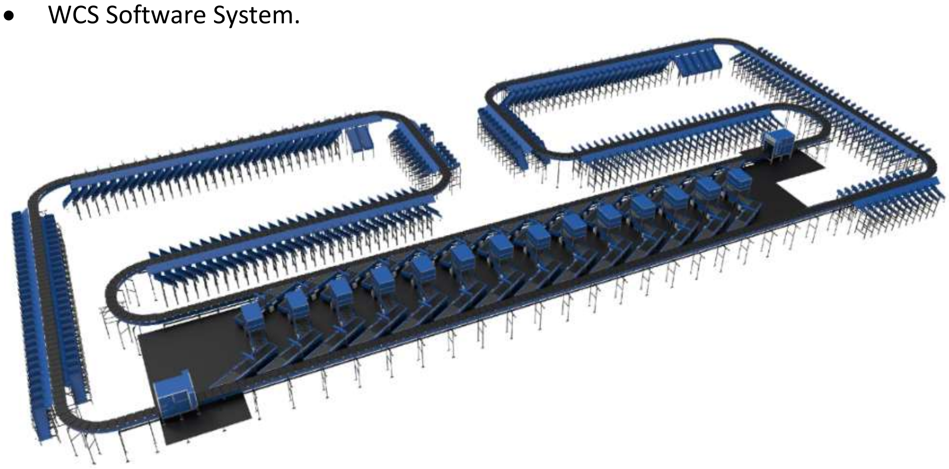
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

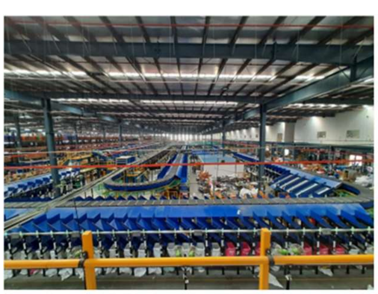
Solution Specifications –

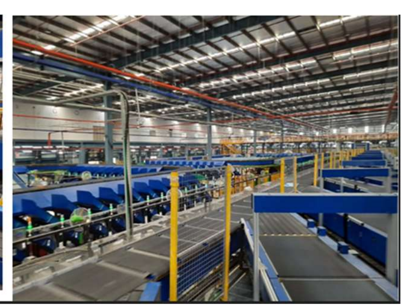
- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.







5.3 Project 3- (Ecommerce Client, India)

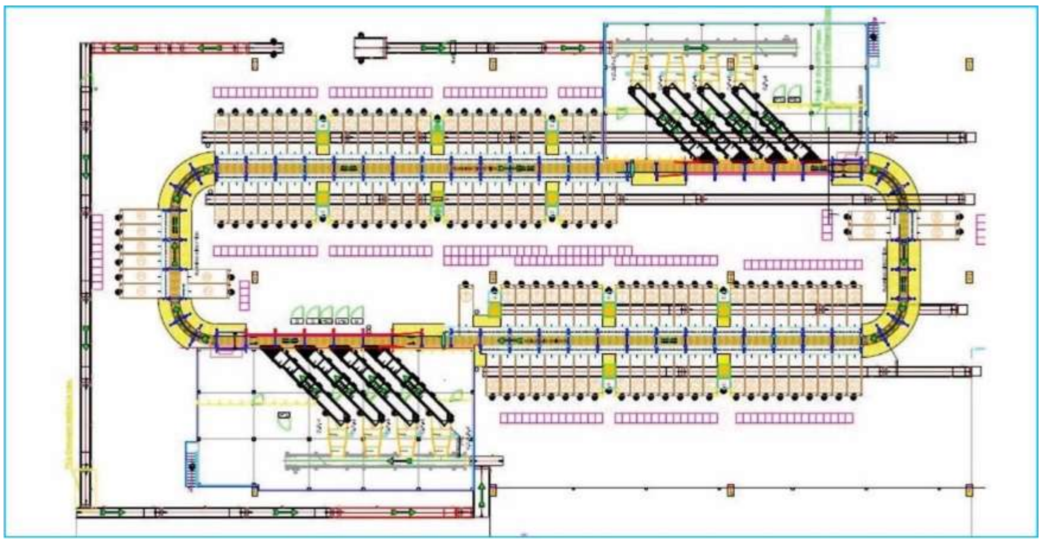
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.4 Project 4- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –

5.5 Project 5- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

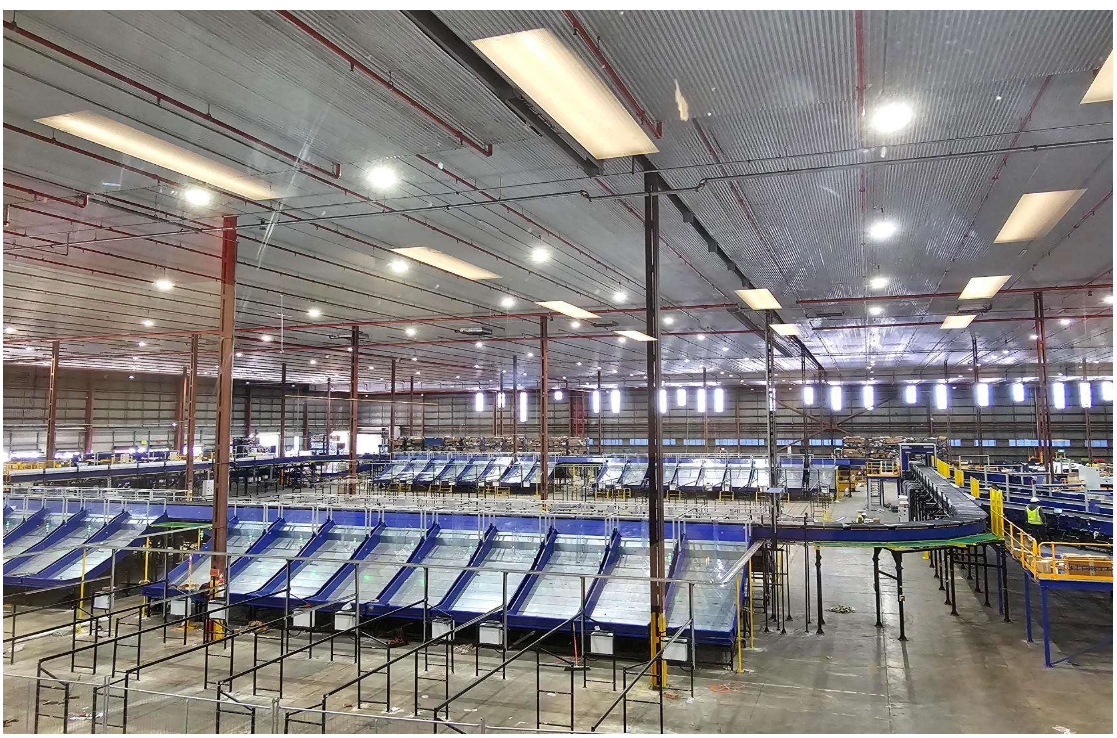
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



6. Handled Shipment Spectrum

As per the shipment spectrum data provided in the RFP documents, Falcon has studied and analysed the shipment spectrum in detail.

Falcon purposes to use its "Dual Belt Regular Configuration" for shipment sorter & "Single Belt XL Configuration" for bag & Semi-large sorter to provide the maximum benefits to Flipkart in-terms of handling various sizes and weight.

6.1 Shipment (Non-Large) size loadable on the shipment sorter

Falcon's Dual belt regular Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	600
Max Width	mm	400
Max Height	mm	300
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	100
Min Height	mm	3
Min Weight	gm	100

6.2 Bag size loadable on the Bag & Semi-large sorter

Falcon's Single belt XL Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	1000
Max Width	mm	600
Max Height	mm	600
Max Weight	Kg	40
Min length	mm	200
Min Width	mm	200
Min Height	mm	10
Min Weight	gm	1000

6.3 Semi-large shipment size loadable on the Bag & Semi-large sorter

Falcon's Single belt XL Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	800
Max Width	mm	800
Max Height	mm	500
Max Weight	Kg	25

6.4 Shipments to be loaded on Sorter shall have the following characteristics:

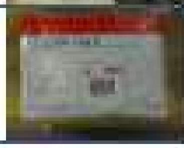
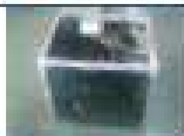
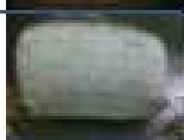
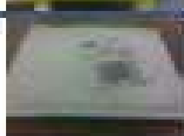
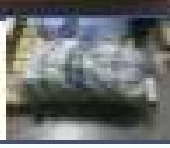
1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.5 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.6 Pictures of Sortable Shipments

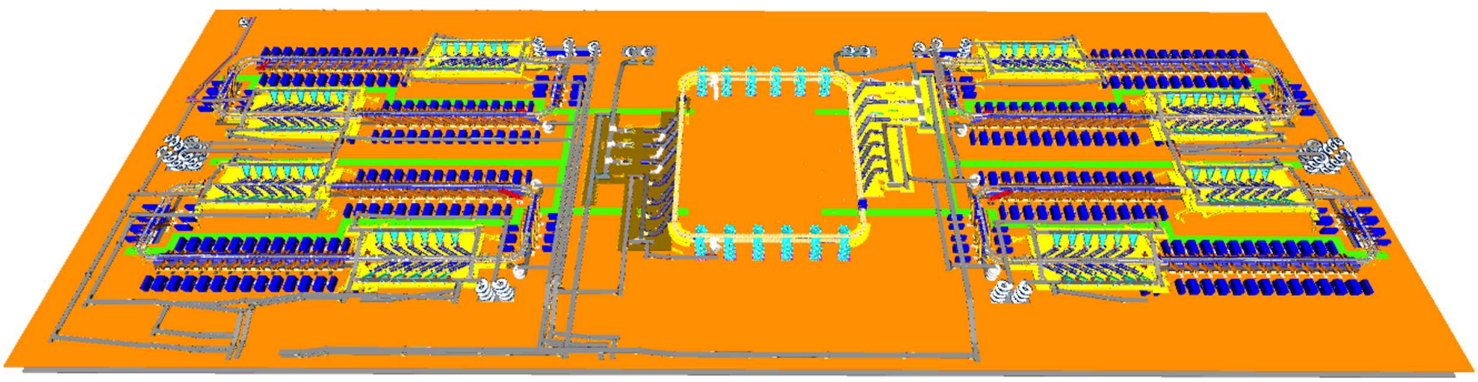
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Loop CBS for sorting shipment, Bags & Semi-Large Shipments as per Flipkart's requirements..

7.2 Summary of the System



Process flow of the Non-Large Shipment Loop CBS System-

1. **Infeed System: -** Shipments Bags are unsealed and dumped in bulk onto the infeed lines of the Cross belt sorter. The shipments ascend to a higher level and arrive at the VDS system. Once the shipments are within the VDS loop, they are evenly distributed among all inducts using arm based VDS technology.
2. **Inducts: -** After the shipments are collected in the VDS chute, an operator picks and positions each shipment on the induct line, ensuring that the shipment is properly aligned and that its barcode is facing upwards. The feedlines then automatically induct the shipments onto the Cross-Belt Sorter Loop.
3. **Loop CBS:** - Once the shipments have entered the main loop, the Cross-Belt Sorter (CBS) efficiently sorts the shipments into their respective output chutes by utilizing the data provided by Flipkart's sorting logic.
4. **Output Chutes: -** The shipments are discharged into two types of chutes.
    a. **Direct Bagging Chutes (L-type) -** Within a double decker loop CBS system, there are a total of 60 L-Type direct bagging chutes. Shipments collected within these direct bagging chutes are bagged.
    b. **Secondary Chutes (L-type) -** Within a double decker loop CBS system, there are a total of 60 L-Type Secondary chutes. Shipments collected within these secondary chutes Further undergo sortation via PTL setup (In FK Scope).

c. **Rejection Chute-** Four rejection chutes are present to handle rejected Shipments.

5. **Bag Takeaway Conveyor: -** Following the direct bagging process & secondary sorting process, the shipments are placed into bags and then manually loaded onto a bag takeaway conveyor located beneath the CBS loop. This conveyor transports the bags out of shipment sorter to bag & Semi-large sorter located in the centre of the facility.

Process flow of the Bag & Semi-large Loop CBS System-

6. **Infeed System: -** Bags from Shipment sorter & Cross dock Bags are dumped onto the infeed lines of the Bag & Semi-large CBS sorter.
   Semi-large shipments from Fulfilment centre & Cross dock are also loaded on the conveyors & spirals, the Semi-large Shipments ascend to a higher level and arrive at the VDS system of the Bag & Semi-large sorter.
   Once the Bags & semi-large Shipments are within the VDS loop, they are evenly distributed among all inducts of the loop CBS.
7. **Inducts: -** After the bags reaches the VDS loops, bags are collected from the VDS loop manually, an operator picks, positions & scan each Bag on the induct line of the sorter whereas Semi-large shipments are diverted with the help of SWEDI diverters & then these shipment are automatically inducted via fully auto inducts onto the Cross-Belt Sorter Loop.
8. **Loop CBS:** - Once the Bags & Semi-large shipments have entered the main XL loop, the Cross-Belt Sorter (CBS) efficiently sorts the bags & Semi-large shipment into their respective output spiral chutes by utilizing the data provided by Flipkart.
9. **Output Chutes: -** The shipments are discharged into two types of chutes.
   a. **Spiral Chutes -** Within a single-loop CBS system, there are a total of 24 spiral chutes. Bags & Semi-large shipments are sorted into these spiral chutes and descend down to mezzanine level, where it further gets sorted via PTL setup (In FK scope).
   b. **Rejection Chute-** Two rejection chute is present to handle rejected Shipments.

   Refeed line: - Bags & shipments collected in these spirals further follow the refeed conveyor & reaches induction zone of the Loop CBS for refeeding.

7.3 Main benefits of the proposed solution

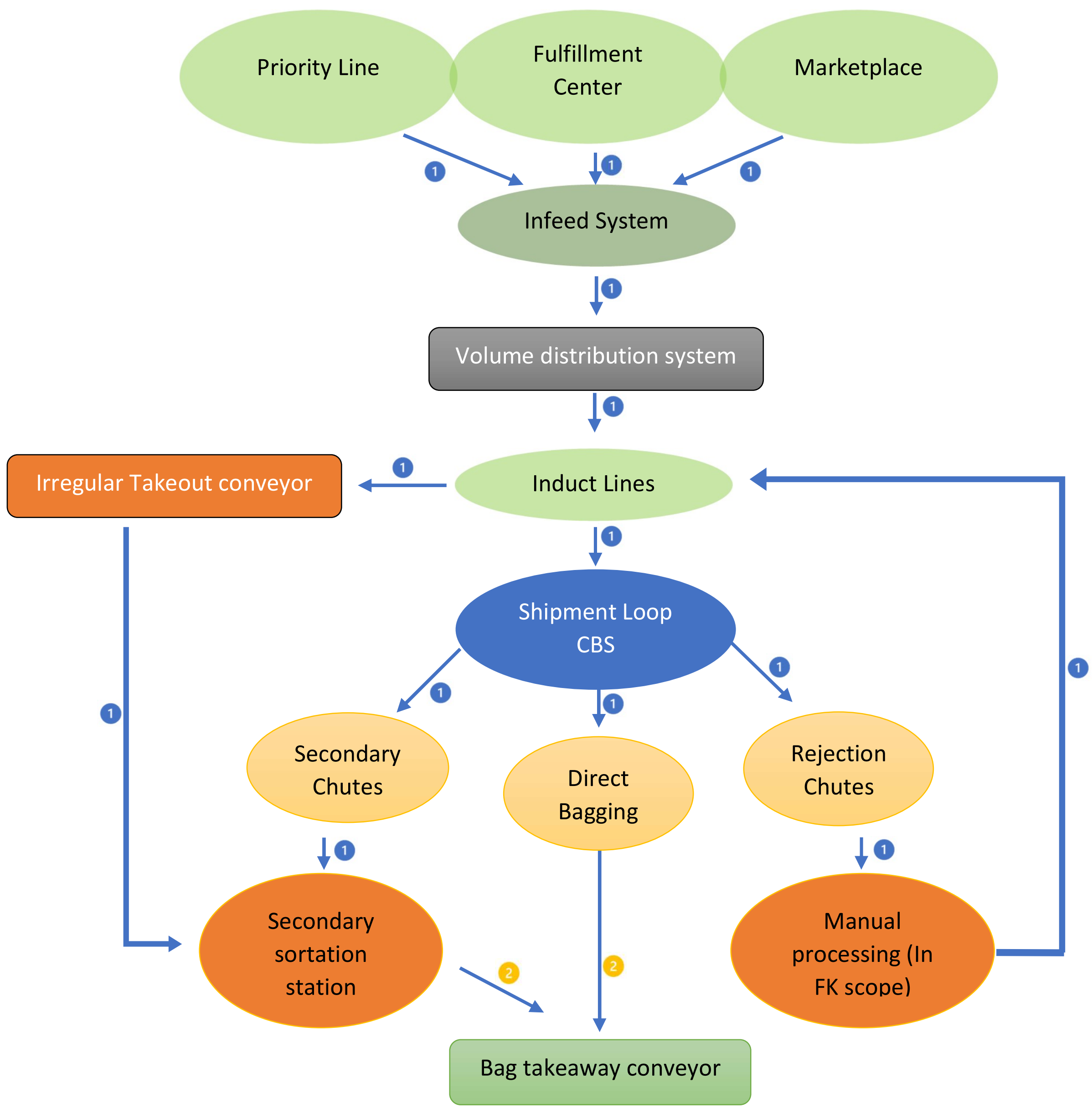
The proposed solution has the following advantages:

1. High operational throughput
2. Ability to serve entire range of conveyable Shipments with a sorter's speed of up to 2.1 m/s for NL Shipment sorter & 2.3 m/s for SL & Bag sorter.
3. Low occupancy of floor space in the building.
4. Narrow discharge centers for the increased number of splits in limited space.
5. FALCON's CBS can adapt to changing business requirements by adjusting its speed.
   to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

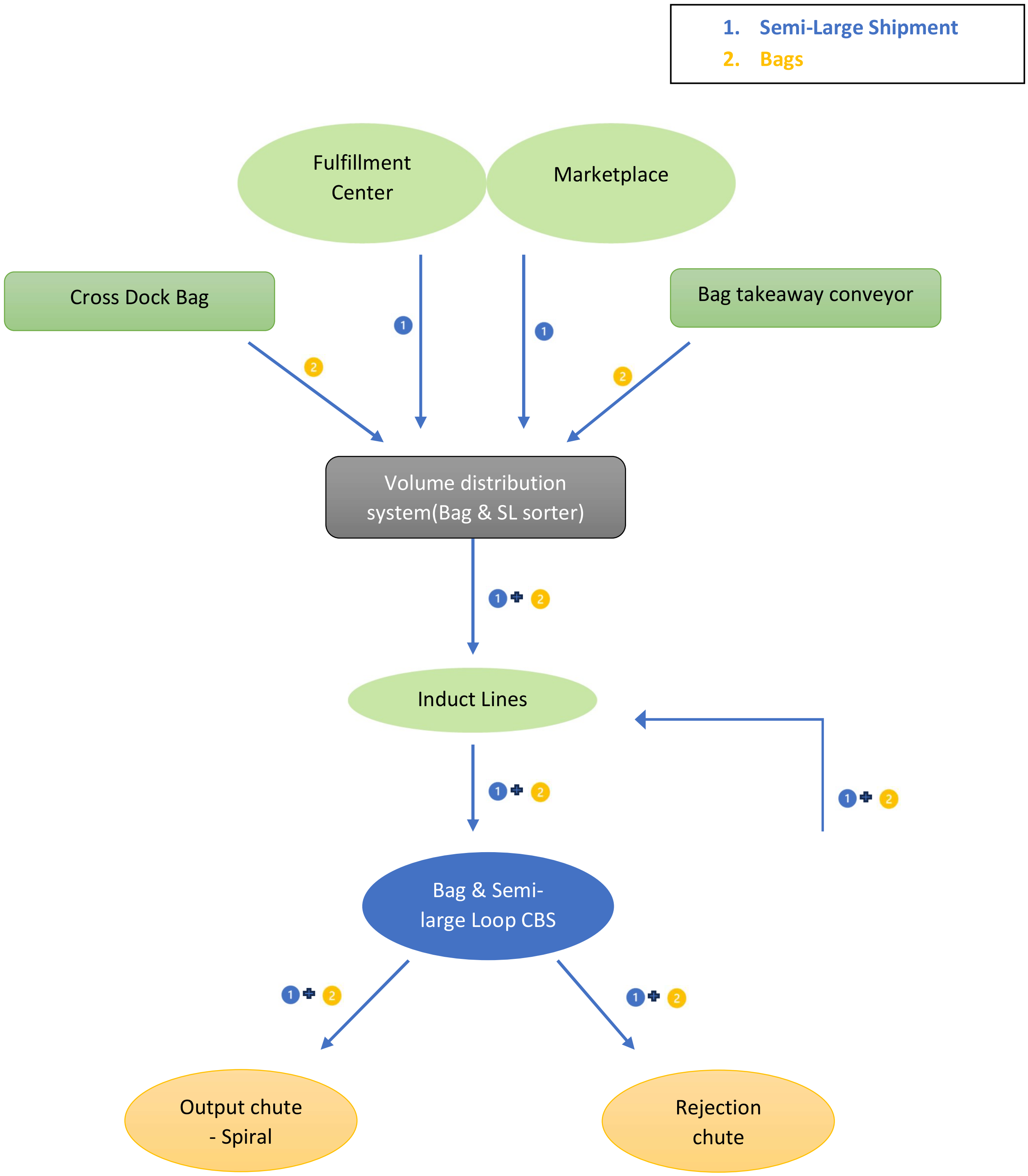
8. Concept Description

8.1 Shipment Sorter

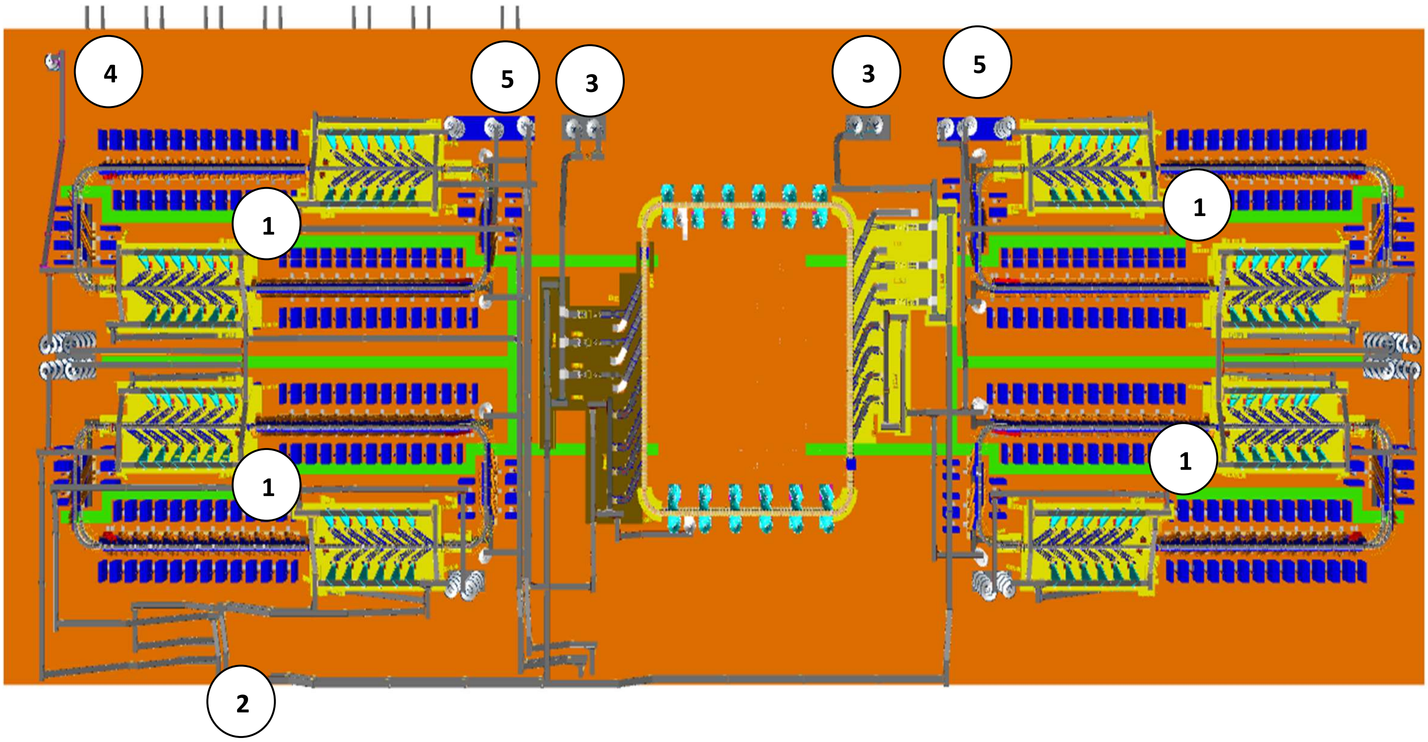
1. Non-Large Shipments
2. Bags



8.2 Semi-Large + Bag Sorter

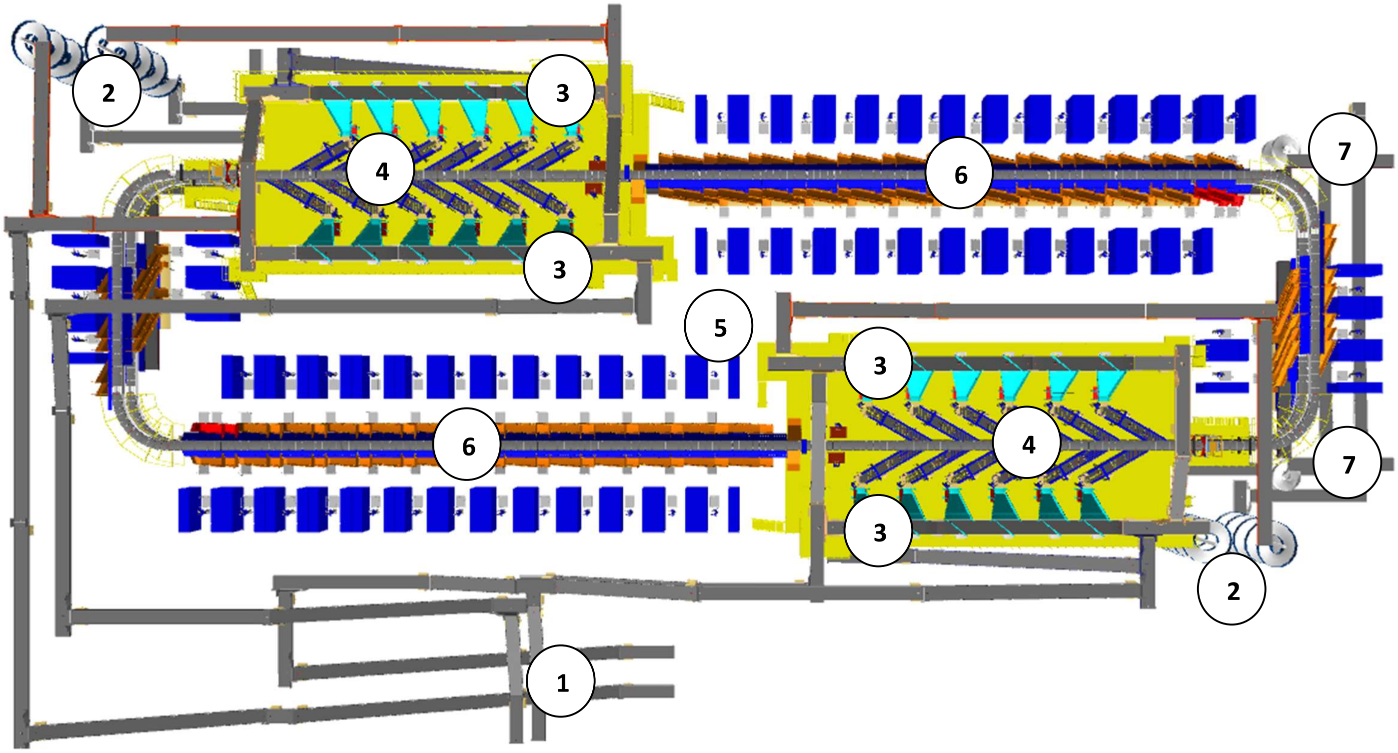


9. Layout Overview



Legend
1 – Shipment Sorters
2 – Fulfilment Centre Lines
3 – Cross Dock Lines
4 – Priority Line
5 – Market Place Lines
6 – Bags & Semi-Large Sorter

10. Layout View of Shipment Sorter



Legend

1 – Infeed Conveyor

2 – Infeed Powered Spiral

3 – VDS System

4 – Induct Lines

5 – Main Loop CBS

6 – Output Chutes (Direct bagging & Secondary Chutes)

7 – Bag Takeout Conveyor

11.Proposed System Capacity Calculations

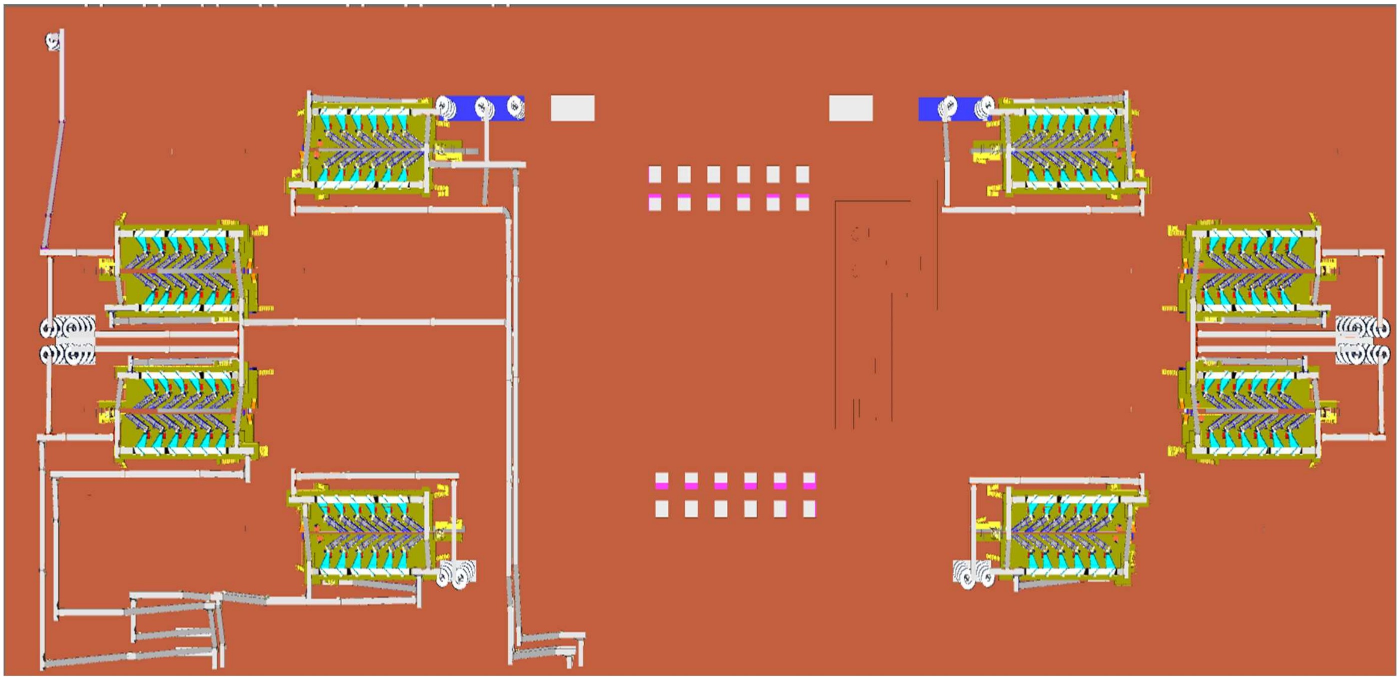
11.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on Flipkart RFP requirements.

SPECIFICATION	VALUE
Sorter Type	**Loop CBS**
Speed	**2.1 m/s**
Pitch	**1175 mm**
Carrier per hour	**6,434 CPH**
Belts per hour	**12,868 BPH**
Effective Designed Throughput of 1 loop CBS (A)	**25,736 Shipments per hour**
Per Feedline Capacity	**2,000 Shipments per hour**
No of Feedlines (Both Zones)	**12 Nos.**
Total Feedline designed capacity (B)	**24,000 Shipments per hour**
System designed throughput (Min of A&B)	**24,000 Shipments per hour**
System designed throughput from 1 Double deck	**48,000 Shipments per hour**
Designed throughput from 4 Double Decks	**1,92,000 Shipments per hour**

12. System description

12.1 Infeed System (Non-Large Shipment Sorter)



The layout presents entire conveyors used as infeed conveyors of the non-large shipment sorters.

Each Zone is connected with conveyor network to receive shipment volume from Market place & Fulfilment centre.

12.1.1 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	PVC Belt Conveyor	10149	12252	8.3	1210	1
2	PVC Belt Conveyor	12750	12750	19.68	1210	1
3	PVC Belt Conveyor	11000	11000	6.24	1210	1
4	PVC Belt Conveyor	11001	11232	18.24	1210	1
5	PVC Belt Conveyor	12501	12501	18.24	1210	1
6	PVC Belt Conveyor	12250	12250	9.6	1210	1
7	PVC Belt Conveyor	11001	11001	8.64	1210	1
8	PVC Belt Conveyor	12751	13060	15.103	1210	1
9	PVC Belt Conveyor	11000	11000	6.72	1210	1
10	PVC Belt Conveyor	11001	11232	18.24	1210	1
11	PVC Belt Conveyor	13251	13251	5.28	1210	1
12	PVC Belt Conveyor	12218	12218	19.68	1210	1
13	PVC Belt Conveyor	10251	10251	11.52	1210	1
14	PVC Belt Conveyor	12218	12218	19.68	1210	1
15	PVC Belt Conveyor	12500	12500	13.92	1210	1
16	PVC Belt Conveyor	11000	11000	19.68	1210	1
17	PVC Belt Conveyor	12108	13544	14.55	1210	1
18	PVC Belt Conveyor	12140	13075	10.289	1210	1
19	PVC Belt Conveyor	11000	11000	19.68	1210	1
20	PVC Belt Conveyor	12390	13536	12.61	1210	1
21	PVC Belt Conveyor	11001	11001	8.64	1210	1
22	PVC Belt Conveyor	12751	12751	7.68	1210	1
23	PVC Belt Conveyor	12637	13057	12.63	1210	1
24	PVC Belt Conveyor	12501	12501	11.52	1210	1
25	PVC Belt Conveyor	12652	11370	5.75	1210	1
26	PVC Belt Conveyor	11008	11214	1.5	1210	1
27	PVC Belt Conveyor	12751	12751	9.6	1210	1
28	PVC Belt Conveyor	13001	13001	3.36	1210	1
29	PVC Belt Conveyor	10901	13350	9.335	1210	1
30	PVC Belt Conveyor	12652	11370	5.75	1210	1
31	PVC Belt Conveyor	13001	13001	6.72	1210	1
32	PVC Belt Conveyor	13001	13001	3.36	1210	1
33	PVC Belt Conveyor	10901	13350	9.335	1210	1
34	PVC Belt Conveyor	12652	11370	5.75	1210	1
35	PVC Belt Conveyor	13001	13001	3.84	1210	1
36	PVC Belt Conveyor	11001	11001	9.12	1210	1
37	PVC Belt Conveyor	13251	13251	2.4	1210	1
38	PVC Belt Conveyor	11008	11214	1.5	1210	1
39	PVC Belt Conveyor	12751	12751	9.6	1210	1
40	PVC Belt Conveyor	13001	13001	3.36	1210	1
41	PVC Belt Conveyor	10901	13350	9.335	1210	1

42	PVC Belt Conveyor	11001	11001	6.24	1210	1
43	PVC Belt Conveyor	13251	13251	2.4	1210	1
44	PVC Belt Conveyor	11001	11001	7.68	1210	1
45	PVC Belt Conveyor	13001	13001	6.72	1210	1
46	PVC Belt Conveyor	12652	11370	5.75	1210	1
47	PVC Belt Conveyor	10901	13350	9.335	1210	1
48	PVC Belt Conveyor	13001	13001	3.36	1210	1
49	PVC Belt Conveyor	12751	12751	9.6	1210	1
50	PVC Belt Conveyor	11008	11214	1.5	1210	1
51	PVC Belt Conveyor	11001	11001	8.64	1210	1
52	PVC Belt Conveyor	13251	13251	2.4	1210	1
53	PVC Belt Conveyor	11001	11001	7.68	1210	1
54	PVC Belt Conveyor	11008	11214	1.5	1210	1
55	PVC Belt Conveyor	13251	13251	5.28	1210	1
56	PVC Belt Conveyor	13251	13251	2.4	1210	1
57	PVC Belt Conveyor	11001	11001	5.76	1210	1
58	PVC Belt Conveyor	13001	13001	6.72	1210	1
59	PVC Belt Conveyor	12751	12751	9.6	1210	1
60	PVC Belt Conveyor	6197	10922	19.68	1210	1
61	PVC Belt Conveyor	6300	6300	18.72	1210	1
62	PVC Belt Conveyor	800	800	11.04	1210	1
63	PVC Belt Conveyor	6544	6339	1.9	1210	1
64	PVC Belt Conveyor	10947	12515	6.5	1210	1
65	PVC Belt Conveyor	10195	10195	6.72	1210	1
66	PVC Belt Conveyor	8428	10275	7.15	1210	1
67	PVC Belt Conveyor	9600	9600	6.72	1410	1
68	PVC Belt Conveyor	9600	9600	19.68	1410	1
69	PVC Belt Conveyor	9851	9851	19.68	1410	1
70	PVC Belt Conveyor	7930	9931	7.68	1410	1
71	PVC Belt Conveyor	9600	9600	19.68	1410	1
72	PVC Belt Conveyor	8020	8020	8.64	1210	1
73	PVC Belt Conveyor	486	3134	10.181	1210	1
74	PVC Belt Conveyor	11896	13030	5.46	1210	1
75	PVC Belt Conveyor	12810	11574	19.338	1210	1
76	PVC Belt Conveyor	9498	13021	13.56	1210	1
77	PVC Belt Conveyor	7667	9684	8.495	1210	1
78	PVC Belt Conveyor	8530	8530	12.96	1210	1
79	PVC Belt Conveyor	5710	8775	11.073	1210	1
80	PVC Belt Conveyor	6060	6060	2.4	1210	1
81	PVC Belt Conveyor	11500	11500	17.76	1210	1
82	PVC Belt Conveyor	8200	8200	3.84	1210	1
83	PVC Belt Conveyor	11500	11500	19.68	1210	1
84	PVC Belt Conveyor	5898	8447	9.8	1210	1
85	PVC Belt Conveyor	10180	10180	11.52	1210	1

86	PVC Belt Conveyor	11600	11600	19.68	1210	1
87	PVC Belt Conveyor	7360	10425	11.073	1210	1
88	PVC Belt Conveyor	7710	7710	2.4	1210	1
89	PVC Belt Conveyor	10195	10195	19.68	1210	1
90	PVC Belt Conveyor	10195	10195	19.68	1210	1
91	PVC Belt Conveyor	7848	11371	13.56	1210	1
92	PVC Belt Conveyor	12750	12750	17.28	1210	1
93	PVC Belt Conveyor	12750	12750	19.68	1210	1
94	PVC Belt Conveyor	12639	13482	12	1210	1
95	PVC Belt Conveyor	8031	8031	3	1210	1
96	PVC Belt Conveyor	6250	6250	4.32	1210	1
97	PVC Belt Conveyor	7950	7950	3.36	1210	1
98	PVC Belt Conveyor	6250	6250	4.8	1210	1
99	PVC Belt Conveyor	12752	12965	5.28	1210	1
100	PVC Belt Conveyor	486	3134	10.181	1210	1
101	PVC Belt Conveyor	8188	8188	2.4	1210	1
102	PVC Belt Conveyor	12750	12750	19.68	1210	1
103	PVC Belt Conveyor	12750	12750	19.68	1210	1
104	PVC Belt Conveyor	8086	12998	18.927	1210	1
105	PVC Belt Conveyor	3162	8268	19.68	1210	1
106	PVC Belt Conveyor	600	600	5.28	1210	1
107	PVC Belt Conveyor	6137	8280	8.231	1210	1
108	PVC Belt Conveyor	8200	8200	1.2	1210	1
109	PVC Belt Conveyor	6137	8280	8.231	1210	1
110	PVC Belt Conveyor	7930	12830	18.882	1210	1
111	PVC Belt Conveyor	12750	12750	19.68	1210	1
112	PVC Belt Conveyor	12500	12500	19.68	1210	1
113	PVC Belt Conveyor	12503	13023	8.16	1210	1
114	PVC Belt Conveyor	11251	11251	9.6	1210	1
115	PVC Belt Conveyor	10085	11260	12.332	1210	1
116	PVC Belt Conveyor	10195	10195	16.8	1210	1
117	PVC Belt Conveyor	7460	7460	5.28	1210	1
118	PVC Belt Conveyor	11600	11600	18.72	1210	1
119	PVC Belt Conveyor	11600	11600	19.68	1210	1
120	PVC Belt Conveyor	10078	11682	6.317	1210	1
121	PVC Belt Conveyor	6000	6000	7.2	1210	1
122	PVC Belt Conveyor	11500	11500	17.28	1210	1
123	PVC Belt Conveyor	11500	11500	19.68	1210	1
124	PVC Belt Conveyor	8098	11581	13.428	1210	1
125	PVC Belt Conveyor	12000	12000	19.68	1210	1
126	PVC Belt Conveyor	11398	12242	3.259	1210	1
127	PVC Belt Conveyor	5810	5810	5.28	1210	1
128	PVC Belt Conveyor	10195	10195	19.68	1210	1
129	PVC Belt Conveyor	6250	6250	5.28	1210	1

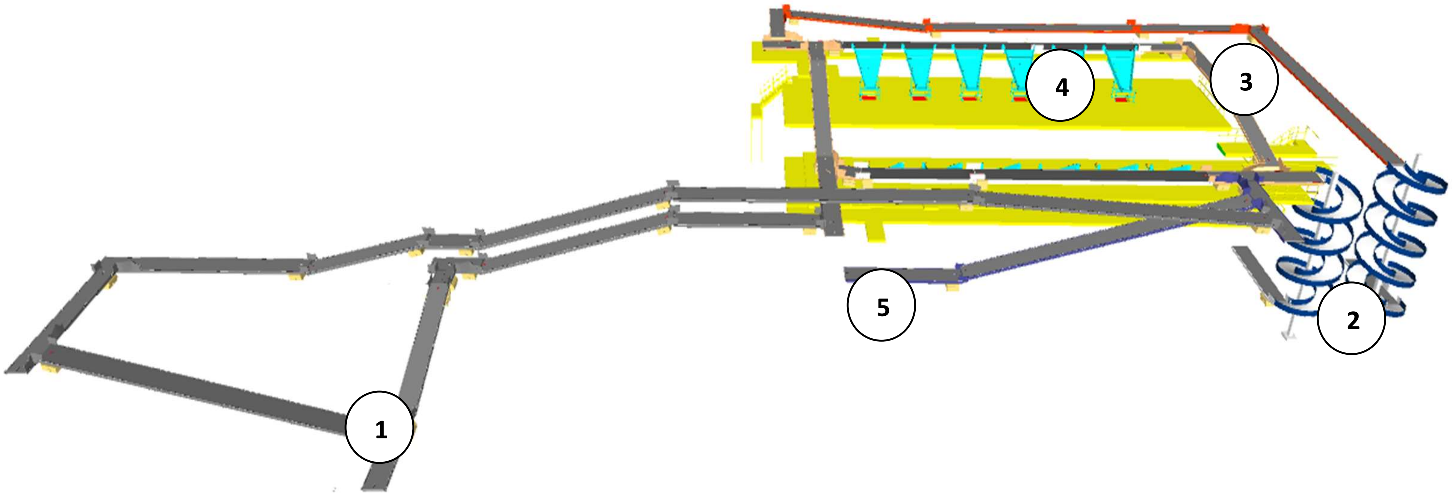
130	PVC Belt Conveyor	9600	9600	3.36	1210	1
131	PVC Belt Conveyor	12901	12901	19.68	1210	1
132	PVC Belt Conveyor	11488	12303	12.18	1210	1
133	PVC Belt Conveyor	3162	8268	19.68	1210	1
134	PVC Belt Conveyor	600	600	5.28	1210	1
135	PVC Belt Conveyor	7770	7770	14.4	1210	1
136	PVC Belt Conveyor	8032	8032	5.28	1410	1
137	PVC Belt Conveyor	9851	9851	19.68	1410	1
138	PVC Belt Conveyor	9851	9851	19.68	1410	1
139	PVC Belt Conveyor	9503	8863	3.38	1410	1
140	PVC Belt Conveyor	10195	10195	19.68	1210	1
141	PVC Belt Conveyor	13251	13251	5.28	1210	1
142	PVC Belt Conveyor	600	600	4.8	1210	1
143	PVC Belt Conveyor	3200	3200	9.6	1210	1
144	PVC Belt Conveyor	2700	2700	7.68	1210	1
145	PVC Belt Conveyor	498	3280	10.698	1210	1
146	PVC Belt Conveyor	2950	2950	6.24	1210	1
147	PVC Belt Conveyor	600	600	4.8	1210	1
148	PVC Belt Conveyor	3200	3200	19.68	1210	1
149	PVC Belt Conveyor	2950	2950	19.68	1210	1
150	PVC Belt Conveyor	3200	3200	7.68	1210	1
151	PVC Belt Conveyor	498	3030	9.73	1210	1
152	PVC Belt Conveyor	2700	2700	5.28	1210	1
153	PVC Belt Conveyor	498	3030	9.73	1210	1
154	PVC Belt Conveyor	2700	2700	12.48	1210	1
155	PVC Belt Conveyor	600	600	4.8	1210	1
156	PVC Belt Conveyor	2950	2950	19.2	1210	1
157	PVC Belt Conveyor	2700	2700	19.68	1210	1
158	PVC Belt Conveyor	600	600	4.8	1210	1
159	PVC Belt Conveyor	2950	2950	19.68	1210	1
160	PVC Belt Conveyor	2700	2700	19.68	1210	1
161	PVC Belt Conveyor	2702	3257	16.287	1210	1
162	PVC Belt Conveyor	2790	2790	17.28	1210	1
163	PVC Belt Conveyor	2700	2700	7.2	1210	1
164	PVC Belt Conveyor	600	600	4.8	1210	1
165	PVC Belt Conveyor	3200	3200	19.68	1210	1
166	PVC Belt Conveyor	2950	2950	4.32	1210	1
167	PVC Belt Conveyor	498	3280	10.698	1210	1
168	PVC Belt Conveyor	2950	2950	3.36	1210	1
169	PVC Belt Conveyor	600	600	4.8	1210	1
170	PVC Belt Conveyor	3200	3200	9.6	1210	1
171	PVC Belt Conveyor	2700	2700	8.16	1210	1
172	PVC Belt Conveyor	498	3280	10.698	1210	1
173	PVC Belt Conveyor	2950	2950	11.52	1210	1

174	PVC Belt Conveyor	3200	3200	4.8	1210	1
175	PVC Belt Conveyor	2700	2700	7.2	1210	1
176	PVC Belt Conveyor	600	600	4.8	1210	1
177	PVC Belt Conveyor	2950	2950	7.68	1210	1
178	PVC Belt Conveyor	600	600	4.8	1210	1
179	PVC Belt Conveyor	2950	2950	17.76	1210	1
180	PVC Belt Conveyor	600	600	4.8	1210	1
181	PVC Belt Conveyor	2950	2950	19.2	1210	1
182	PVC Belt Conveyor	2700	2700	19.68	1210	1
183	PVC Belt Conveyor	600	600	4.8	1210	1
184	PVC Belt Conveyor	2950	2950	19.68	1210	1
185	PVC Belt Conveyor	2700	2700	19.68	1210	1
186	PVC Belt Conveyor	2702	3257	16.287	1210	1
187	PVC Belt Conveyor	2790	2790	15.84	1210	1
188	PVC Belt Conveyor	2700	2700	7.2	1210	1
189	PVC Belt Conveyor	600	600	4.8	1210	1
190	PVC Belt Conveyor	3200	3200	19.68	1210	1
191	PVC Belt Conveyor	2950	2950	7.68	1210	1
192	PVC Belt Conveyor	0	2700	4.32	1210	1
193	PVC Belt Conveyor	0	2950	11.52	1210	1
194	PVC Belt Conveyor	2950	2950	4.32	1210	1
195	PVC Belt Conveyor	2950	2950	1.2	1210	1
196	PVC Belt Conveyor	498	3280	10.698	1210	1
197	PVC Belt Conveyor	2700	2700	4.8	1210	1
198	PVC Belt Conveyor	2950	2950	12	1210	1
199	PVC Belt Conveyor	2702	3095	9.71	1210	1
200	PVC Belt Conveyor	2700	2700	19.68	1210	1
201	PVC Belt Conveyor	2700	2700	19.68	1210	1
202	PVC Belt Conveyor	498	3030	9.73	1210	1
203	PVC Belt Conveyor	2700	2700	19.68	1210	1
204	PVC Belt Conveyor	2700	2700	19.68	1210	1
205	PVC Belt Conveyor	498	3030	9.73	1210	1
206	PVC Belt Conveyor	2700	2700	12.48	1210	1
207	PVC Belt Conveyor	498	3030	9.73	1210	1
208	PVC Belt Conveyor	2700	2700	5.28	1210	1
209	PVC Belt Conveyor	498	3030	9.73	1210	1
210	PVC Belt Conveyor	3200	3200	7.68	1210	1
211	PVC Belt Conveyor	2950	2950	19.68	1210	1
212	PVC Belt Conveyor	2700	2700	4.32	1210	1
213	PVC Belt Conveyor	3200	3200	19.68	1210	1
214	PVC Belt Conveyor	600	600	4.8	1210	1
215	PVC Belt Conveyor	2950	2950	6.24	1210	1
216	PVC Belt Conveyor	498	3280	10.698	1210	1
217	PVC Belt Conveyor	2700	2700	7.68	1210	1

218	PVC Belt Conveyor	3200	3200	9.6	1210	1
219	PVC Belt Conveyor	600	600	4.8	1210	1
220	PVC Belt Conveyor	2950	2950	7.68	1210	1
221	PVC Belt Conveyor	2950	2950	1.2	1210	1
222	PVC Belt Conveyor	498	3280	10.698	1210	1
223	PVC Belt Conveyor	2700	2700	4.8	1210	1
224	PVC Belt Conveyor	2950	2950	12.48	1210	1
225	PVC Belt Conveyor	2702	3095	9.71	1210	1
226	PVC Belt Conveyor	2700	2700	19.68	1210	1
227	PVC Belt Conveyor	2700	2700	19.68	1210	1
228	PVC Belt Conveyor	498	3030	9.73	1210	1
229	PVC Belt Conveyor	2700	2700	19.68	1210	1
230	PVC Belt Conveyor	2700	2700	19.68	1210	1
231	PVC Belt Conveyor	498	3030	9.73	1210	1
232	PVC Belt Conveyor	2950	2950	18.24	1210	1
233	PVC Belt Conveyor	600	600	4.8	1210	1
234	PVC Belt Conveyor	2950	2950	7.68	1210	1
235	PVC Belt Conveyor	600	600	4.8	1210	1
236	PVC Belt Conveyor	2700	2700	7.2	1210	1
237	PVC Belt Conveyor	3200	3200	4.8	1210	1
238	PVC Belt Conveyor	498	3280	10.698	1210	1
239	PVC Belt Conveyor	2700	2700	8.16	1210	1
240	PVC Belt Conveyor	3200	3200	9.6	1210	1
241	PVC Belt Conveyor	600	600	4.8	1210	1
242	PVC Belt Conveyor	2950	2950	3.36	1210	1
243	PVC Belt Conveyor	498	3280	10.698	1210	1
244	PVC Belt Conveyor	12652	11370	5.75	1210	1
245	PVC Belt Conveyor	11001	11001	9.12	1210	1
246	PVC Belt Conveyor	12750	12750	18.72	1210	1
247	PVC Belt Conveyor	12652	11370	5.75	1210	1
248	PVC Belt Conveyor	11008	11214	1.5	1210	1
249	PVC Belt Conveyor	12751	12751	9.6	1210	1
250	PVC Belt Conveyor	13001	13001	3.36	1210	1
251	PVC Belt Conveyor	10901	13350	9.335	1210	1
252	PVC Belt Conveyor	12390	13536	12.61	1210	1
253	PVC Belt Conveyor	11000	11000	19.68	1210	1
254	PVC Belt Conveyor	13251	13251	4.32	1210	1
255	PVC Belt Conveyor	11008	11214	1.5	1210	1
256	PVC Belt Conveyor	12751	12751	10.56	1210	1
257	PVC Belt Conveyor	13251	13251	2.4	1210	1
258	PVC Belt Conveyor	12652	11370	5.75	1210	1
259	PVC Belt Conveyor	12250	12250	6.72	1210	1
260	PVC Belt Conveyor	13001	13001	3.84	1210	1
261	PVC Belt Conveyor	11001	11001	9.12	1210	1

262	PVC Belt Conveyor	13251	13251	2.4	1210	1
263	PVC Belt Conveyor	11001	11001	8.64	1210	1
264	PVC Belt Conveyor	12751	13060	15.103	1210	1
265	PVC Belt Conveyor	11000	11000	4.8	1210	1
266	PVC Belt Conveyor	11001	11238	18.72	1210	1
267	PVC Belt Conveyor	13251	13251	5.28	1210	1
268	PVC Belt Conveyor	11998	11998	19.68	1210	1
269	PVC Belt Conveyor	13001	13001	6.72	1210	1
270	PVC Belt Conveyor	11001	11001	9.12	1210	1
271	PVC Belt Conveyor	13251	13251	2.4	1210	1
272	PVC Belt Conveyor	11001	11001	6.24	1210	1
273	PVC Belt Conveyor	10149	12249	8.064	1210	1
274	PVC Belt Conveyor	10251	10251	3.36	1210	1
275	PVC Belt Conveyor	12251	12251	8.64	1210	1
276	PVC Belt Conveyor	10901	13350	9.335	1210	1
277	PVC Belt Conveyor	13001	13001	3.36	1210	1
278	PVC Belt Conveyor	12751	12751	9.6	1210	1
279	PVC Belt Conveyor	11008	11214	1.5	1210	1
280	PVC Belt Conveyor	11998	11998	19.68	1210	1
281	PVC Belt Conveyor	12500	12500	13.92	1210	1
282	PVC Belt Conveyor	11000	11000	19.68	1210	1
283	PVC Belt Conveyor	11890	13532	12.15	1210	1
284	PVC Belt Conveyor	12652	11370	5.75	1210	1
285	PVC Belt Conveyor	10901	13350	9.335	1210	1
286	PVC Belt Conveyor	13001	13001	3.36	1210	1
287	PVC Belt Conveyor	12751	12751	9.6	1210	1
288	PVC Belt Conveyor	11008	11214	1.5	1210	1
289	PVC Belt Conveyor	12140	13075	10.289	1210	1
290	PVC Belt Conveyor	10901	13350	9.335	1210	1
291	PVC Belt Conveyor	13001	13001	3.36	1210	1
292	PVC Belt Conveyor	13001	13001	6.72	1210	1
293	PVC Belt Conveyor	12501	12501	18.24	1210	1
294	PVC Belt Conveyor	11001	11238	18.72	1210	1
295	PVC Belt Conveyor	11000	11000	4.8	1210	1
296	PVC Belt Conveyor	11001	11001	8.64	1210	1
297	PVC Belt Conveyor	13251	13251	2.4	1210	1
298	PVC Belt Conveyor	11001	11001	5.76	1210	1
299	PVC Belt Conveyor	13001	13001	3.84	1210	1
300	PVC Belt Conveyor	12751	12751	4.32	1210	1
301	PVC Belt Conveyor	12637	13057	12.63	1210	1
302	PVC Belt Conveyor	12501	12501	13.44	1210	1
303	PVC Belt Conveyor	11001	11001	3.36	1210	1
304	Telescopic Belt Conveyor			0		14
305	Belt turn 90°				1410	2.0

12.2 Infeed System



Legend

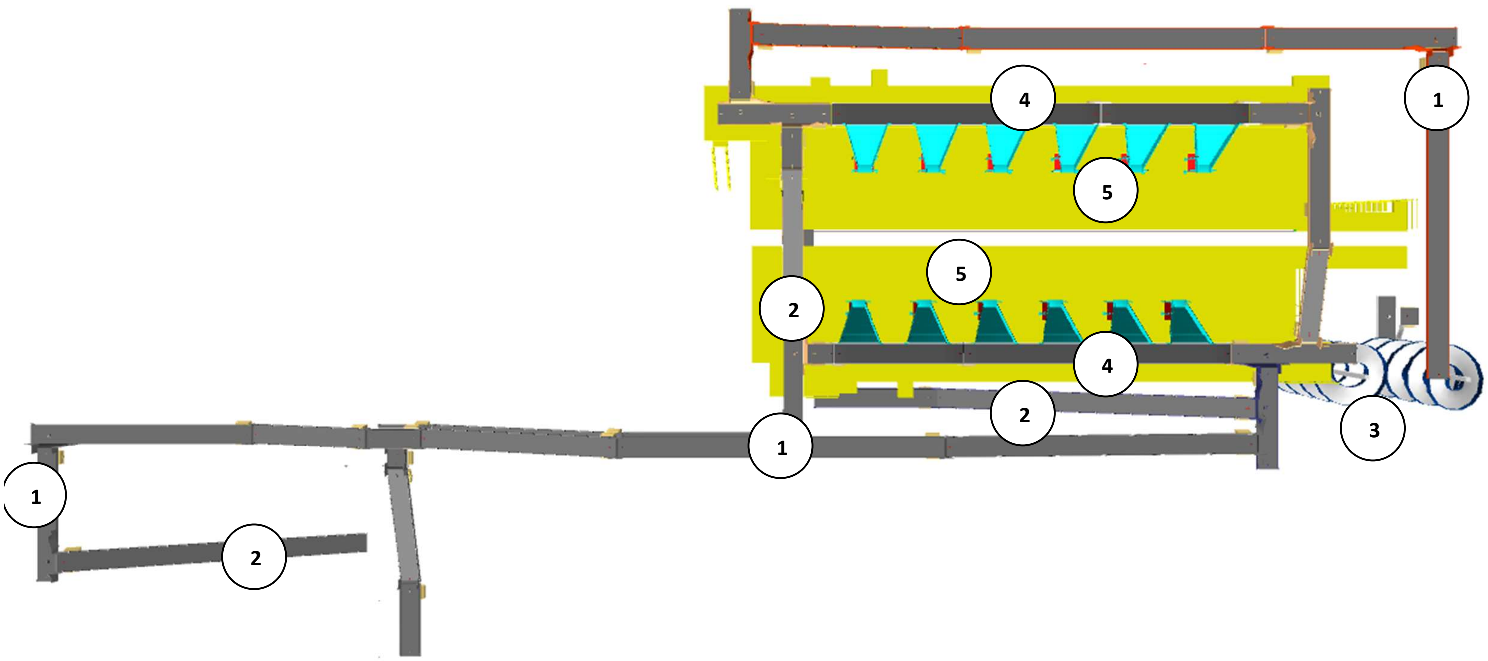
1- Infeed Conveyor (FC Volume)
2- Infeed Powered Spiral (Market Place Volume)
3- VDS Loop
4- VDS Chutes
5- Non-Conveyable refeed Conveyor.

Every CBS loop is equipped with two Infeed conveyor facilitating the movement of FC shipments and Two Powered Spiral for facilitating the movement of Marketplace shipment volume to VDS loop. Additionally, each loop features two Rejection/DBO refeed lines for transporting processed shipments back to Inducts.

12.2.1 Infeed System

Infeed conveyor system consists of below components:

1. Telescopic Belt Conveyor
2. Straight PVC Conveyor
3. Inclined PVC Conveyor
4. Powered Spiral Conveyor
5. VDS System (Arm based/ Tilted Belt)
6. VDS Chute



12.2.1.1 Telescopic Belt Conveyor

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Base Length	Mm	6000
Extended Length (Total Length)	Mm	18000
Motor Rating	Type	0.75 Kw, 415V AC, 50 Hz

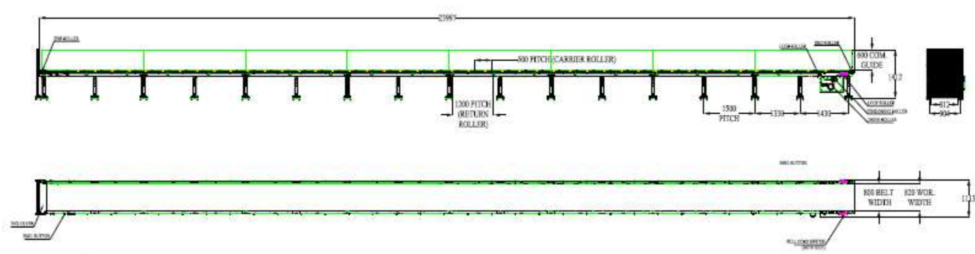
12.2.1.2 Straight and Inclined Powered Belt Conveyor

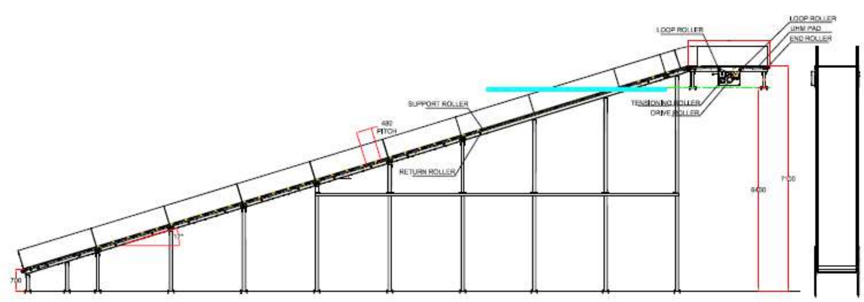
Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.





Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

12.2.1.3 Powered Spiral Conveyor

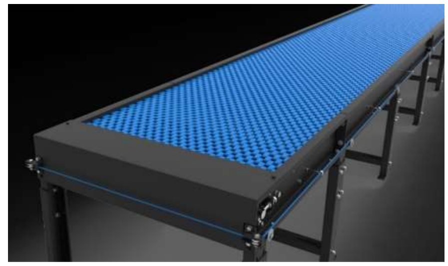
Specification	UOM	Remark
Manufacturer Name	Name	Apollo/Ambaflex
Overall height	mm	As per Layout requirement
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	mm	approx. 600
Roller Diameter	mm	approx. 50
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Yes
End Stopper	yes/no	Yes

12.2.1.4 ZTR Curves (Pic for ref.)

12.2.1.5 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

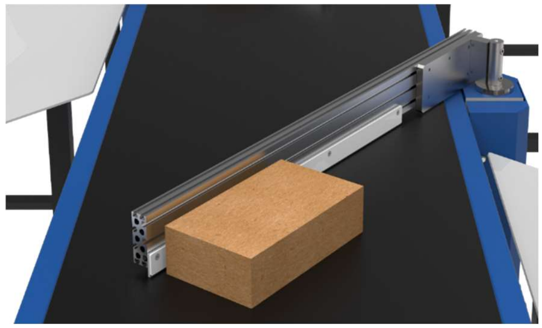


- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make	Make	Movex
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Wear Strip Material	MOC	UHMW PE1000
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Driven Shaft	Type	MOC-EN9 Solid Shaft hardened to BHN 240, Size-60SQ MM, Zinc Plated -25 microns, Bearings-UCF-210D1
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

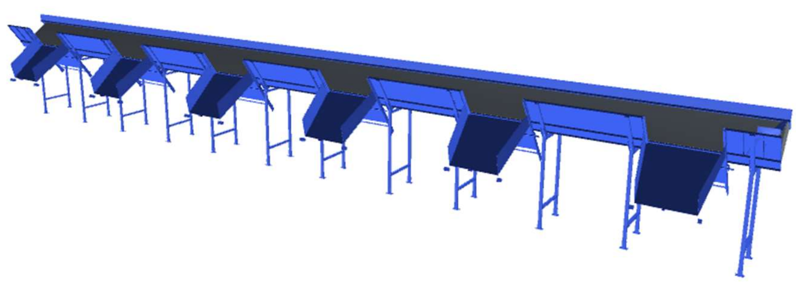
12.2.1.6 Diverter ARM

Arm diverter System of Falcon Autotech is an extremely cost-effective Sortation system. Arm diverter's simple and modular design gives it advantage of low cost, space saving, and modularity.



Specification	ARM XL
Property	Value
Actuation method	Electric induction motor
Belt type	Flat top modular
Belt width (mm)	1200
Arm length (mm)	2400
Max. product weight (kg)	40

12.2.1.7 Tilted Belt VDS



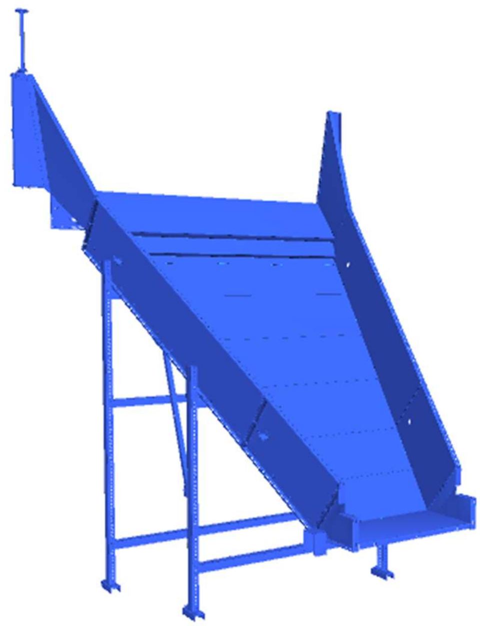
Specification	Tilted belt
Actuation method	Electric induction motor
Belt type	Tilted modular top
Belt width (mm)	1200
Max. product weight (kg)	40

12.2.1.8 Pros & Cones of ARM & Tilted Belt VDS System

	ARM VDS	Tilted Belt VDS
Advantages	Positive discharge	Positive Blocking
	Economical	All type of Parcels can Pass
	Ease of maintenance	
Disadvantage	Negative blocking	Negative discharge
	Small parcel may stuck	Light Parcels might be stuck on belt itself
		Parcel might remain on the conveyor Guides

12.2.1.9 VDS Chutes

A VDS chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level (at an operator level). It utilizes base metal to guide and support the flow of items along the chute & collect the shipments within its area.



Calculation for Design: We have considered an average parcel size of 300 X 300 X 300 mm while designing the capacity of chutes.

	L(m)	B(m)	H(m)	Volume (m3)
Average parcel size considered for chute volume calculation	0.3	0.3	0.3	0.027
Collection type Chute	3.3	1.5	0.4	2.07

Assuming a 60% occupancy rate, about **45 parcels** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

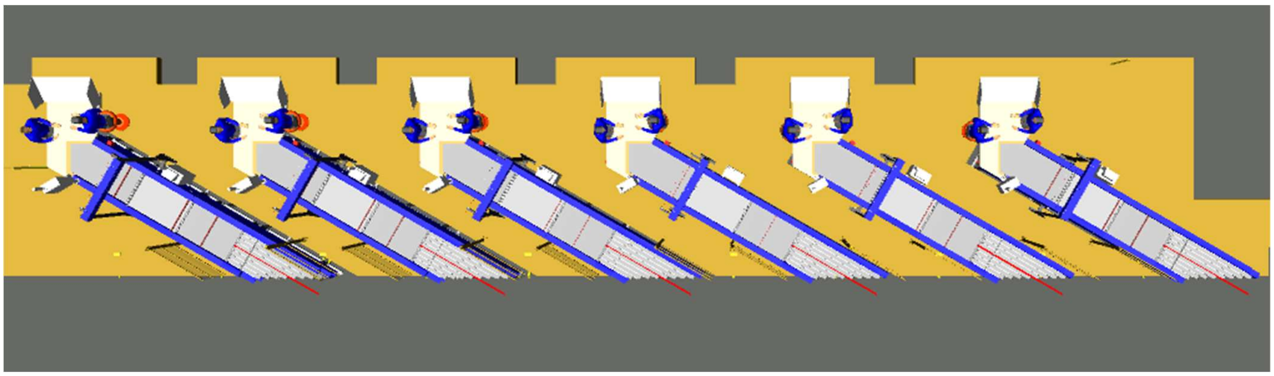
- Chute Full Sensor
- Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.3 Induction to Sorter: Feedlines (Bespoke Flipkart Design)

The main purpose of the feedlines is to transport the shipments from the VDS chutes to the sorter by means Loading conveyors, Weighing, Buffer/Spacing Conveyor and Angle Merge conveyors. The flow of shipments on the Inducts are controlled by sensors in such a way that the required gap is ensured between the shipments.

These Feedlines are designed especially for Flipkart Group considering Operator Ease and ergonmics. Where the operator ensures proper placement of the shipments on the Induct.



Induction Zone

12.3.1 Feedlines

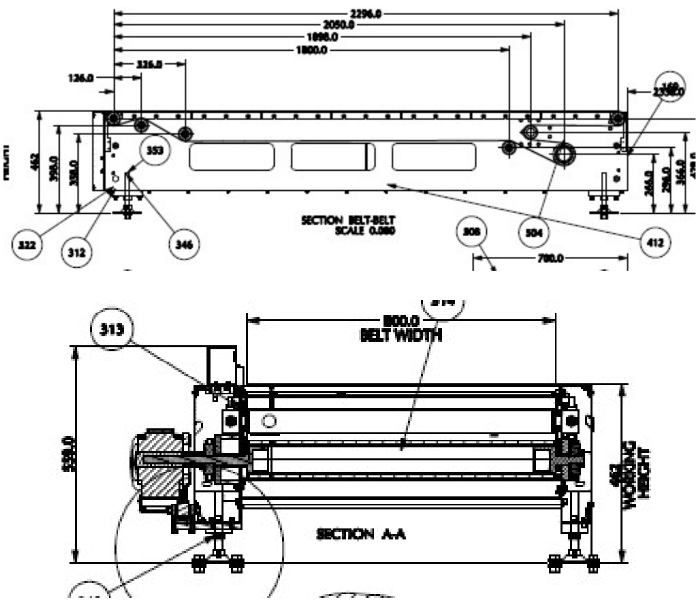
Each Loop CBS has a total 12 Automatic Feedlines with Designed IPP of 2000 PPH.

1. Each Feedline has Four Conveyor Modules-
    - Orientation/Loading Conveyor- 1 Nos.
    - Weighing Conveyor- 1 Nos.
    - Buffer Conveyor- 1 Nos.
    - Intelligent Merge- 1 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
3. Over the feedline, the shipments are evenly spaced and smoothly transferred to the main Loop.

12.3.1.1 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.



12.3.1.2 Weighing Conveyor

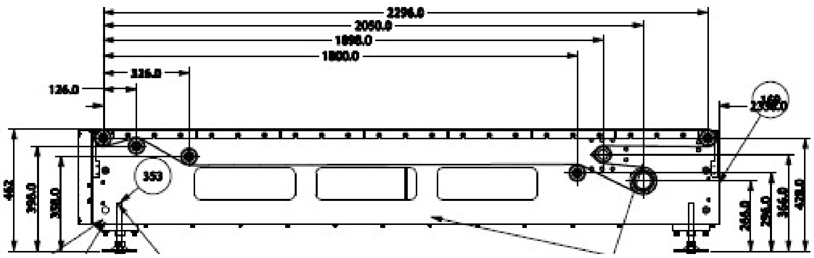
A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

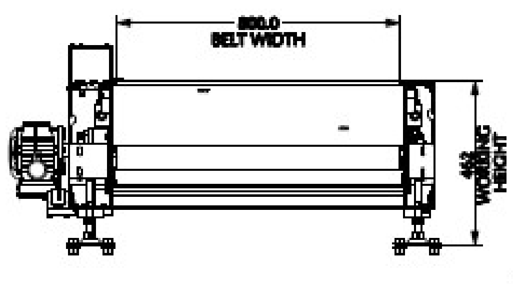
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

Make- Bizerba/ Mettler Toledo

12.3.1.3 Orientation Conveyor

An Orientation conveyor is a type of conveyor system used to align & Release the products for induction onto CBS. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.

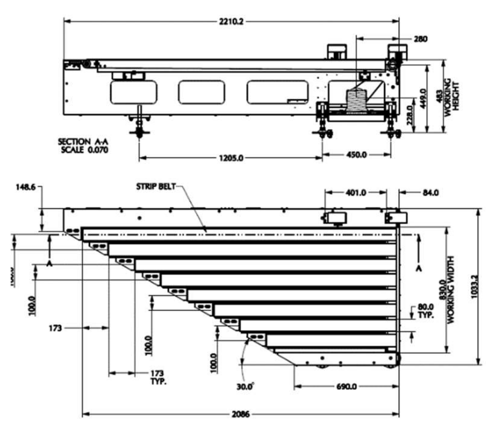




12.3.1.4 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



12.3.1.5 New Bespoke Ergonomic Feedline

- Ergonomic Operator & Remote Panel
- Strip type belt conveyors
- One touch bidirectional conveyors
- Lean design
- MLG sensor
- Two Roller conveyor design

Ergonomic Operator & Remote Panel: Easily reachable operator panel allowing quick & convenient access from operator position. Relocation of remote panel below feedline.

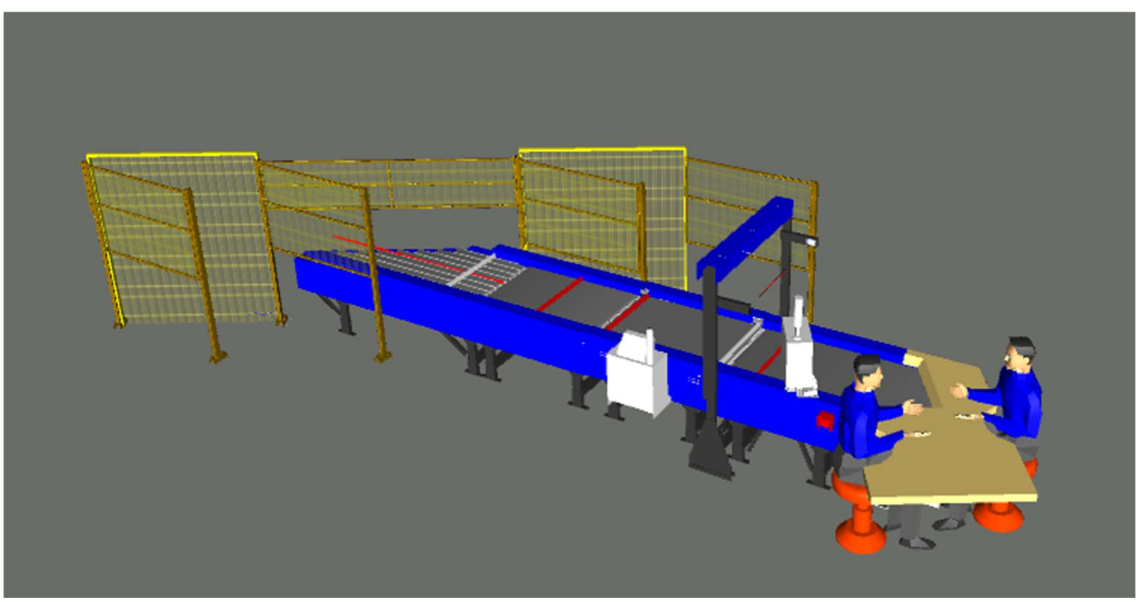
Lean Design: Under conveyor motor placement make the feedline leaner & increase possibility of free man movement.

Bidirectional Conveyor: Thanks to bidirectional conveyors that move in reverse direction when overweight and/or over sized parcel needs to be brought back, hence the operator movement close to CBS loop is eliminated **increasing operator safety.**

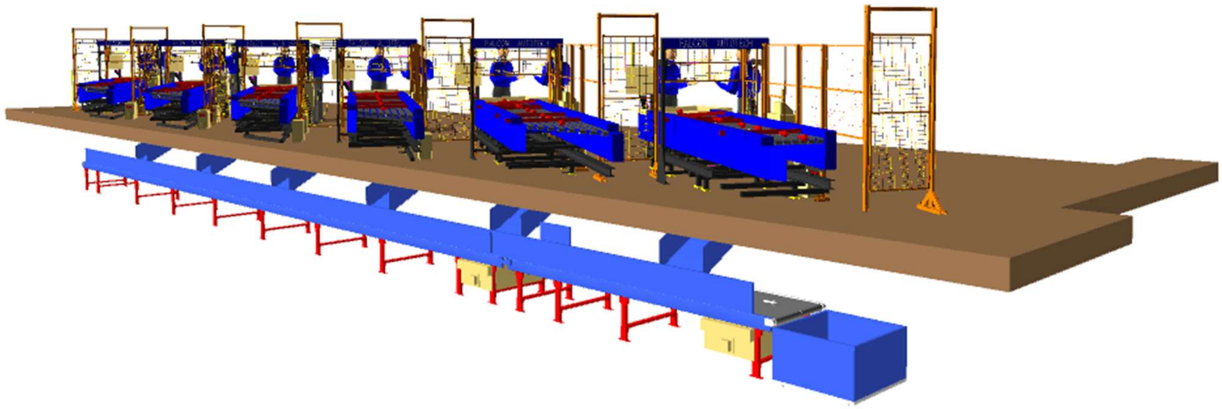
MLG: Now with MLG we have light grids at 5 mm pitch, therefore more precise position of parcel can be determined.

12.3.1.6 New Induct Fencing

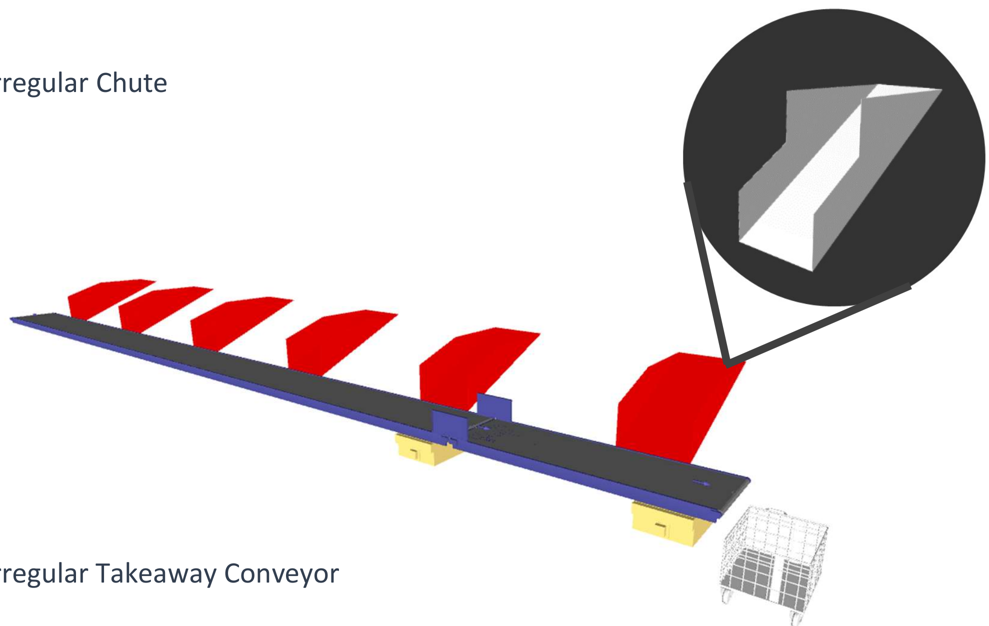
Setting a new Safety paramount. The new Induct Fencing helps prevent unauthorized operators from entering the induction zone, eliminating the risk of accidents and injuries.



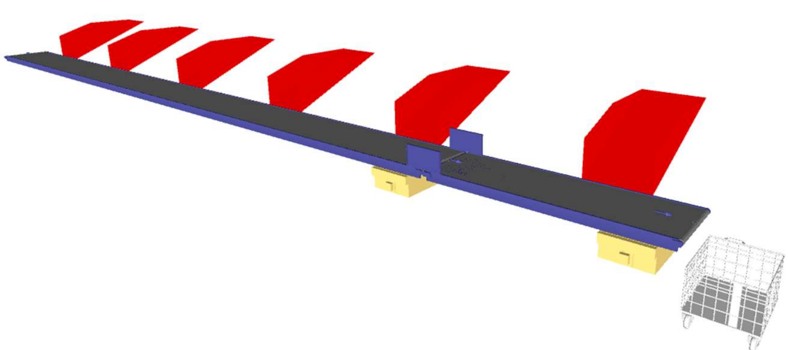
12.3.2 Irregular Takeout Conveyor



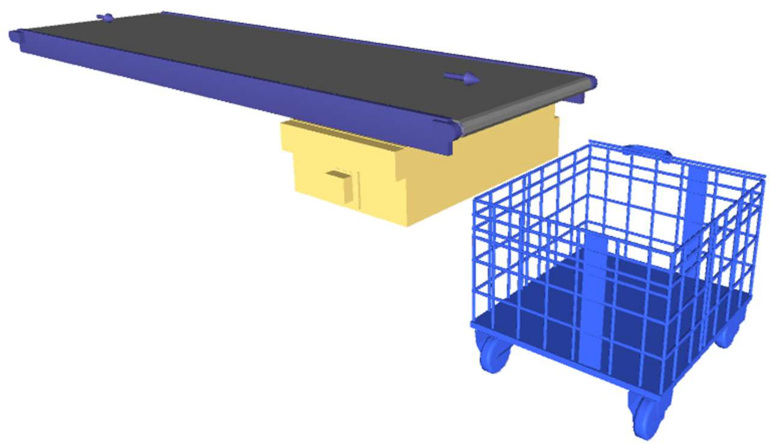
12.3.2.1 Irregular Chute



12.3.2.2 Irregular Takeaway Conveyor



12.3.2.3 Irregular Collection Bin (In Flipkart's scope)

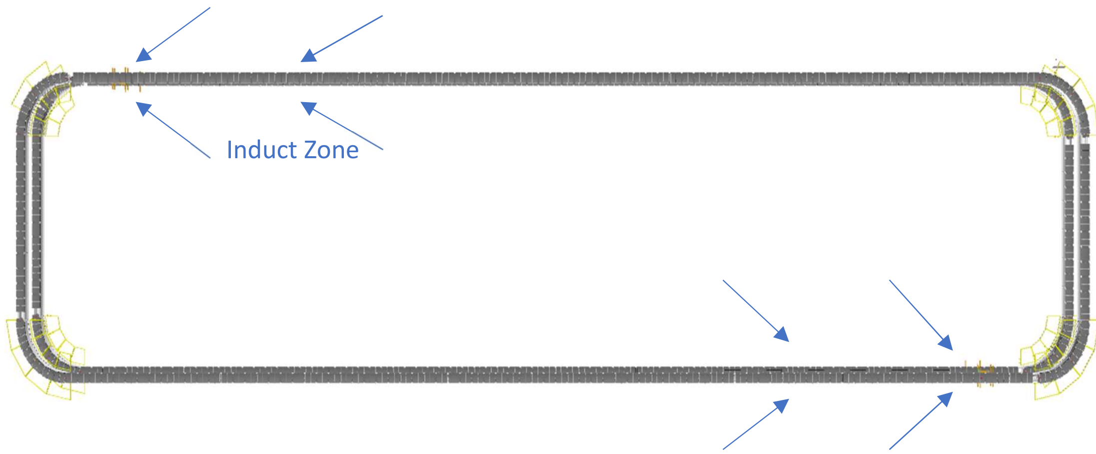


12.3.3 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	PVC Belt Conveyor	6801	6801	4.8	1210	1
2	PVC Belt Conveyor	8801	8801	4.32	1210	1
3	PVC Belt Conveyor	7300	7300	4.32	1210	1
4	PVC Belt Conveyor	6801	6801	4.8	1210	1
5	PVC Belt Conveyor	7300	7300	3.84	1210	1
6	PVC Belt Conveyor	6000	6000	6.72	1210	1
7	PVC Belt Conveyor	8801	8801	4.32	1210	1
8	PVC Belt Conveyor	6801	6801	7.68	1210	1
9	PVC Belt Conveyor	10684	11505	3.35	1210	1
10	PVC Belt Conveyor	6000	6000	8.64	1210	1
11	PVC Belt Conveyor	6000	6000	8.16	1210	1
12	PVC Belt Conveyor	11251	11251	6.24	1210	1
13	PVC Belt Conveyor	8801	8801	19.68	1210	1
14	PVC Belt Conveyor	6801	6801	19.68	1210	1
15	PVC Belt Conveyor	11262	11262	3	1210	1
16	PVC Belt Conveyor	5898	10635	19.58	1210	1
17	PVC Belt Conveyor	8801	8801	19.68	1210	1
18	PVC Belt Conveyor	8801	8801	4.32	1210	1
19	PVC Belt Conveyor	6000	6000	3.36	1210	1
20	PVC Belt Conveyor	10660	11510	3.47	1210	1
21	PVC Belt Conveyor	6801	6801	4.32	1210	1
22	PVC Belt Conveyor	8801	8801	4.32	1210	1
23	PVC Belt Conveyor	6199	11479	19.68	1210	1
24	PVC Belt Conveyor	5897	10659	19.68	1210	1
25	PVC Belt Conveyor	11250	11250	3	1210	1

26	PVC Belt Conveyor	6801	6801	16.8	1210	1
27	PVC Belt Conveyor	8801	8801	19.68	1210	1
28	PVC Belt Conveyor	5898	10656	19.68	1210	1
29	PVC Belt Conveyor	10681	11296	2.5	1210	1
30	PVC Belt Conveyor	6801	6801	19.68	1210	1
31	PVC Belt Conveyor	7300	7300	4.8	1210	1
32	PVC Belt Conveyor	8801	8801	19.68	1210	1
33	PVC Belt Conveyor	6801	6801	19.68	1210	1
34	PVC Belt Conveyor	7300	7300	4.8	1210	1
35	PVC Belt Conveyor	7300	7300	4.8	1210	1
36	PVC Belt Conveyor	6801	6801	19.68	1210	1
37	PVC Belt Conveyor	8801	8801	19.68	1210	1
38	PVC Belt Conveyor	7300	7300	4.8	1210	1
39	PVC Belt Conveyor	6801	6801	19.68	1210	1
40	PVC Belt Conveyor	10681	11296	2.5	1210	1
41	PVC Belt Conveyor	5898	10656	19.68	1210	1
42	PVC Belt Conveyor	8801	8801	19.68	1210	1
43	PVC Belt Conveyor	6801	6801	16.8	1210	1
44	PVC Belt Conveyor	11250	11250	3	1210	1
45	PVC Belt Conveyor	5897	10659	19.68	1210	1
46	PVC Belt Conveyor	6199	11479	19.68	1210	1
47	PVC Belt Conveyor	8801	8801	4.32	1210	1
48	PVC Belt Conveyor	6801	6801	4.32	1210	1
49	PVC Belt Conveyor	10660	11510	3.47	1210	1
50	PVC Belt Conveyor	6000	6000	3.36	1210	1
51	PVC Belt Conveyor	8801	8801	4.32	1210	1
52	PVC Belt Conveyor	8801	8801	19.68	1210	1
53	PVC Belt Conveyor	5898	10635	19.58	1210	1
54	PVC Belt Conveyor	11262	11262	3	1210	1
55	PVC Belt Conveyor	6801	6801	19.68	1210	1
56	PVC Belt Conveyor	8801	8801	19.68	1210	1
57	PVC Belt Conveyor	11251	11251	3	1210	1
58	PVC Belt Conveyor	6000	6000	7.2	1210	1
59	PVC Belt Conveyor	6000	6000	8.64	1210	1
60	PVC Belt Conveyor	10684	11505	3.35	1210	1
61	PVC Belt Conveyor	6801	6801	7.68	1210	1
62	PVC Belt Conveyor	8801	8801	4.32	1210	1
63	PVC Belt Conveyor	6000	6000	6.72	1210	1
64	PVC Belt Conveyor	7300	7300	3.84	1210	1
65	PVC Belt Conveyor	6801	6801	4.8	1210	1
66	PVC Belt Conveyor	7300	7300	4.32	1210	1
67	PVC Belt Conveyor	8801	8801	4.32	1210	1
68	PVC Belt Conveyor	6801	6801	4.8	1210	1

12.3.4 Main Loop



1. The proposed Double Deck CBS Sorter will be installed on Mezzanine level. The Sorter runs at 3500 mm (Lower Loop) & 5500 mm (Upper Loop) from Mezzanine level.

2. We have provided loop with Dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.

3. One single deck loop CBS has a total length of 254 m & 217 number of carriers.

4. As Shipments pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches the respective chute, the carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

12.3.5 Sorting Output

12.3.5.1 Direct Bagging Chute (Bespoke Flipkart design to be used)

A Direct bagging chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & collect the shipments into bags placed at the end of each chute.

Each Direct bagging chute has capacity to accumulate 40 shipments, The sorted shipments will travel along these collection chutes and get accumulated at the bottom of the chute.

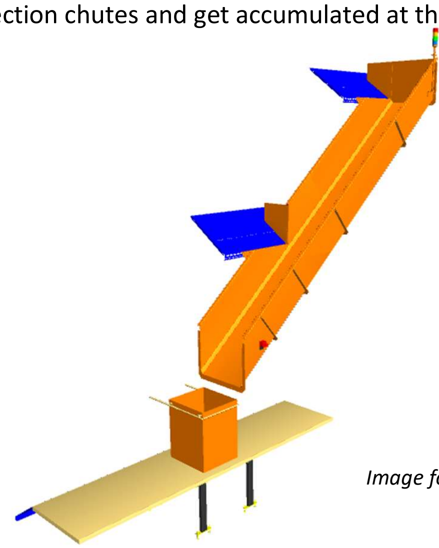


Image for representation purpose only

Calculation for Design: We have considered an average shipment size of 300 X 300 X 300 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type Chute	5.2	0.7	0.5	1.82

Assuming a 60% occupancy rate, about **40 shipments** will readily fit into these chutes. 67

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.3.5.2 Secondary Chute (L-Type) (Bespoke Flipkart design to be used)

Secondary Chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & end gate can be closed to collect the shipments within its area.

Rejection chute has capacity to accumulate 40 shipments, The sorted shipments will directly deposit into roll cage trolley place below.

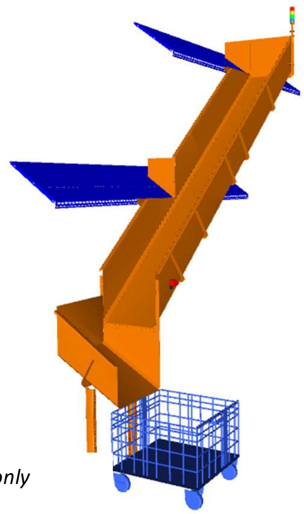


Image for representation purpose only

Calculation for Design: We have considered an average shipment size of 300 X 300 X 300 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type PTL Chute	5.2	0.7	0.5	1.82

Assuming a 60% occupancy rate, about **40 shipments** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.3.5.3 Rejection Chute (L-Type) (Bespoke Flipkart design to be used)

Rejection Chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & end gate can be closed to collect the shipments within its area.

Rejection chute has capacity to accumulate 40 shipments, The sorted shipments will directly deposit into roll cage trolley place below.

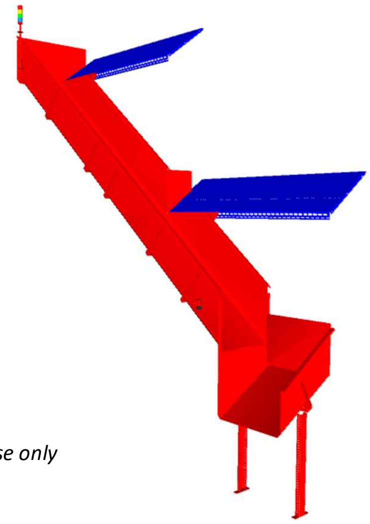


Image for representation purpose only

Calculation for Design: We have considered an average shipment size of 300 X 300 X 300 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type PTL Chute	5.2	0.7	0.5	1.82

Assuming a 60% occupancy rate, about **40 shipments** will readily fit into these chutes.

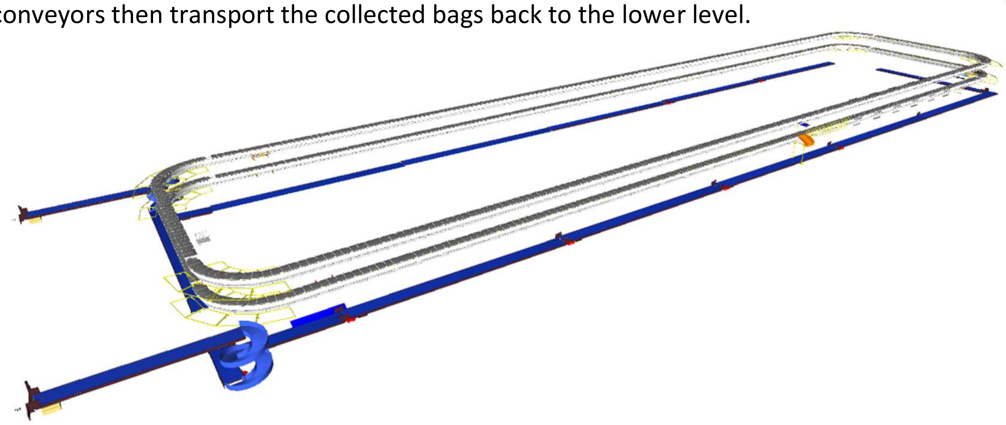
Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.3.6 Bag Take away Conveyor

In each Loop CBS, there is a series of Bag Takeaway Conveyors, spanning a combined length of 1180 meters. Following the secondary sorting process, bags are securely fastened and manually placed onto these conveyors, which are positioned below the Loop CBS to cover all chute locations. These conveyors then transport the collected bags back to the lower level.



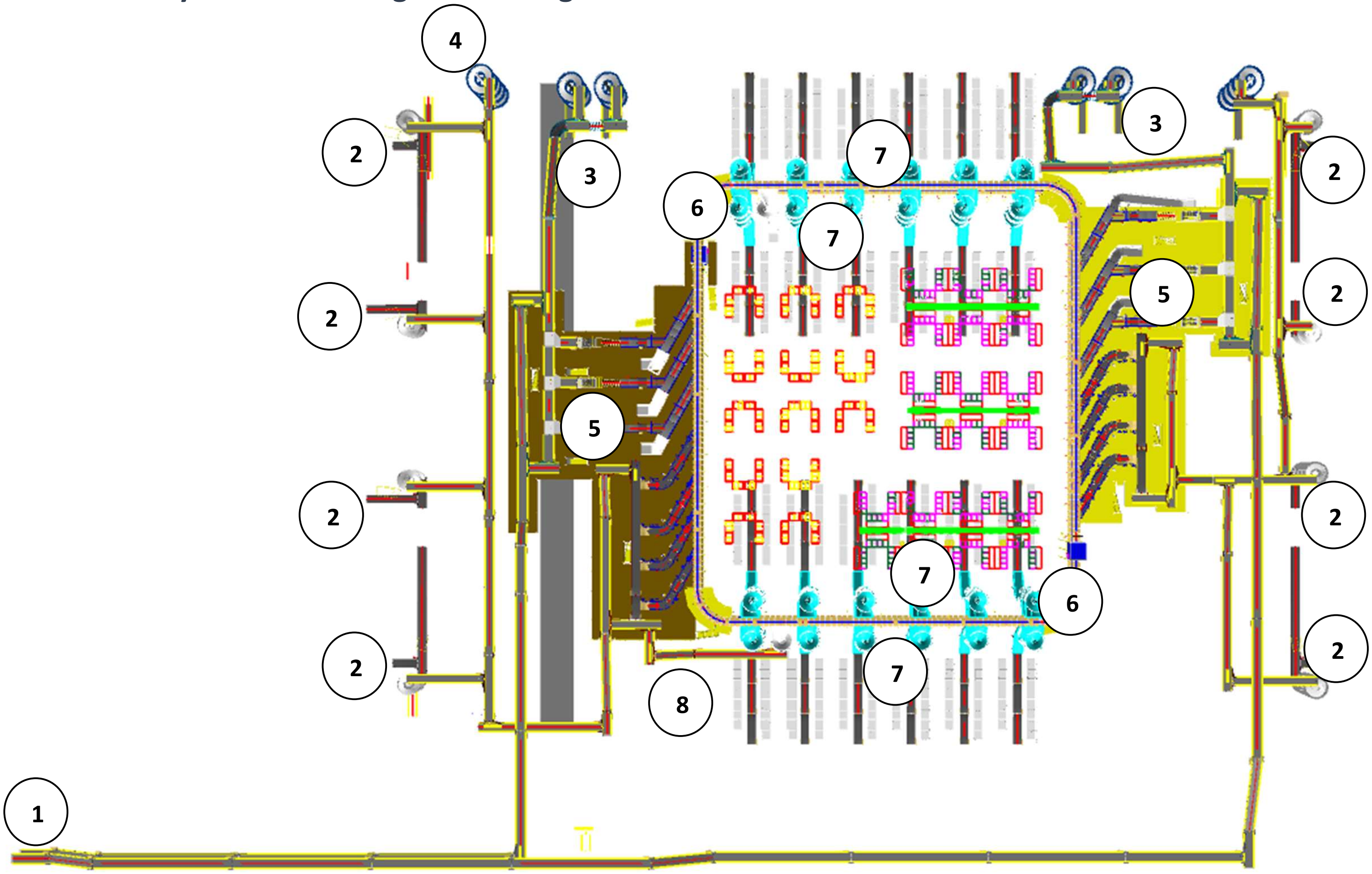
12.3.6.1 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	PVC Belt Conveyor	5800	5900	19.94	1210	1
2	PVC Belt Conveyor	5800	5900	19.94	1210	1
3	PVC Belt Conveyor	5800	5900	19.94	1210	1
4	PVC Belt Conveyor	5800	5900	19.94	1210	1
5	PVC Belt Conveyor	5800	5900	3.6	1210	1
6	PVC Belt Conveyor	5800	5900	16.58	1210	1
7	PVC Belt Conveyor	5738	5838	4.2	1210	1
8	PVC Belt Conveyor	5800	5900	7.46	1210	1
9	PVC Belt Conveyor	5800	5900	19.94	1210	1
10	PVC Belt Conveyor	5800	5900	19.94	1210	1
11	PVC Belt Conveyor	5800	5900	19.94	1210	1
12	PVC Belt Conveyor	5800	5900	19.94	1210	1
13	PVC Belt Conveyor	5800	5900	3.6	1210	1
14	PVC Belt Conveyor	5800	5900	16.58	1210	1
15	PVC Belt Conveyor	5738	5838	4.2	1210	1
16	PVC Belt Conveyor	5800	5900	7.46	1210	1
17	PVC Belt Conveyor	5800	5900	19.94	1210	1
18	PVC Belt Conveyor	5738	5838	19.94	1210	1
19	PVC Belt Conveyor	5738	5838	19.94	1210	1
20	PVC Belt Conveyor	5800	5900	19.94	1210	1
21	PVC Belt Conveyor	5800	5900	19.94	1210	1
22	PVC Belt Conveyor	5800	5900	19.94	1210	1
23	PVC Belt Conveyor	5800	5900	19.94	1210	1
24	PVC Belt Conveyor	5800	5900	19.94	1210	1
25	PVC Belt Conveyor	5738	5838	19.94	1210	1
26	PVC Belt Conveyor	5738	5838	19.94	1210	1

27	PVC Belt Conveyor	5800	5900	19.94	1210	1
28	PVC Belt Conveyor	5800	5900	19.94	1210	1
29	PVC Belt Conveyor	5800	5900	19.94	1210	1
30	PVC Belt Conveyor	5800	5900	19.94	1210	1
31	PVC Belt Conveyor	8200	8200	9.6	1210	1
32	PVC Belt Conveyor	8200	8200	19.68	1210	1
33	PVC Belt Conveyor	8450	8450	11.04	1210	1
34	PVC Belt Conveyor	7588	8940	11.465	1210	1
35	PVC Belt Conveyor	7947	7947	17.76	1210	1
36	PVC Belt Conveyor	8450	8450	11.04	1210	1
37	PVC Belt Conveyor	8200	8200	19.68	1210	1
38	PVC Belt Conveyor	8200	8200	4.32	1210	1
39	PVC Belt Conveyor	8200	8200	19.68	1210	1
40	PVC Belt Conveyor	8450	8450	11.04	1210	1
41	PVC Belt Conveyor	450	450	5.76	1210	1
42	PVC Belt Conveyor	7697	7697	3.36	1210	1
43	PVC Belt Conveyor	8450	8450	11.04	1210	1
44	PVC Belt Conveyor	8200	8200	19.68	1210	1
45	PVC Belt Conveyor	5800	5900	19.94	1210	1
46	PVC Belt Conveyor	5800	5900	19.94	1210	1
47	PVC Belt Conveyor	5800	5900	19.94	1210	1
48	PVC Belt Conveyor	5800	5900	19.94	1210	1
49	PVC Belt Conveyor	5800	5900	3.6	1210	1
50	PVC Belt Conveyor	5800	5900	16.58	1210	1
51	PVC Belt Conveyor	5738	5838	4.2	1210	1
52	PVC Belt Conveyor	5800	5900	7.46	1210	1
53	PVC Belt Conveyor	5800	5900	19.94	1210	1
54	PVC Belt Conveyor	5738	5838	19.94	1210	1
55	PVC Belt Conveyor	5738	5838	19.94	1210	1
56	PVC Belt Conveyor	5800	5900	19.94	1210	1
57	PVC Belt Conveyor	5800	5900	19.94	1210	1
58	PVC Belt Conveyor	5800	5900	19.94	1210	1
59	PVC Belt Conveyor	5800	5900	19.94	1210	1
60	PVC Belt Conveyor	5800	5900	19.94	1210	1
61	PVC Belt Conveyor	5800	5900	19.94	1210	1
62	PVC Belt Conveyor	5800	5900	19.94	1210	1
63	PVC Belt Conveyor	5800	5900	19.94	1210	1
64	PVC Belt Conveyor	5800	5900	3.6	1210	1
65	PVC Belt Conveyor	5800	5900	16.58	1210	1
66	PVC Belt Conveyor	5738	5838	5.06	1210	1
67	PVC Belt Conveyor	5800	5900	7.46	1210	1
68	PVC Belt Conveyor	5800	5900	19.94	1210	1
69	PVC Belt Conveyor	5738	5838	19.94	1210	1
70	PVC Belt Conveyor	5738	5838	19.94	1210	1

71	PVC Belt Conveyor	5800	5900	19.94	1210	1
72	PVC Belt Conveyor	5800	5900	19.94	1210	1
73	PVC Belt Conveyor	5800	5900	19.94	1210	1
74	PVC Belt Conveyor	5800	5900	19.94	1210	1

13.Layout View of Bag & Semi-large Sorter



Legend

1 – Infeed Conveyor (FC)

2 – Infeed Powered Spiral (From Shipment sorters)

3 – Infeed Powered Spiral (Cross Dock Semi-large shipments)

4 - Infeed Powered Spiral (Cross Dock Bags)

5 – Induct Lines

6 – Main Loop CBS

7 – Output Spiral Chutes

8 – Refeeding Line

14. Proposed System Capacity Calculations

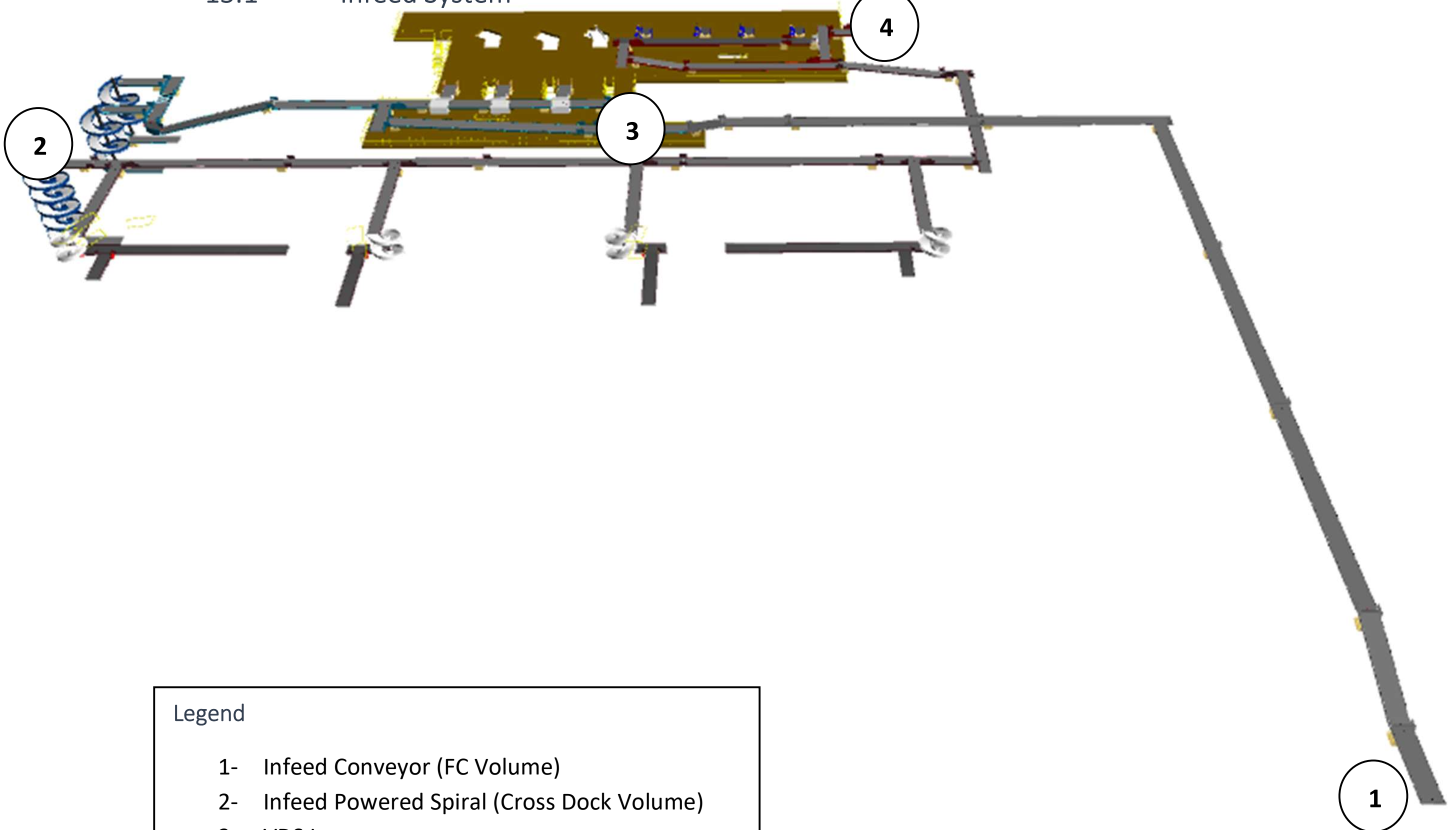
14.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on Flipkart RFP requirements.

SPECIFICATION	VALUE
Sorter Type	**Loop CBS**
Speed	**2.3 m/s**
Pitch	**800 mm**
Carrier per hour	**10,350 CPH**
Belts per hour	**10,350 BPH**
Effective Designed Throughput of 1 loop CBS (A)	**10,350 Shipments per hour**
No of Feedlines (Bags + Semi-large) per loop	**7 (4+3) Nos.**
Per Feedline Capacity (Bags)	**900 Bags per hour**
Total feedline capacity (Bags) per loop	**3,600 Bags per hour**
Per Feedline Capacity (Semi-large shipment)	**2400 SL Shipments per hour**
Total feedline capacity (Bags) per loop	**7,200 SL Shipments per hour**
Total Feedline designed capacity (B)	**10,800 Shipments per hour**
System designed throughput	**10,350 Shipments per hour**
Designed throughput from Two loops	**20,700 Shipments per hour**

15. System description

15.1 Infeed System



Legend

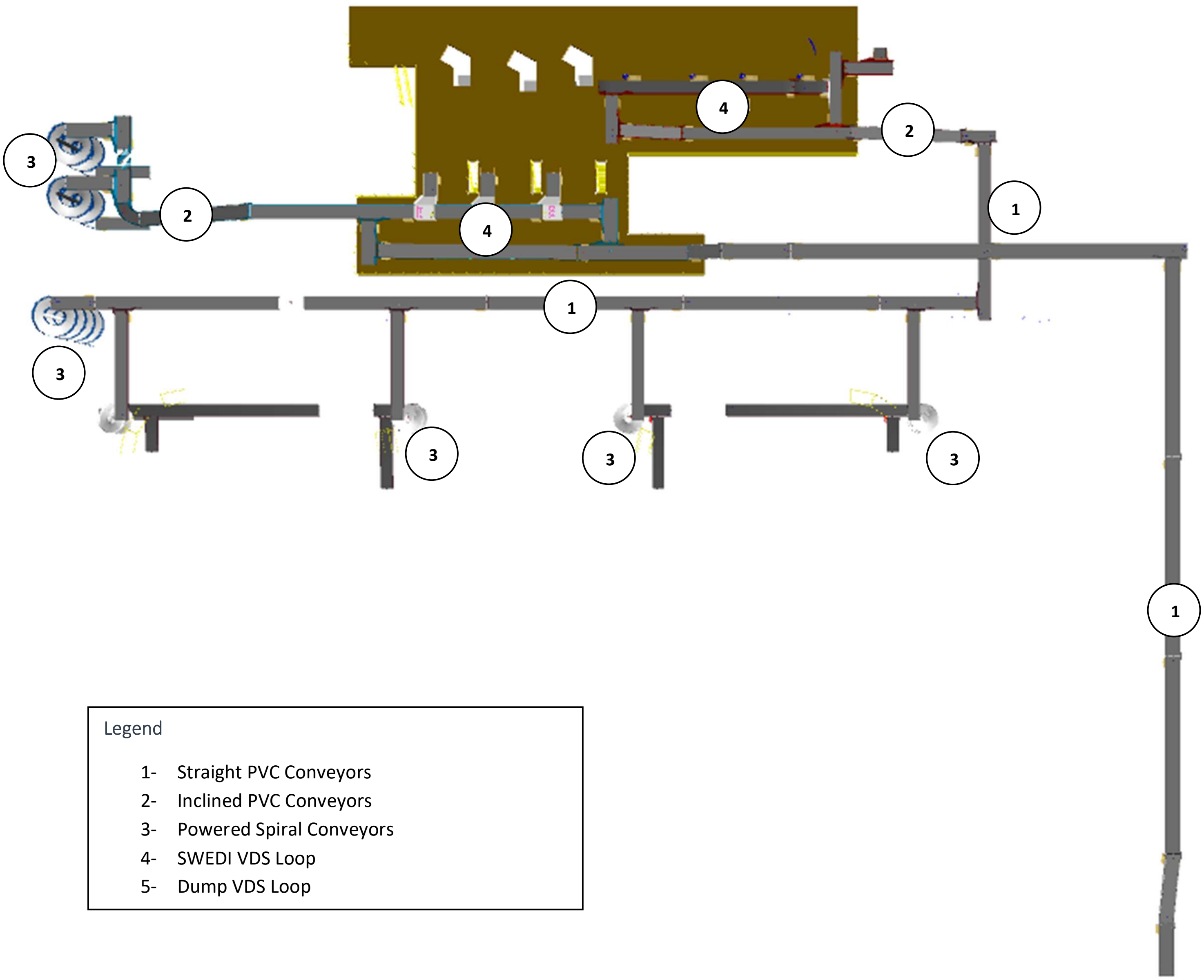
1- Infeed Conveyor (FC Volume)
2- Infeed Powered Spiral (Cross Dock Volume)
3- VDS Loop
4- Refeed Conveyor.

CBS loop is equipped with Infeed conveyor facilitating the movement of FC SL shipment volume to Loop VDS and in parallel, Two Powered Spiral for facilitating the movement of Marketplace SL shipment volume to the same VDS loop. Additionally, Bags transported from NL Shipment & Cross Dock Bags are connected to separate VDS loop for further induction.

15.1.1 Infeed System

Infeed conveyor system consists of below components:

1. Straight PVC Conveyor
2. Inclined PVC Conveyor
3. Powered Spiral Conveyor
4. VDS System (Arm based/ Tilted Belt)

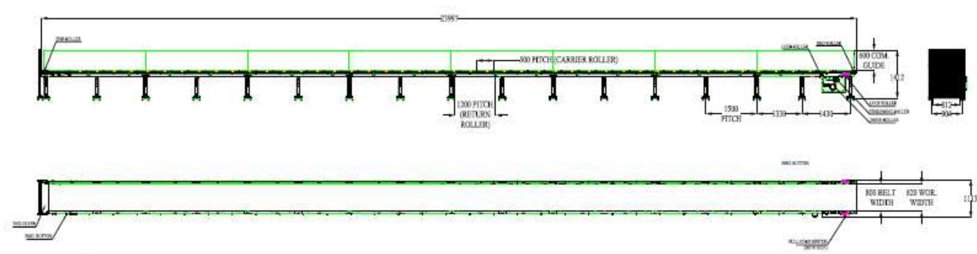


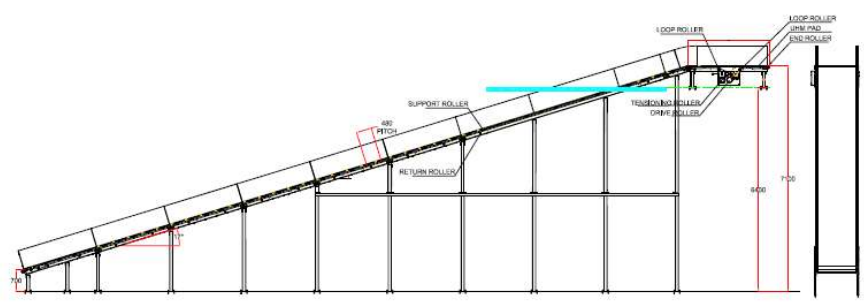
15.1.1.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.
Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.





Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

15.1.1.2 Powered Spiral Conveyor

Specification	UOM	Remark
Manufacturer Name	Name	Apollo/Ambaflex
Overall height	mm	As per Layout requirement
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	mm	approx. 600
Roller Diameter	mm	approx. 50
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Yes
End Stopper	yes/no	Yes

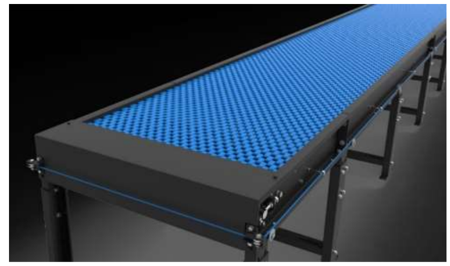
15.1.1.3 Roller Aligner conveyor

15.1.1.4 ZTR Curves (Pic for ref.)

15.1.1.5 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:



- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make	Make	Movex
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Wear Strip Material	MOC	UHMW PE1000
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Driven Shaft	Type	MOC-EN9 Solid Shaft hardened to BHN 240, Size-60SQ MM, Zinc Plated -25 microns, Bearings-UCF-210D1
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

15.1.1.6 Swivel Wheel Divert

Swivel Wheel Diverter (SWEDI) is a non-impact sortation module for linear sortation lines, ideal for medium volume sort centres. SWEDI's modular design allows it to save space while also allowing for future expansion of sorting point

Technical specifications of SWEDI Unit

Items	Make
SWEDI Unit's Dimensions	2000 mm (L) x 1400 mm (W)
Load Capacity	35 KG
Throughput	2400 PPH
Conveying Drive Unit	48V, 50 Watt DC powered smart roller
Diverting Drive Unit	Servo Motor; 220V, 50HZ, 750W, Single Phase
Conveying Speed	Up to 120 m/min
Diverting Angle	90 Degrees at double sides
Frame Material	MS Powder Coated Steel Frame
Stand Support	MS Powder Coated Steel Frame
PLC	Siemens
Servo Motor	Delta
Noise Level	<75 Db when measured from 1 m in a silent warehouse.

15.1.1.7 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	Width (mm)	Set
1	60° Belt turn	9000	9000	NA	1010	4
2	60° Belt turn	9000	9000	NA	1010	4
3	90° Belt turn	6300	6300	NA	1410	1
4	90° Belt turn	6300	6300	NA	1410	1
5	PVC Belt Conveyor	9000	9000	2	1010	4
6	PVC Belt Conveyor	9000	9000	2	1010	4
7	Aligning Conveyor	9000	9000	3	1410	0
8	Aligning Conveyor	9000	9000	3	1410	0
9	Aligning Conveyor	9000	9000	2	1410	1
10	Aligning Conveyor	9000	9000	2	1410	1
11	Singulator (With center aligning)	NA	NA	NA	NA	3
12	Singulator (With center aligning)	NA	NA	NA	NA	3
13	PVC Belt Conveyor	11500	11500	2	1410	1
14	PVC Belt Conveyor	11500	11500	2	1410	1
15	PVC Belt Conveyor	11500	11500	2	1410	1
16	PVC Belt Conveyor	9000	9000	2	1410	1
17	PVC Belt Conveyor	9000	9000	2	1410	1
18	PVC Belt Conveyor	9000	9000	2	1410	1
19	PVC Belt Conveyor	11250	11250	3	1410	1
20	PVC Belt Conveyor	NA	NA	6	1210	1
21	PVC Belt Conveyor	NA	NA	20	1210	1
22	PVC Belt Conveyor	NA	NA	6	1210	1
23	PVC Belt Conveyor	NA	NA	12	1210	1
24	PVC Belt Conveyor	NA	NA	8	1210	1
25	PVC Belt Conveyor	NA	NA	4	1210	1
26	PVC Belt Conveyor	NA	NA	18	1210	1
27	PVC Belt Conveyor	NA	NA	9	1410	1
28	PVC Belt Conveyor	NA	NA	20	1410	1
29	PVC Belt Conveyor	NA	NA	20	1410	1
30	PVC Belt Conveyor	NA	NA	20	1410	1
31	PVC Belt Conveyor	NA	NA	20	1410	1
32	PVC Belt Conveyor	NA	NA	20	1410	1
33	PVC Belt Conveyor	NA	NA	5	1410	1
34	PVC Belt Conveyor	11750	11750	8	1410	1
35	PVC Belt Conveyor	NA	NA	10	1410	1
36	PVC Belt Conveyor	NA	NA	3	1410	1
37	PVC Belt Conveyor	NA	NA	1	1410	1
38	PVC Belt Conveyor	NA	NA	6	1210	1
39	PVC Belt Conveyor	11750	11750	2	1410	1
40	PVC Belt Conveyor	NA	NA	18	1410	1

41	PVC Belt Conveyor	11750	11750	6	1410	1
42	PVC Belt Conveyor	11500	11500	5	1210	1
43	PVC Belt Conveyor	9250	9250	4	1410	1
44	PVC Belt Conveyor	8393	8393	3	1210	1
45	PVC Belt Conveyor	8143	8143	13	1210	1
46	PVC Belt Conveyor	NA	NA	3	1410	1
47	PVC Belt Conveyor	NA	NA	5	1410	1
48	PVC Belt Conveyor	9000	9000	4	1410	1
49	PVC Belt Conveyor	NA	NA	16	1410	1
50	PVC Belt Conveyor	NA	NA	5	1410	1
51	PVC Belt Conveyor	NA	NA	6	1210	1
52	PVC Belt Conveyor	8750	8750	17	1210	1
53	PVC Belt Conveyor	8644	9756	6	1210	1
54	PVC Belt Conveyor	9000	9000	7	1210	1
55	PVC Belt Conveyor	NA	NA	6	1210	1
56	PVC Belt Conveyor	9500	9500	5	1210	1
57	PVC Belt Conveyor	NA	NA	20	1410	1
58	PVC Belt Conveyor	NA	NA	4	1410	1
59	PVC Belt Conveyor	NA	NA	11	1410	1
60	PVC Belt Conveyor	NA	NA	3	1410	1
61	PVC Belt Conveyor	NA	NA	13	1410	1
62	PVC Belt Conveyor	8041	9465	6	1210	1
63	PVC Belt Conveyor	9209	9209	5	1210	1
64	PVC Belt Conveyor	9250	9250	4	1410	1
65	PVC Belt Conveyor	9250	9250	4	1410	1
66	PVC Belt Conveyor	11144	12256	6	1210	1
67	PVC Belt Conveyor	11250	11250	16	1210	1
68	PVC Belt Conveyor	12000	12000	5	1210	1
69	PVC Belt Conveyor	11750	11750	5	1410	1
70	PVC Belt Conveyor	NA	NA	4	1410	1
71	PVC Belt Conveyor	11500	11500	4	1410	1
72	PVC Belt Conveyor	NA	NA	6	1210	1
73	PVC Belt Conveyor	NA	NA	1	1410	1
74	PVC Belt Conveyor	NA	NA	3	1410	1
75	PVC Belt Conveyor	NA	NA	9	1410	1
76	PVC Belt Conveyor	NA	NA	15	1410	1
77	PVC Belt Conveyor	NA	NA	5	1410	1
78	PVC Belt Conveyor	NA	NA	8	1410	1
79	PVC Belt Conveyor	NA	NA	20	1410	1
80	PVC Belt Conveyor	NA	NA	20	1410	1
81	PVC Belt Conveyor	NA	NA	20	1410	1
82	PVC Belt Conveyor	NA	NA	20	1410	1
83	PVC Belt Conveyor	NA	NA	17	1410	1
84	PVC Belt Conveyor	NA	NA	2	1210	1

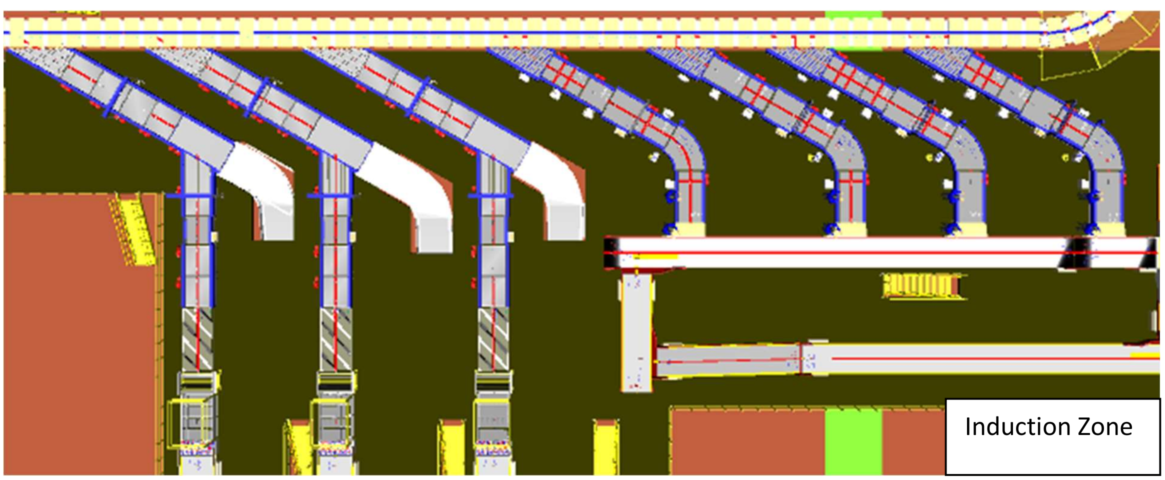
85	PVC Belt Conveyor	NA	NA	20	1210	1
86	PVC Belt Conveyor	NA	NA	4	1210	1
87	PVC Belt Conveyor	NA	NA	6	1210	1
88	PVC Belt Conveyor	NA	NA	15	1210	1
89	PVC Belt Conveyor	NA	NA	9	1210	1
90	PVC Belt Conveyor	NA	NA	10	1210	1
91	PVC Belt Conveyor	NA	NA	12	1410	1
92	PVC Belt Conveyor	NA	NA	18	1410	1
93	Modular belt conveyor	9250	9250	3	1210	1
94	Modular belt conveyor	9250	9250	16	1210	1
95	Modular belt conveyor	9250	9250	6	1210	1
96	Modular belt conveyor	9250	9250	20	1210	1

15.2 Induction to Bag & Semi-Large Sorter

The main purpose of the feedlines is to transport the Semi-large shipments & Bag from the infeed points to the sorter by means of Singulator, straight and curved belt conveyors, 30° merging conveyors and 30° induction lines. The flow of shipments on the belt conveyors is controlled by sensors in such a way that a gap is ensured between the shipments.

Weighing Scales installed on the feedlines will capture the weight of each shipment and will perform weight control to detect out-of-spectrum shipments.

The system's design ensures that the maximum capacity of each feed, considered as the sum of the flows from the connected inductions, is always less than the planned induction capacity of 2400 for Semi-Large shipments/hour and 900 bags/hour.



15.2.1 Automatic Feedlines for Semi-large Shipment

Each Loop CBS has a total 3 Automatic Feedlines for Semi-Large Shipment with Designed IPP of 2400 PPH.

1. Each Feedline has 13 Conveyor Modules-

    - Singulator- 1 Nos.
    - Aligning Conveyor- 1 Nos.
    - Weighing Conveyor- 1 Nos.
    - Receiving Conveyor- 1 Nos.
    - Spacing Conveyor- 4 Nos.
    - Buffer conveyor- 3 Nos.
    - Intelligent Merge- 2 Nos.
    - Oversize/Overweight Chute- 1 Nos

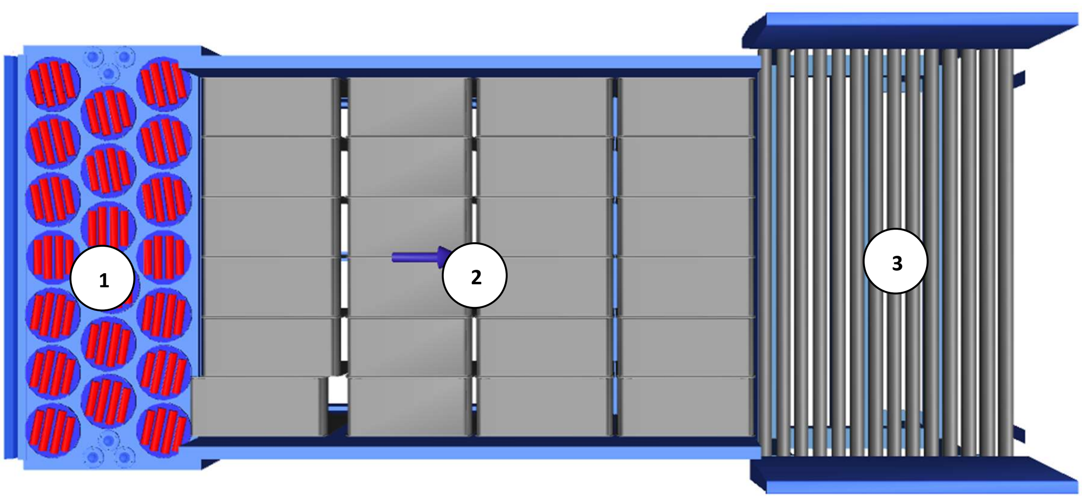
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.

3. Over the feedline, shipments are evenly spaced and smoothly transferred to the main Loop.

15.2.1.1 Singulator

To maintain singulated flow for semi-large parcels, Singulators are provided at Automatic Induct Lines. Each Singulator is a combination of three conveyor set.

- Divergence Module
- Separation Module
- Roller Section



15.2.1.2 Aligning Conveyors

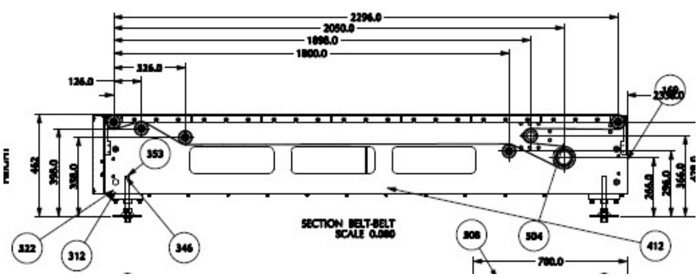
An Aligning conveyor is a type of conveyor system used to align the shipments to a particular side of the conveyor.

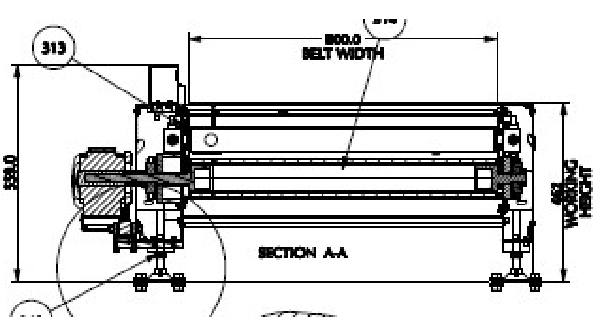
This Conveyors required to maintain the product alignment on the line.

15.2.1.3 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.





15.2.1.4 Weighing Conveyor

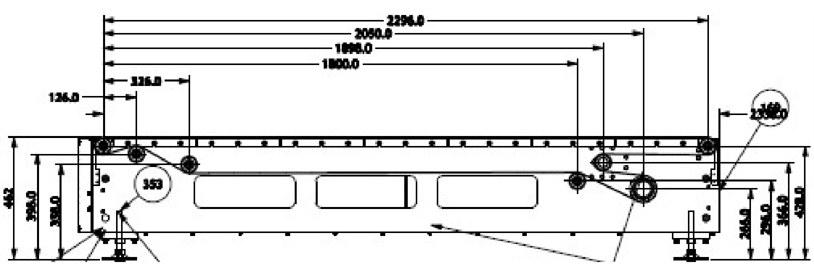
A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

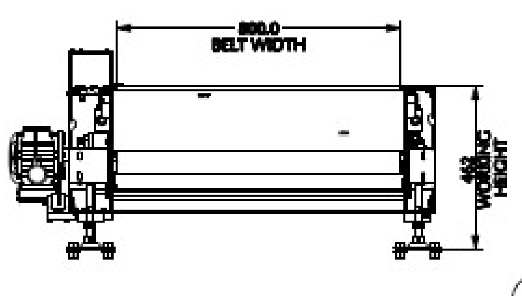
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

Make- Bizerba/ Mettler Toledo

15.2.1.5 Receiving Conveyor

A Receiving conveyor is a type of conveyor system used to receive & Release the products for induction onto CBS. It serves as the connection point at turn point of entry where products are collected and conveyed to subsequent stages of the process.

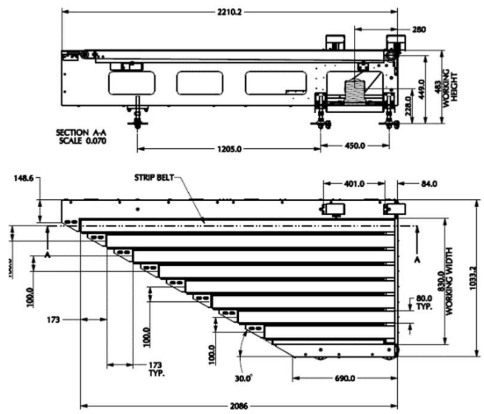




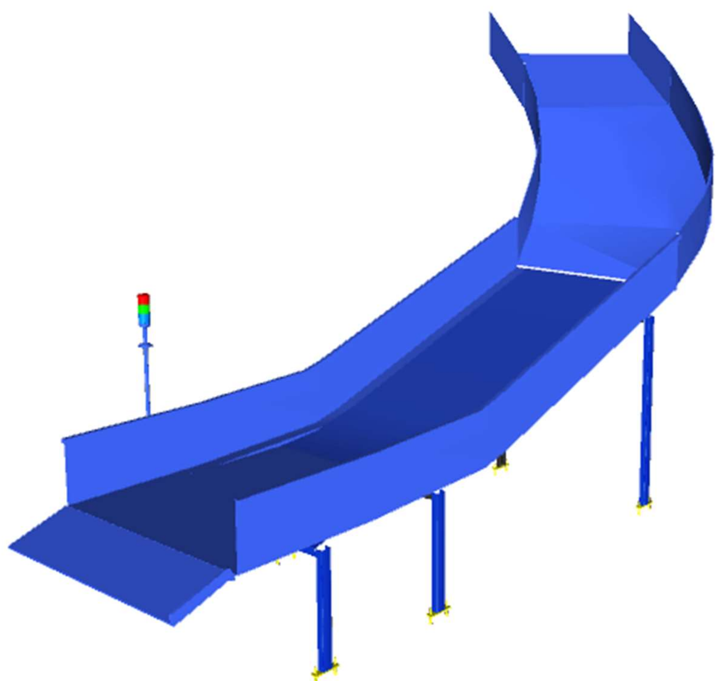
15.2.1.6 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



15.2.2 Irregular Collection Chute



15.2.3 Semi-Automatic Feedlines for Bags

Each Loop CBS has a total 4 Semi-Automatic Feedlines for bags with Designed IPP of 900 PPH

1. Each Feedline has Four Conveyor Modules-

- Scanning Conveyor- 1 Nos.
- Weighing Conveyor- 1 Nos.
- Buffer conveyor- 2 Nos.
- Intelligent Merge- 1 Nos.

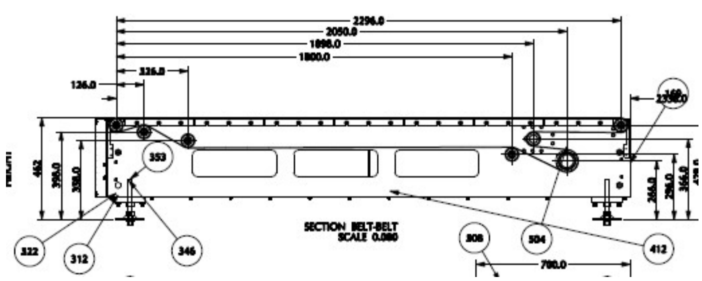
4. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.

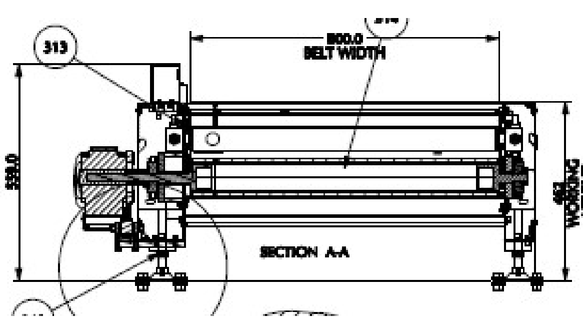
5. Over the feedline, shipments are evenly spaced and smoothly transferred to the main Loop.

15.2.3.1 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.





15.2.3.2 Weighing Conveyor

A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

Make- Bizerba/ Mettler Toledo

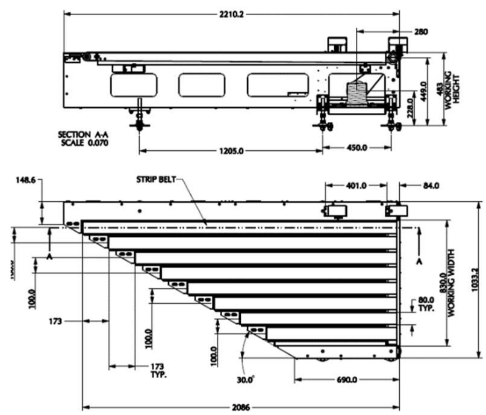
15.2.3.3 Scanning Conveyor

A Scanning conveyor is a type of conveyor system used to Pause, Scan & Release the bags for induction onto CBS. It serves as the Static scanning of product where products are paused for a while and conveyed to subsequent stages of the process after scanning

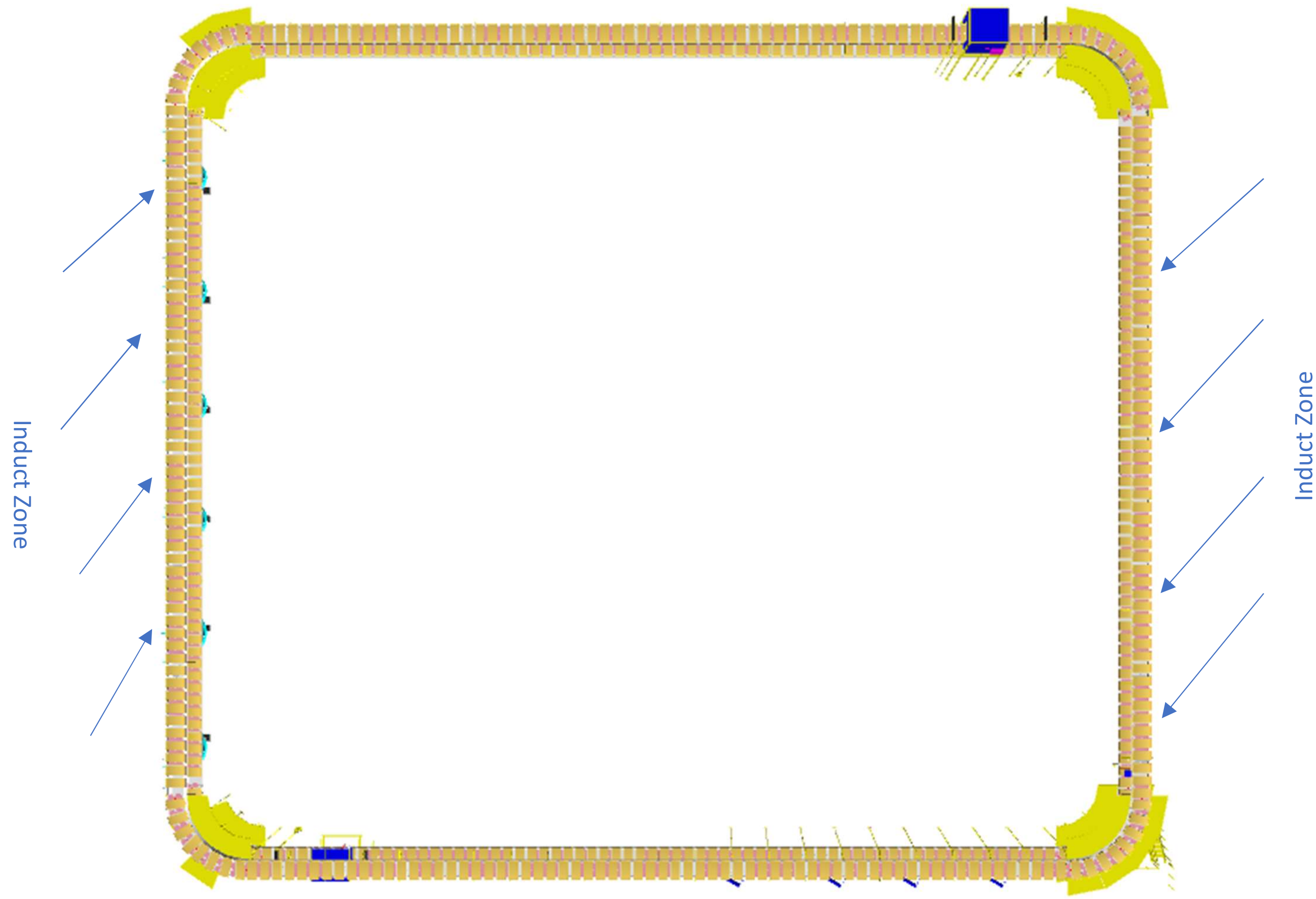
15.2.3.4 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



15.2.4 Main Loop



1. The proposed Double Deck CBS Sorter will be installed on Mezzanine level. The Sorter runs at 3500 mm & 6000 mm from Mezzanine level.

2. We have provided loop with Single Belt Carriers with 800 mm pitch and Belt Size of 600 x 1200mm.

3. One loop CBS has a total length of 223 m & 280 number of carriers.

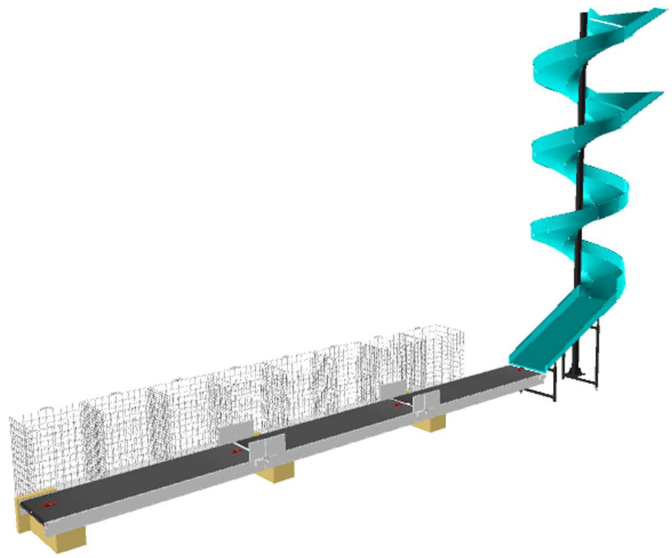
4. As SL Shipments passes through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, Bags are scanned manually at inducts, After the scanning chutes are assigned & once the shipment reaches the respective chute, the carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

15.2.5 Sorting Output

15.2.5.1 Spiral Chutes

A spiral chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level with compact footprint. It utilizes base metal to guide and support the flow of items along the chute length & drops the bags or Semi-large shipment onto Belt conveyor that further follows the roll cage trolley.

Each Spiral chute is further linked to 8 roll cage trolley via modular conveyor line (roll cage trolley is In FK scope).



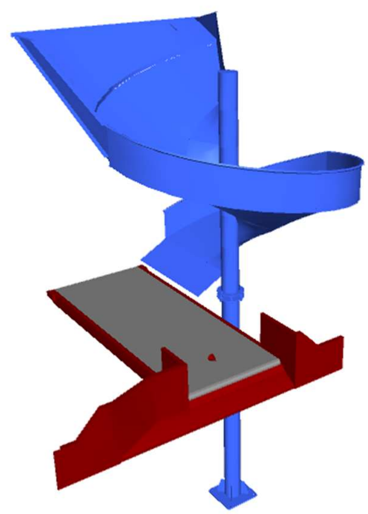
Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

15.2.5.2 Rejection Chute

Rejection Chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & transfer the Shipment/Bags to Powered belt conveyor



Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Four Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Four Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

15.2.5.3 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	Modular belt conveyor	1002	1002	4.62	1210	1
2	Modular belt conveyor	1002	1002	4.62	1210	1
3	Modular belt conveyor	1002	1002	4.62	1210	1
4	Modular belt conveyor	1002	1002	4.62	1210	1
5	Modular belt conveyor	1002	1002	4.62	1210	1
6	Modular belt conveyor	1002	1002	4.62	1210	1
7	Modular belt conveyor	1002	1002	4.62	1210	1
8	Modular belt conveyor	1002	1002	4.62	1210	1
9	Modular belt conveyor	1002	1002	4.62	1210	1
10	Modular belt conveyor	1002	1002	4.62	1210	1
11	Modular belt conveyor	1002	1002	4.62	1210	1
12	Modular belt conveyor	1002	1002	4.62	1210	1
13	Modular belt conveyor	1002	1002	4.62	1210	1
14	Modular belt conveyor	1002	1002	4.62	1210	1
15	Modular belt conveyor	1002	1002	4.62	1210	1
16	Modular belt conveyor	1002	1002	4.62	1210	1
17	Modular belt conveyor	1002	1002	4.62	1210	1
18	Modular belt conveyor	1002	1002	4.62	1210	1
19	Modular belt conveyor	1002	1002	4.62	1210	1
20	Modular belt conveyor	1002	1002	4.62	1210	1
21	Modular belt conveyor	1002	1002	4.62	1210	1
22	Modular belt conveyor	1002	1002	4.62	1210	1
23	Modular belt conveyor	1002	1002	4.62	1210	1
24	Modular belt conveyor	1002	1002	4.62	1210	1
25	Modular belt conveyor	1002	1002	4.62	1210	1
26	Modular belt conveyor	1002	1002	4.62	1210	1
27	Modular belt conveyor	1002	1002	4.62	1210	1
28	Modular belt conveyor	1002	1002	4.62	1210	1
29	Modular belt conveyor	1002	1002	4.62	1210	1
30	Modular belt conveyor	1002	1002	4.62	1210	1
31	Modular belt conveyor	1002	1002	4.62	1210	1
32	Modular belt conveyor	1002	1002	4.62	1210	1
33	Modular belt conveyor	1002	1002	4.62	1210	1
34	Modular belt conveyor	1002	1002	4.62	1210	1
35	Modular belt conveyor	1002	1002	4.62	1210	1
36	Modular belt conveyor	1002	1002	4.62	1210	1
37	Modular belt conveyor	1002	1002	4.62	1210	1
38	Modular belt conveyor	1002	1002	4.62	1210	1
39	Modular belt conveyor	1002	1002	4.62	1210	1
40	Modular belt conveyor	1002	1002	4.62	1210	1
41	Modular belt conveyor	1002	1002	4.62	1210	1

42	Modular belt conveyor	1002	1002	4.62	1210	1
43	Modular belt conveyor	1002	1002	4.62	1210	1
44	Modular belt conveyor	1002	1002	4.62	1210	1
45	Modular belt conveyor	1002	1002	4.62	1210	1
46	Modular belt conveyor	1002	1002	4.62	1210	1
47	Modular belt conveyor	1002	1002	4.62	1210	1
48	Modular belt conveyor	1002	1002	4.62	1210	1
49	Modular belt conveyor	1002	1002	4.62	1210	1
50	Modular belt conveyor	1002	1002	4.62	1210	1
51	Modular belt conveyor	1002	1002	4.62	1210	1
52	Modular belt conveyor	1002	1002	4.62	1210	1
53	Modular belt conveyor	1002	1002	4.62	1210	1
54	Modular belt conveyor	1002	1002	4.62	1210	1
55	Modular belt conveyor	1002	1002	4.62	1210	1
56	Modular belt conveyor	1002	1002	4.62	1210	1
57	Modular belt conveyor	1002	1002	4.62	1210	1
58	Modular belt conveyor	1002	1002	4.62	1210	1
59	Modular belt conveyor	1002	1002	4.62	1210	1
60	Modular belt conveyor	1002	1002	4.62	1210	1
61	Modular belt conveyor	1002	1002	4.62	1210	1
62	Modular belt conveyor	1002	1002	4.62	1210	1
63	Modular belt conveyor	1002	1002	4.62	1210	1
64	Modular belt conveyor	1002	1002	4.62	1210	1
65	Modular belt conveyor	1002	1002	4.62	1210	1
66	Modular belt conveyor	1002	1002	4.62	1210	1
67	Modular belt conveyor	1002	1002	4.62	1210	1
68	Modular belt conveyor	1002	1002	4.62	1210	1
69	Modular belt conveyor	1002	1002	4.62	1210	1
70	Modular belt conveyor	1002	1002	4.62	1210	1
71	Modular belt conveyor	1002	1002	4.62	1210	1
72	Modular belt conveyor	1002	1002	4.62	1210	1

16. Description of Components of Equipment

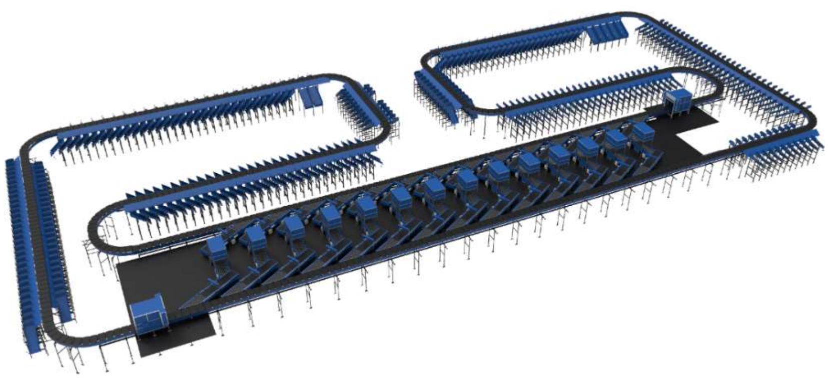
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

16.1 Cross Belt Sorter -

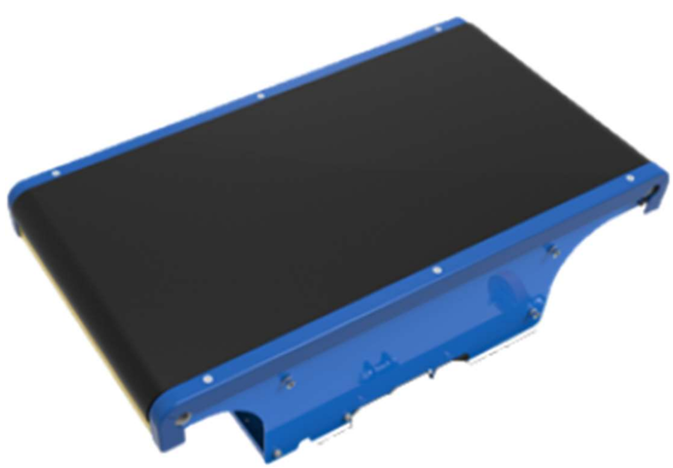
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



16.1.1 CBS Carrier

Falcon's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners "and non-sortable shipment recirculation.



16.1.2 Servo Roller

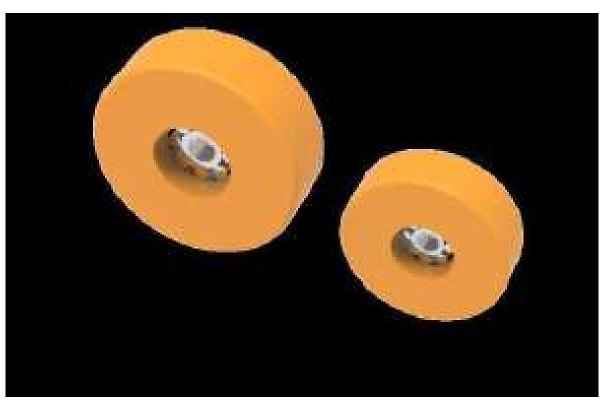
High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.

16.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.

16.1.4 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle.



16.1.5 Friction Wheel Drive

Extremely High Energy Efficient FWD that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed. Using servo motors to power friction wheels, FWD makes it better in terms of controlled speed, acceleration & deceleration

16.1.6 Non-contact based Linear motor drive

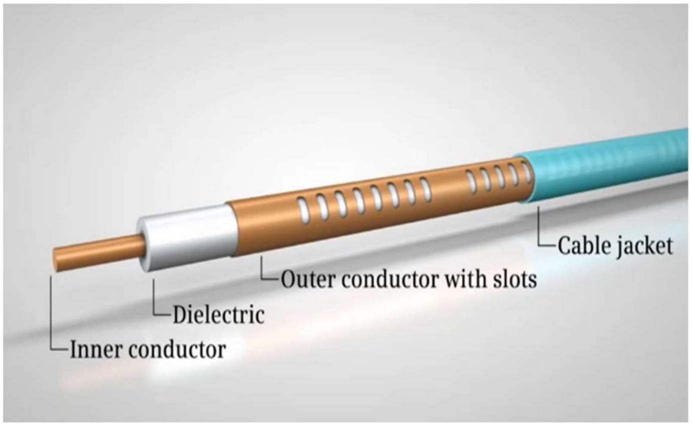
Extremely High Energy Efficient Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.

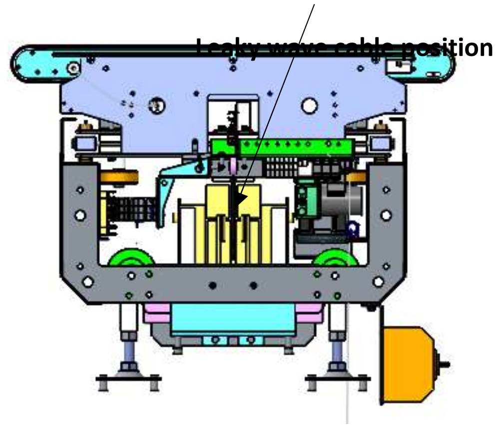
16.1.7 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.

16.1.8 Data Transmission

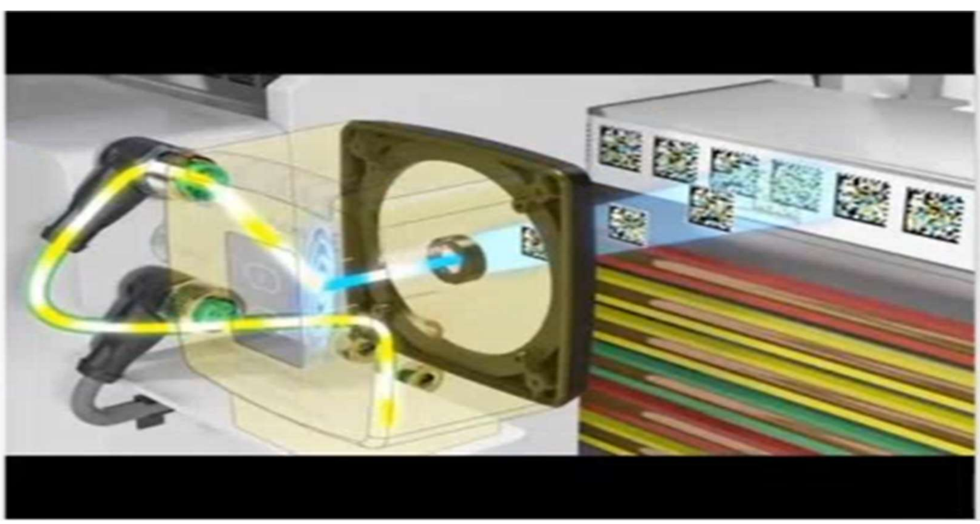
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.





16.1.9 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



16.1.10 Technical specifications of Sorter (Shipment Sorter)

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length (Upper Deck)	~254 m
Sorter Loop Length (Lower Deck)	~254 m
Sorter Actuation Technology	Electric
Chutes	
Direct Bagging chute	60
Secondary Chutes	60
VDS Chutes	24
Irregular chutes	24
Irregular chutes (02)	4
Rejection Chutes	4
DBO chutes	4
Height of Sorter	~5500 mm / 3500 mm
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	434 Nos.
Drives	
Motor/Drive type	LIM
Power Consumption	Will be specified during DAP
Feedlines	24 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	04 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

16.1.11 Technical specifications of Sorter (SL & Bag sorter)

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.3 m/s
Sorter Loop Length	~224 m
Sorter Actuation Technology	Electric
Chutes	
Spiral Chute	24
Sliding Plate	4
Overweight Chute	3
Height of Sorter	3500 mm
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	800
No. of Carriers	280 Nos.
Drives	
Motor/Drive type	LIM
Power Consumption	Will be specified during DAP
Feedlines	7 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

16.2 Scanning & Sensing on Main Sorter Loop

16.2.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan shipment 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. As per FLIPKART requirements, Scanning is in FLIPKART scope.

Falcon needs to supply Volume Measurement System. VMS 4200 quoted as option for Dimensioning of Shipments.

16.2.2 Shipment Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple shipment positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Shipment on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).

16.2.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the shipments on the empty carriers. (Camera based)

16.3 Steel Works – Mezzanine, Maintenance Platforms & Staircases-

16.3.1 Mezzanine (Non-large Shipment Sorter)

Plan view of Mezzanine

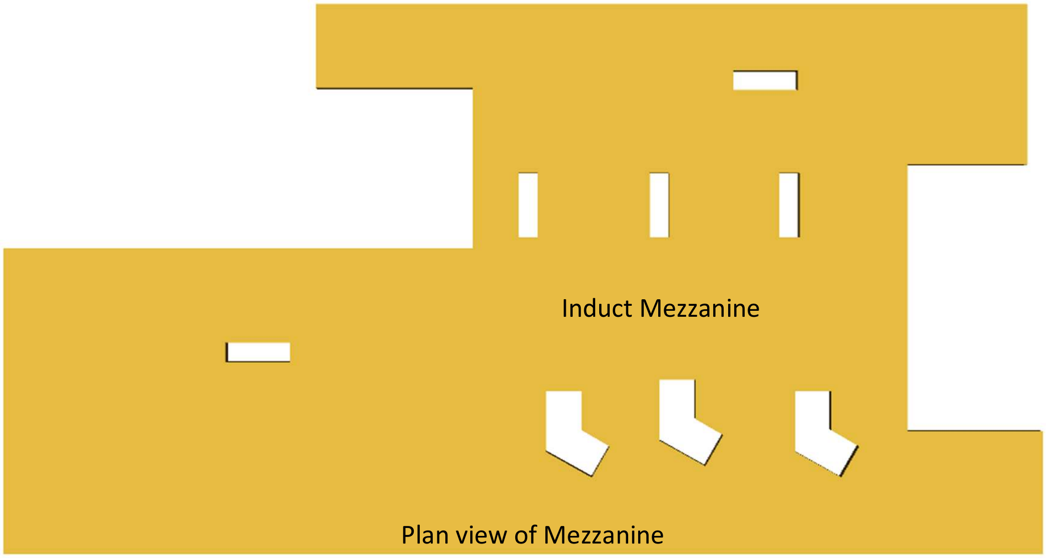
- Total Mezzanine Area: 6088 Sq. Metres
- Clear height below Mezzanine: 4400 mm / 2400 mm
- No of Stairs: 72 Nos

16.3.2 Maintenance Platform & Staircase

- 4 Sq. Meter: 324 Nos
- Safety doors: 324 Nos
- Monkey ladder: 324 Nos

Maintenance platform with lockable gate

16.3.3 Mezzanine (SL & Bag Sorter)

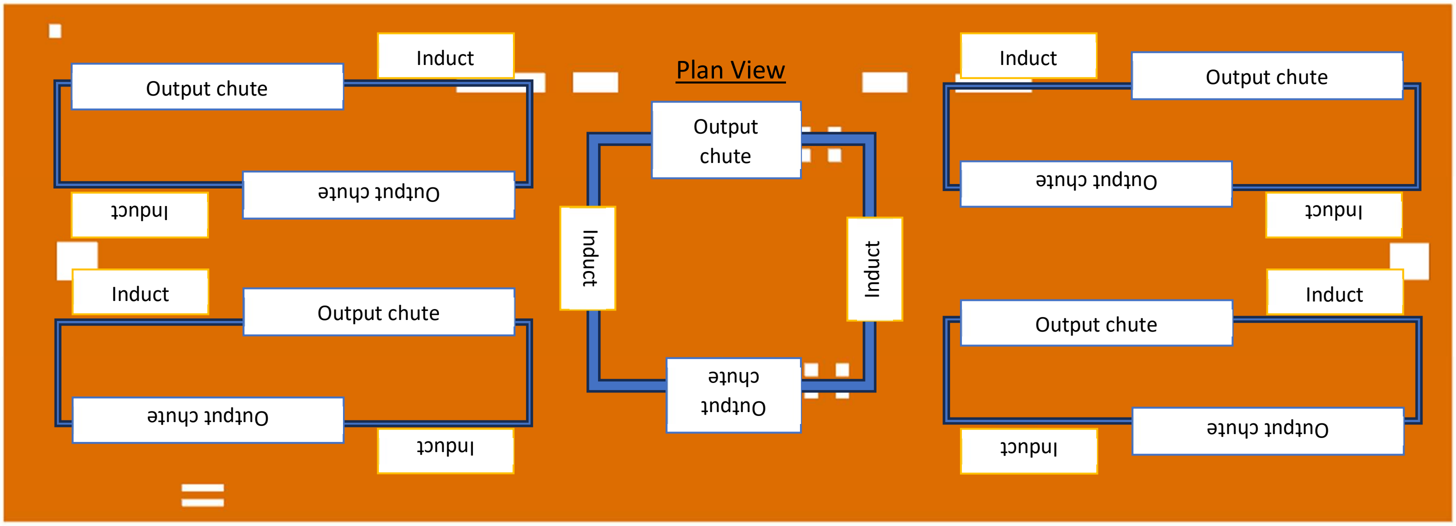


- Total Mezzanine Area: 1952 Sq. Metres
- Clear height below Mezzanine: 2400 mm / 4900 mm
- Stair Platform: ~12 Sq.metres
- No of Stairs: 20 Nos

16.3.4 Maintenance Platform & Staircase

- 4 Sq. Meter: 36 Nos
- Monkey ladder: 36 Nos
- Safety Gate: 36 Nos

17. System Layout & Stagging Area Description



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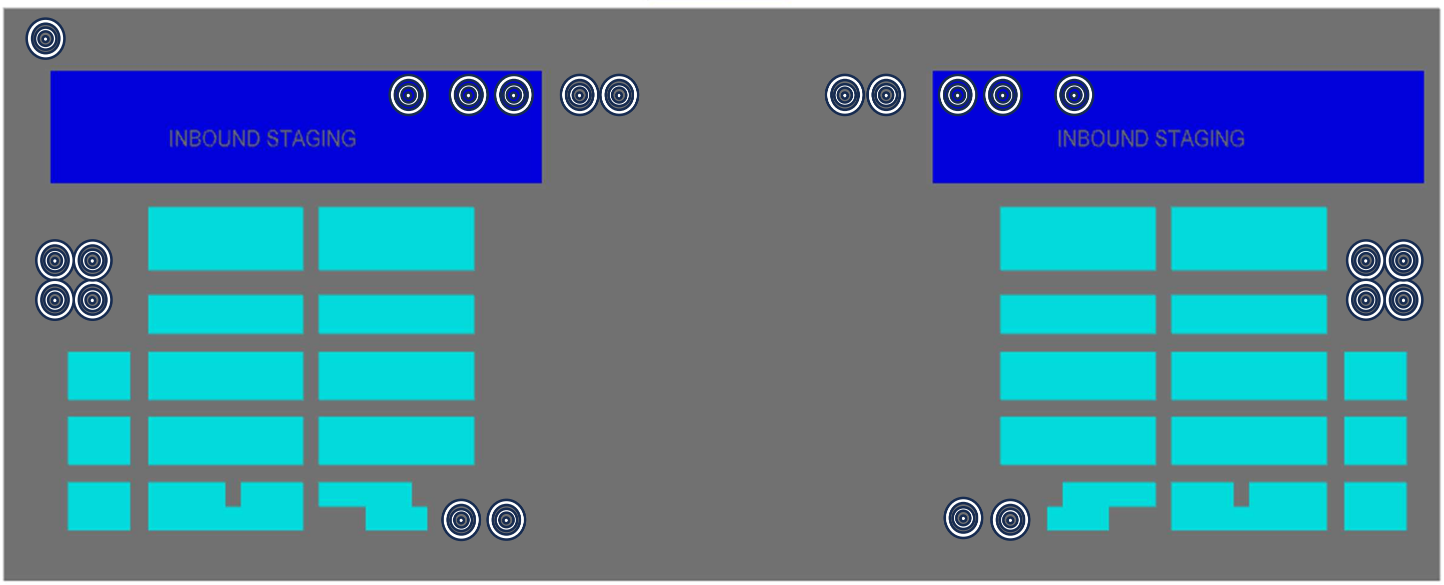

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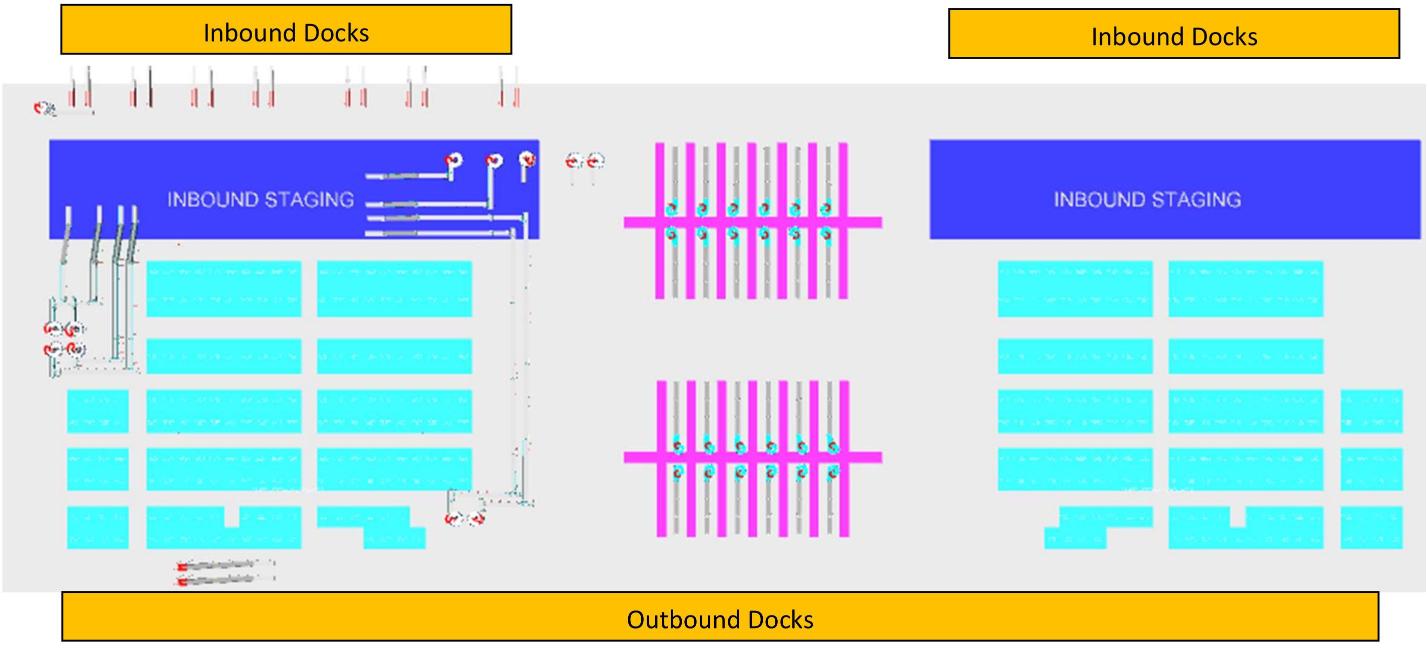
System is marked with different type of Staging areas.

Inbound Staging Area – A total area of 6,834 Sq. Meter is dedicated for Inbound staging of Bags & Shipments.

Outbound Stagging Area – A total area of 9636 Sq. Meter is dedicated for Outbound bag & Shipment Stagging.

Note – The dedicated Areas are considered basis the specification shared by Flipkart.

18. Dock Calculation



Inbound Network cluster as per RFQ	% volume	Truck /cap	Volume /hr	No of trucks	Truck TAT	Truck filling rate /hr	No of truck position	No of docks
Local	31%	90	4574	51	**42**	128	36	18
Trans	41%	120	6050	50	**52**	137	44	44
Zonal	7%	600	1033	2	**63**	571	2	2
National	22%	600	3246	5	**63**	571	6	6
Total Number of Inbound Docks								**69**

Outbound Network cluster as per RFQ	% volume	Truck /cap	Volume /hr	No of trucks	Truck TAT	Truck filling rate /hr	No of truck position	No of docks
Local	31%	90	2745	30	60	90	30	15
Trans	41%	120	3630	30	75	96	38	38
Zonal	7%	600	620	1	90	400	2	2
National	22%	600	1948	3	90	400	5	5
Total Number of Outbound Docks								**59**

The tabular data represent the total number of Inbound & Outbound dock = 128 Docks

19. Proposed System Technical Details (Shipment Sorter)

19.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	**Infeed System** • Powered Belt Conveyors • Modular Belt Conveyor • 90 deg turns • Telescopic Belt Conveyor • XL ARM Diverters/Tilted VDS gates • VDS Chutes • Powered Spiral Conveyors	 ~3083 metres (303 Modules) ~396 metres (32 Modules) 02 Nos 14 Nos 96 Nos 96 Nos 17 Nos
2	96	**Feedlines** Consists of • Receiving Conveyor • Weighing Conveyor • Buffer Conveyor • Intelligent merge	 1 Set 1 Set 1 Set 1 Set
3	1	**Irregular Shipment Handling** • Powered Belt Conveyors • Irregular chutes • Irregular chutes (02)	 ~672 metres (68 Modules) 96 Nos 16 Nos
4	4	**Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch **Including:** • Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Dimension Scanning System • E-Stops • Hooters • Fencing	 3500mm/5500mm ~254 m(Deck 1), 254 m(Deck2) ~2.1 m/s Linear Motor 1175 mm
5	4	**Sorter Outputs** • Secondary chutes • Direct bagging chutes • Bag holding assembly • Rejection chutes • DBO • Transfer Plate • Bag holding assembly table	 60 Nos 60 Nos 60 Nos 04 Nos 04 Nos ~256 m ~229 Sqm

6	**1**	**Bag Takeaway Conveyor** • Powered Belt Conveyors	~1180 metres (74 Modules)
7	**1**	**Steel Works** • Operator and Maintenance Platform • Stairs • Stairs Platform • Motors maintenance platform • Safety gate • Monkey ladder	~6088 SQM 72 Nos 04 Nos 324 Nos 324 Nos 324 Nos

19.2 Electrical Equipment

Electricals			
1	**1**	**Consists of** Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

19.3 Control System

Components			
1	**1**	**Consists of** Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement

20. Proposed System Technical Details (SL & Bag Sorter)

20.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	**Infeed System** • Powered Belt Conveyors • Modular Belt Conveyor • 60 deg turns • 90 deg turns • Aligning conveyor • Sigulators • Powered Spiral Conveyors • Sliding chute for feedline	 ~711 metres (88 Modules) ~45 metres (4 Modules) 08 Nos 02 Nos 02 Nos 06 Nos 14 Nos 06 Nos
2	2	**Feedlines** SL Feedline Consists of (3 Nos) • Receiving Conveyor • Weighing Conveyor • Angle Merge • Short Buffer • Spacing Conveyor Bag Feedline Consists of (4 Nos) • Loading Conveyor • Weighing Conveyor • Angle Merge • Short Buffer	 1 Set 1 Set 2 Set 3 Set 4 Set 1 Set 1 Set 1 Set 2 Set
3	1	**Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch **Including:** • Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Dimension Scanning System • E-Stops • Hooters • Fencing	 3500mm/5500mm ~254 m(Deck 1), 254 m(Deck2) ~2.3 m/s Linear Motor 800 mm
4	1	**Sorter Outputs** • Spiral Chutes • Exceptional handling • DBO chute	 24 Nos 01 Nos 01 Nos
5	1	**Bag Conveyor (followed by spiral chute)** • Modular Belt Conveyors	 ~332 metres (72 Modules)

6	**1**	**Steel Works** • Operator Platform • Stairs • Motors maintenance platform • Safety gate • Monkey ladder	~1952 SQM 20 Nos 36 Nos 36 Nos 36 Nos

20.2 **Electrical Equipment**

Electricals			
1	**1**	**Consists of** Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

20.3 Control System

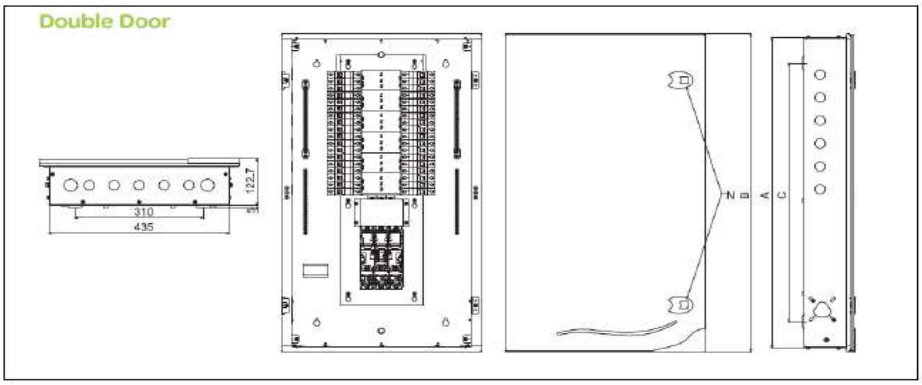
Components			
1	**1**	**Consists of** Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement

21. Electrical System

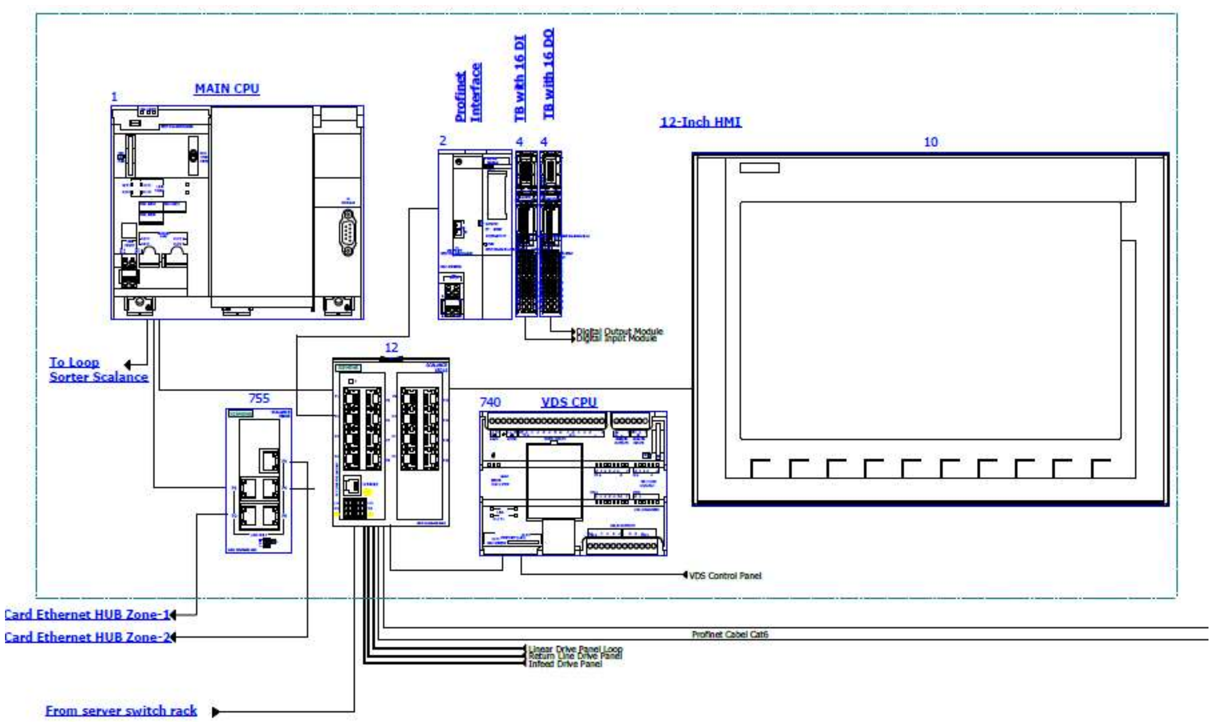
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets. PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

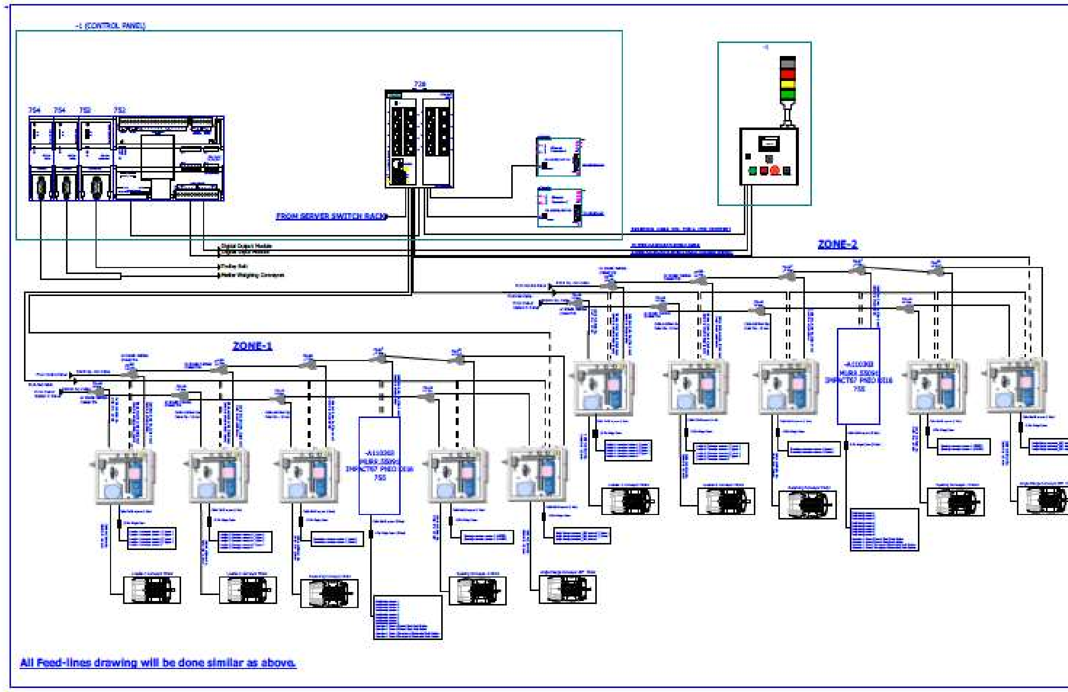
21.1 Reference Picture of Power Distribution Panel



21.2 Main Control Panel (Reference)



21.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter. The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.
All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

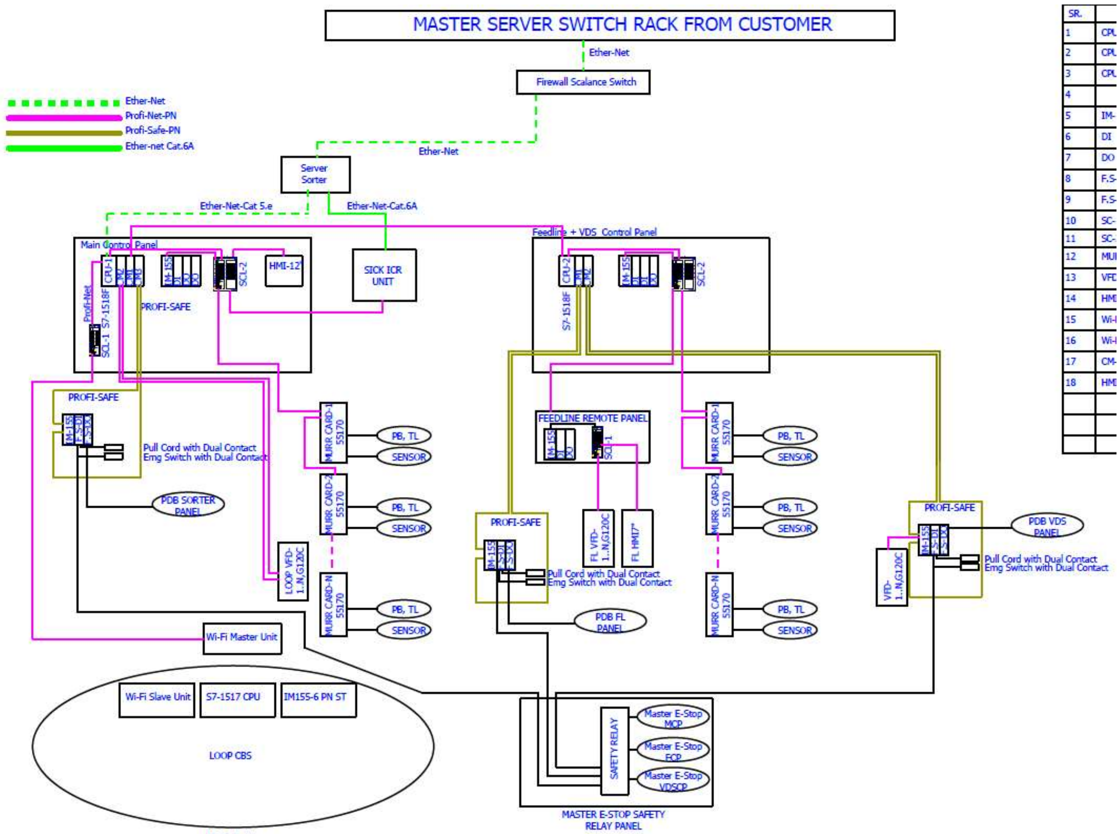
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

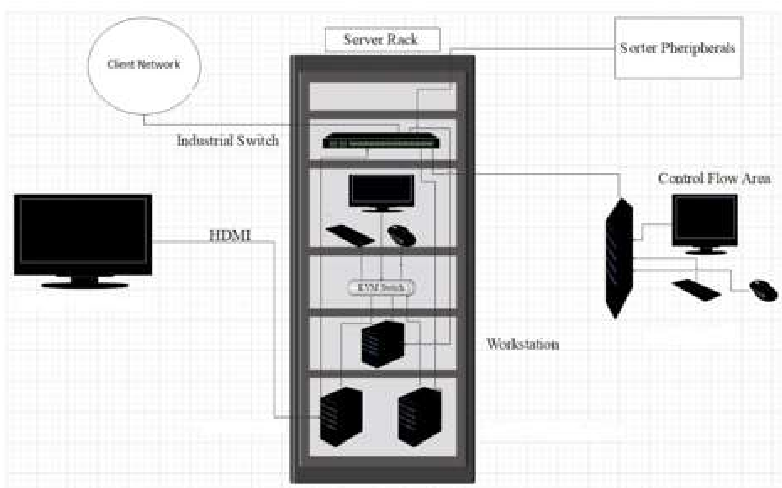
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

22. Controls Architecture (Ref.)



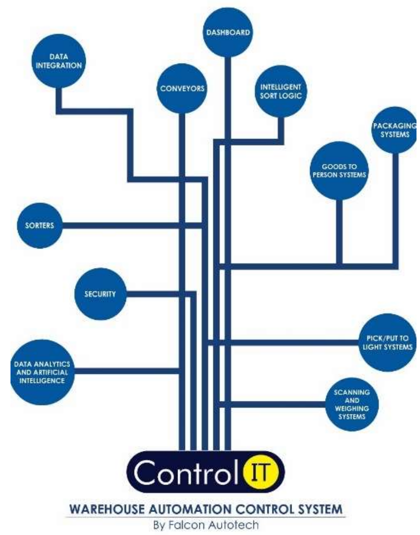
23. IT Architecture (Ref.)



23.1 HAA Server (In Flipkart Scope)

S.no	Description	Qty
	20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40	
1	**Synology_storage_DS723+** 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports, 2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays | 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers	2
2	**Dell Tower Model T440** -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	2
3	**Dell PowerEdge T40** Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3 Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	1
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Netgear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Convertor Cadyce Brand	1

24. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

25. Falcon's Visual Inspection System (SCADA)

SCADA stands for Supervisory Control and Data Acquisition. It is a system of hardware and software components that allows for remote monitoring, control, and data acquisition of industrial processes or facilities

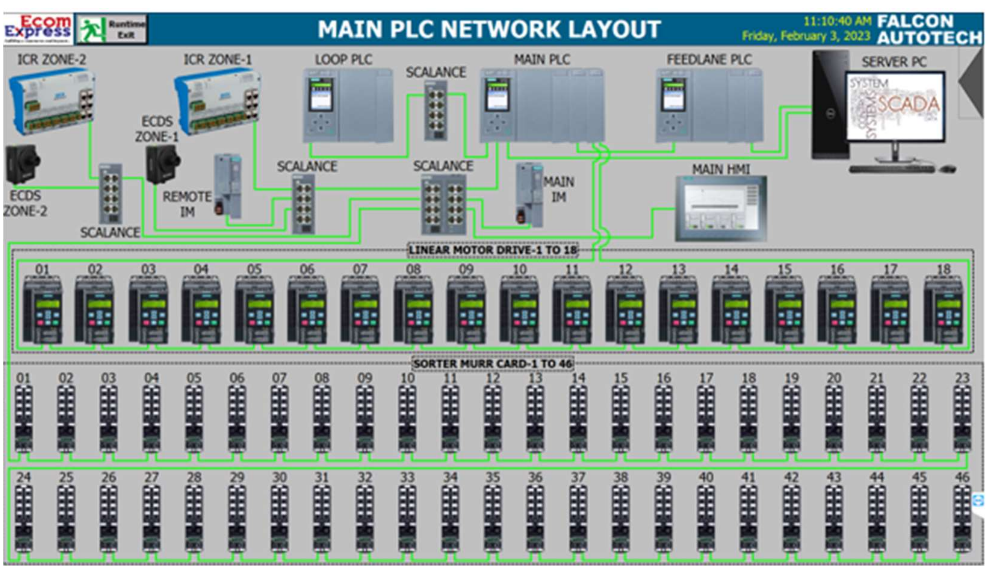
The Visualization system provided by FALCON (or SCADA) allows the monitoring and control of the different systems delivered for Flipkart Kalash. This SCADA system receives from each monitored sub-system all information on their operating status in real time.
At the system monitoring level, the functions performed are:

- Field data acquisition.
- Animated visualization of equipment.
- Representation of the operating mode of the system (nominal, contingency, etc.).
- Alarm management.
- Alarm history management.
- Diagnostic help.
- Failure detection.
- Equipment control.
- Statistics on equipment operation.
- Historical Statistical Report.
- Recording and archiving.
- Safety operator interface.

Falcon Autotech will be introducing an Innovative Service Dashboard to monitor system health real time use that for Predictive Maintenance through Existing sensors and leveraging IOT.

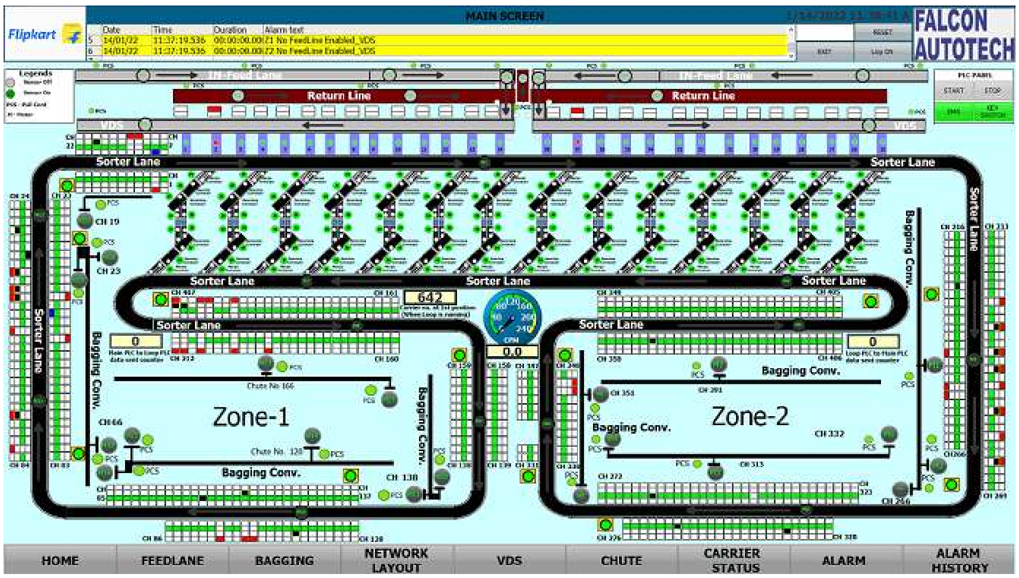
25.1 FIELD DATA ACQUISITION

The field data acquisition function is performed by the SCADA system connected to the sorters' PLC. The communication with the PLCs is done using equipped CPU cards that are able to manage the communication with the PLC on the Industrial Ethernet network, without overloading the server.The acquisition of signals from external systems is carried out via the PLC, using dry contacts.

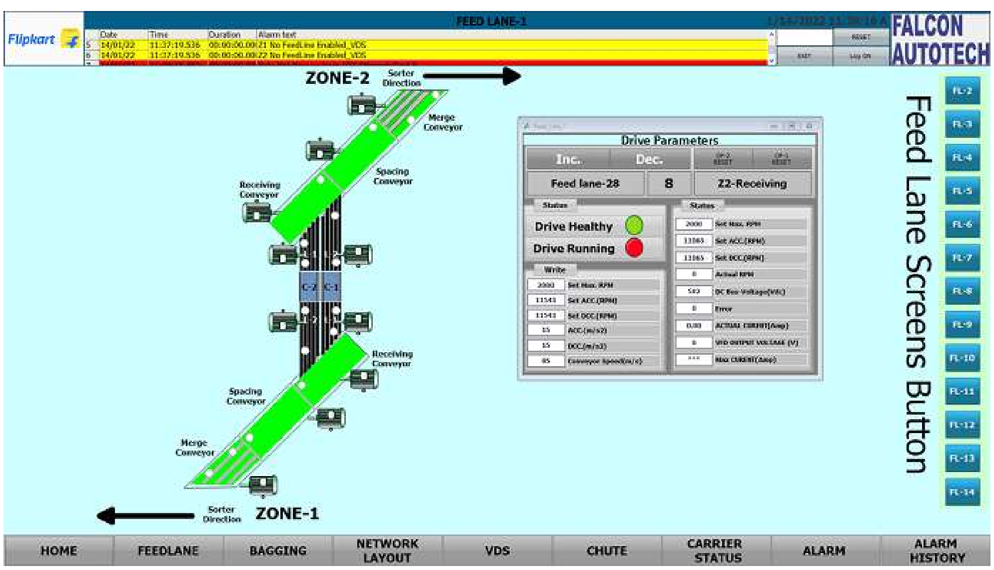


25.2 ANIMATED SYSTEM VISUALIZATION

The animated view represents the dynamic graphical user interface that allows real-time monitoring of the controlled systems and the execution of their control procedures.
The various states of the digital signals are displayed on the screen by graphic symbols. All views are web based with responsive capabilities when required so that various devices can be used to display status and information on PC. The types of information that can be displayed on each type of device will be discussed during the design phase of the project.



Example of a synoptic view



25.3 ALARM MANAGEMENT

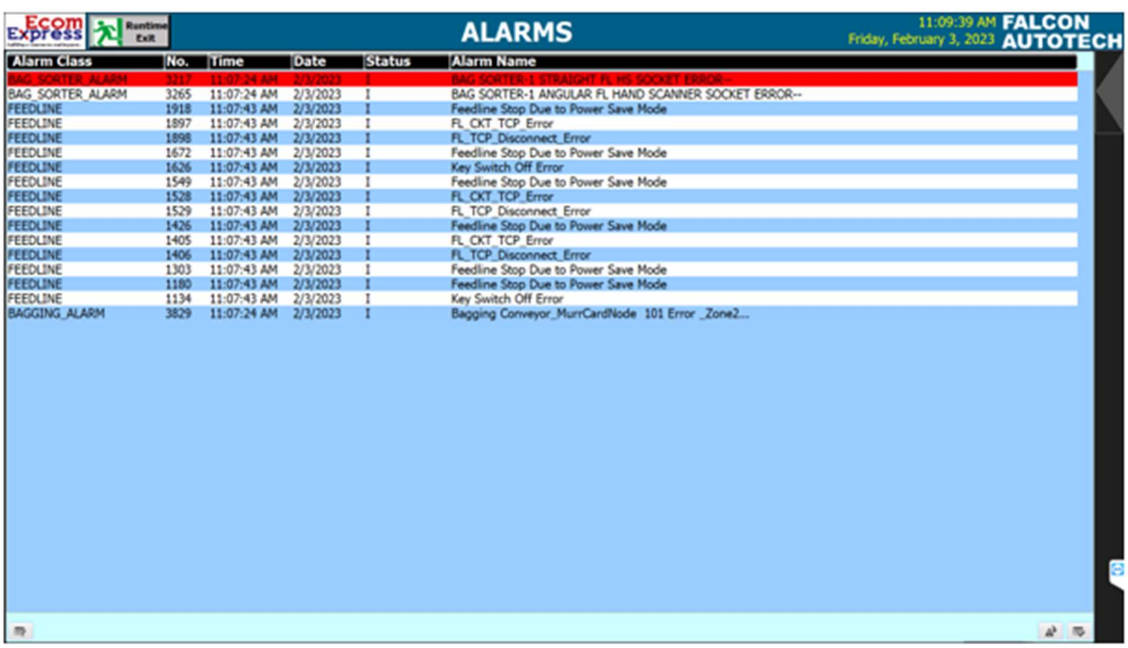
The alarm pages display a series of information to identify the nature of the alarm or event, the elements involved and the time.

The alarm management includes:

- The Alarm name.
- The date on which the above-mentioned alarm was activated.
- The moment the alarm goes off.
- The date on which the alarm is activated again.
- The moment the alarm is activated again.
- The sub-system that activated the alarm.
- The area where the alarm was triggered.
- The name of the beacon that activated the alarm.
- The alarm state.
- The alarm value.
- A complete alarm description.

The historic data of each alarm is stored in the SCADA database. This type of fault detected includes:

- ✓ Sensors.
- ✓ Drive Fault.
- ✓ Lack of monitored equipment.
- ✓ Etc.



Sample Alarm History Screen

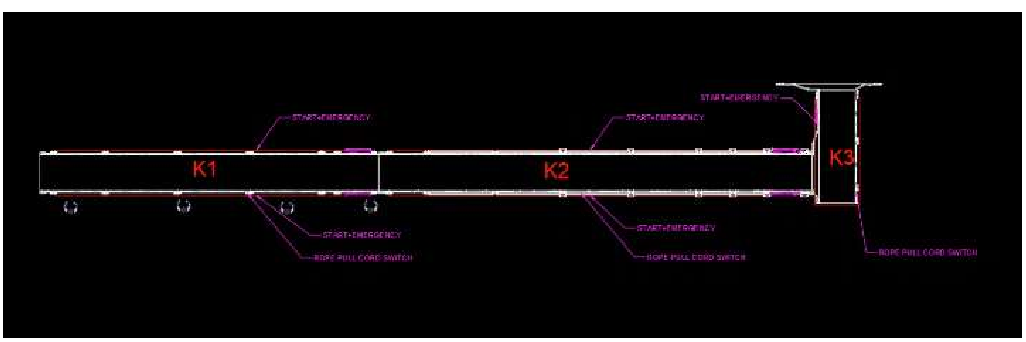
26. Key Components Make

Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Friction Wheel Drive	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Volume Scanners (VMS 4200)	SICK
Weighing Scales	Bizerba/ Mettler Toledo
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens/ Omron
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF/Equivalent
Power Transmission Systems	Vahle
HMIs	Siemens

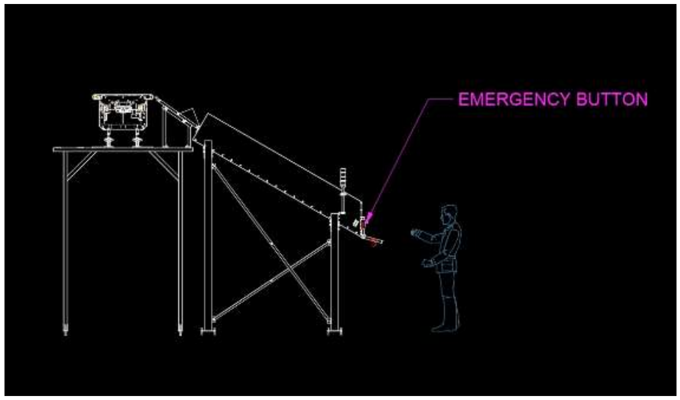
27. Principle of Safety

27.1 E-Stops

a. At Every Conveyor Module – Both sides

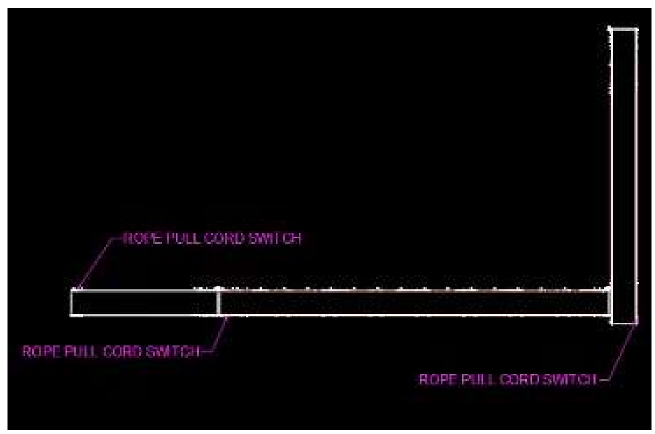


b. VDS Chutes



27.2 Pull Cords Switch-

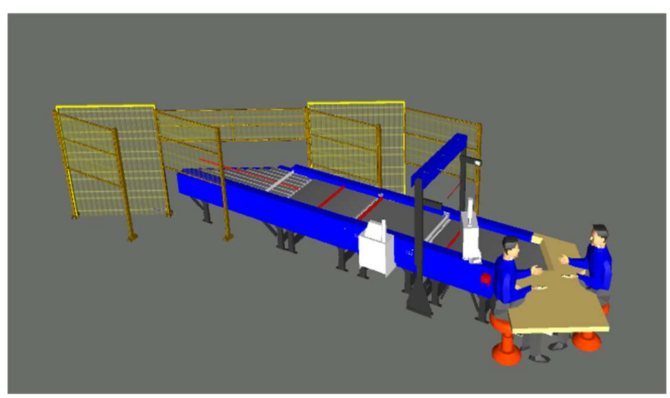
Required for Infeed & Takeout Conveyors



27.3 Fencing

a. Between Inducts

b. Between Inducts and Sorter



28. Additional Scope-

- Key management System at Inducts with Fortress
- FRLS Cables- LAPP/ Polycab/ KEI Make
- Power Saving Sensors for Conveyors
- Safety PLC
- Safety Net between Chutes.
- Energy Meters & surge arresters
- Safety Gate at ICRs

29. Program Organisation

29.1 Program/ Project Schedule

S.No.	Activity	Timeline
1	Receipt of PO	Week 0
2	Project Kick off	Week 2
3	DAP Submission	Week 5
4	DAP Approval	Week 7
5	Detailed Designing	Week 11
6	Manufacturing & Procurement	Week 19
7	Inhouse Assembly	Week 23
8	Dispatch Start	Week 24

**Above plan is tentative, detailed plan will be shared post order confirmation.*

29.2 Program & Account Management

At Falcon Autotech, our commitment revolves around fostering optimal engagement and transparency throughout the project management process. To achieve this objective, we propose the implementation of the following approach:

1. Establishment of a dedicated Flipkart Business Unit within Falcon Autotech, overseeing project management with assigned spokespersons at each hierarchical level to enhance coordination.
2. Implementation of weekly or fortnightly project routines to facilitate meticulous tracking and effective management.
3. Adoption of a robust framework for Change Risk and Opportunity Management, coupled with a governance structure to ensure proactive handling of challenges and opportunities.

30. Flipkart Responsibility

30.1 Flipkart Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- Flipkart to provide HPT/ Forklift for material movement for complete duration.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

30.2 Responsibilities of Flipkart During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

30.3 Flipkart Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.

- Provision of a list of participants for each available training course 3 days before the start of the course.

- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- Check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of FLIPKART.

31.Testing, Training, and Support

31.1Acceptance Testing

This section serves to outline the precise criteria and methodology for Falcon to validate that the installed system complies with the operational and performance requirements outlined in this proposal.

Testing necessitates comprehensive assistance from both Falcon and Flipkart Personnel, including:

- Provision of Flipkart material/products for functional testing (Falcon requires a sufficient quantity and a complete range of sizes for thorough testing).
- Accessibility to software components and applications (including accurate data for testing).
- Engagement of Flipkart labour (and equipment if necessary) to facilitate the testing process.
- Ensuring the availability of system equipment throughout the duration of each test.

31.1.1 Acceptance Test Plan

Based on the criteria outlined and mutually agreed upon by Falcon and Flipkart in this proposal, Falcon will formulate the Acceptance Test Plan (ATP). The ATP will define a systematic approach for the execution of the Acceptance Tests specified in this section. Furthermore, the ATP serves as the instrument employed by Falcon to achieve formal acceptance and obtain sign-off from Flipkart.

Falcon will furnish the ATP to Flipkart before the initiation of any tests. The ATP will encompass the following elements:

- Test prerequisites
- Daily plan/schedule
- Personnel responsibilities
- Computer data and test loads required
- Specific test procedures
- Anticipated test results

31.1.2 Minor / Major Faults

Falcon will classify any encountered faults during testing into either Minor or Major categories, as outlined below:

- Minor Faults are characterized as issues affecting a confined area or a single component with no repercussions on the overall test. These faults will be included in the System Snaglist and will not impede the ongoing testing process.

- Major Faults are defined as issues with a significant impact, leading to the inability to showcase the functionality in question. Addressing Major Faults may necessitate restarting the test.

31.1.2.1 Specific Procedure

- Falcon staff will oversee the upkeep and dissemination of the System Snaglist during Acceptance Testing.
- Upon the resolution of each issue, Flipkart will validate the correction and endorse the resolution on the System Snaglist.

Note: Validation can be conducted independently of Falcon personnel presence and should be documented through the designated signature of Flipkart on the System Snaglist.

31.1.3 Included Acceptance Tests

The following Acceptance Tests are part of the Final Acceptance Testing:

- Physical Acceptance Testing.
- Functional Acceptance Testing.
- System throughput Testing (as specified).

31.1.4 Physical Acceptance Testing

31.1.4.1 Test Definition

The Physical Test involves a walkthrough of the System, attended by both Falcon and Flipkart personnel. They will verify that the physical equipment is complete and installed according to the final drawings and specifications, typically documented by area or subsystem.

31.1.4.2 Specific Practice

- Physical Testing is conducted continuously and concurrently with installation.
- Physical Testing may extend into the commissioning and integration periods and should be completed before other Acceptance Testing.
- Physical Testing is considered complete after all System walkthroughs and the resolution of items recorded in the System Snaglist.

31.1.5 Functional Acceptance Testing

31.1.5.1 Test Definition

Functional Testing follows the completion of Falcon engineering commissioning activities. It verifies the proper function of the System according to the requirements in this Proposal and the detailed specifications developed during the Project.

Note: This testing does not repeat each test conducted during the Commissioning phase of the Project.

Both Falcon and Flipkart personnel will attend and verify that the System functions as described in this Proposal. Defects requiring correction will be logged into the System Snaglist.

Functional Testing typically includes:

- Operation/equipment interface, safety mechanisms, and interaction testing.
- Electrical/control functions such as normal start/stop, flow control, and emergency stopping.
- Controller and computer system interfaces (uploads, downloads, and interlock signals).

31.1.5.2 Specific Practice

Functional Testing is considered complete after the successful conclusion of all Functional Tests and the resolution of items recorded in the System Snaglist.

31.1.6 Throughput Testing

31.1.6.1 Test Definition

To demonstrate System's ability to meet the system throughput in the final Material Flow Throughput section of this Proposal, Falcon proposes to conduct Mechanical System Throughput Testing for each area defined in the ATP before beneficial use of the system.

31.1.6.2 Measurement

System Throughput Testing is completed by Falcon under non-production/controlled conditions, after the completion of related commissioning activities. Defects requiring correction are logged into the System Snaglist.

Tests verify that system throughputs are achieved for the areas specified in the ATP.

31.1.6.3 Specific Practice

System Throughput Testing is considered complete after the successful conclusion of all System Throughput Tests.

31.1.7 Final Acceptance Criteria

Final Acceptance is declared complete when:

- Physical Acceptance Testing, Functional Acceptance Testing, and System Throughput Testing are all complete.
- All items in the System Snaglist are resolved.
- As-built documentation is submitted.

31.1.7.1 Acceptance Test Letter

After successful completion of Acceptance Testing, Falcon will provide Flipkart a letter addressing the following:

- The System has been installed and accepted by Flipkart.
- Note: Acceptance may be conditional on the remedy of agreed-upon System Snaglist items with established completion dates.
- Confirmation of the commencement of the warranty period.
- The final payment amount required and the designated due date.

Note: Falcon has approached the testing for this project based on Falcon standard testing practices.

31.2Training

Training for the recently installed material handling system at Flipkart's facility is an integral part of the implementation process. Falcon employs a business-linked learning approach in the design, development, delivery, and evaluation of its training programs.

Falcon adheres to a well-defined process tailored to create learning content specific to Flipkart's installation and application of their systems.

31.2.1 Operations Training

System Operations Training is provided to offer participants an understanding of their area from an operational perspective. Falcon targets critical operations in the assigned functional area, covering flow control, system operation, and control devices. Discussions also emphasize the impact of individual and area performance on upstream and downstream workers.

31.2.2 Maintenance Training

The Maintenance Training curriculum encompasses mechanical, electrical, and controls aspects of the installed system. Topics include safety, system construction and installation, system operation, maintenance and repair procedures, and technical documentation.

Falcon recommends assigning personnel responsible for system maintenance to work with the installation and commissioning crews during their final weeks on-site. Formal training courses for maintenance personnel will be conducted before system startup, featuring classroom lectures, audio-visual presentations, and hands-on demonstrations with the installed system. Site tours will be organized to highlight common operational issues affecting system uptime.

31.2.3 Information System Training

Information System training covers the computer hardware and software aspects of the Automated Material Handling System. Software Application Engineering topics include system administration, data storage, computer system operation, and other computer-related tasks.

32.Life Cycle Services

Falcon can provide as an option the following Life Cycle services:

- Preventive maintenance.
- Corrective maintenance.
- Hotline service.
- On Site Support
- Supply and management of spare parts.

32.1Preventive Maintenance

A preventive maintenance package refers to planned maintenance activities and services designed to prevent equipment, machinery, or systems from breaking down or deteriorating over time. Preventive maintenance aims to identify and address potential issues before they lead to costly breakdowns or disruptions in operations.

32.2Corrective Maintenance

Corrective maintenance, also known as corrective maintenance, is a type of maintenance activity that focuses on addressing and repairing equipment or system failures, defects, or malfunctions when they occur. Unlike preventive maintenance, which aims to prevent breakdowns through scheduled inspections and maintenance tasks, Corrective maintenance is reactive and comes into play when a problem has already arisen.

32.3Hotline Support

The hotline service access is available 24 X 7

The main function is to dispatch the call to the proper qualified engineer for immediate answer. If the Falcon qualified engineer is not available, the call is transferred to a back-up qualified engineer and finally an answering machine. In the last case, the FLIPKART personnel shall be contacted within the time frame described in the service level.

The Falcon qualified engineer will support FLIPKART maintenance personnel by using a remote access (via VPN) to the equipment/systems.

32.4 On Site Support

To adhere to the committed System uptime, dedicated On site team will be available at Site covering all shifts. Team comprises of Supervisor in general shift and 1 Electrical & 1 Mechanical technician in each Shift.

32.5 Supply and Management of Spare Parts

This paragraph describes the principles and procedures that will be implemented between Falcon and FLIPKART, with regard to the spare parts management policy.

- Spare parts orders.
- On completion of the DAP, when the configuration will be finalized a final and detailed list, which will be extended to include all spare and wear parts, will be presented to FLIPKART.
- The exact number of spare parts will be determined with FLIPKART, according to its stock strategy on the basis of the list of recommended spare parts.

33. Commercial Terms

33.1 Price Sheet

Non-Large Shipment Sorter

S. No	Component	Set	Price (INR)
1	Double Deck Loop CBS System	4	₹ 76,26,64,100
2	Conveyor Automation	1	₹ 40,73,43,521
3	Output Chutes	4	₹ 11,60,72,479
4	Steel Works	1	₹ 12,09,87,778
5	Spiral Conveyors	1	₹ 31,09,80,008
6	Toolkit	4	₹ 3,19,200
7	Software Packages	1	₹ 44,00,000
8	Installation & Commissioning	1	₹ 6,87,34,683
9	Packaging & forwarding	1	₹ 2,57,75,506
10	Project Management	1	₹ 3,63,45,545
Grand Total (INR)			**₹ 1,85,36,22,820**

Semi-Large and Bag Sorter

S. No	Component	Set	Price (INR)
1	XL Loop CBS System	1	₹ 21,45,89,057
2	Conveyor Automation	1	₹ 17,75,55,387
3	Output Chutes	1	₹ 36,25,029
4	Steel Works	1	₹ 2,93,42,486
5	Spiral Conveyors	1	₹ 3,65,79,583
6	Toolkit	1	₹ 79,800
7	Software Packages	1	₹ 13,00,000
8	Installation & Commissioning	1	₹ 1,84,70,854
9	Packaging & forwarding	1	₹ 69,26,570
10	Project Management	1	₹ 97,69,375
Grand Total (INR)			**₹ 49,82,38,141**

The Total Price is to be understood.

- Freight - Excluded
- Taxes as applicable
- Packing and Forwarding - Included
- Price is valid for 60 days from the date of proposal.

*Material Unloading & Laydown area is in customer scope.

33.2Payment Terms

Payment Percentage	Stage
20%	Advance along with PO against ABG
50% + taxes	Material received at Site
20% + 50% 0f I&C	Against erection
10% + 50% 0f I&C	Acceptance and complete handover against PBG

34.Warranty Period

Falcon's offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
    - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
    - Negligence or abnormal use of equipment.
    - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
    - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

35. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Server System**
- **Mobile Carts**
- **Construction Power**
- **Steel Works- If not specified**
- **Collection Trolleys below chutes**
- **Secondary Sortation**
- **PC for SCADA.**
- **Fans at chutes & Inducts**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.